AGENT-BASED MODELLING OF FINANCIAL MARKETS:

THE NEW KIND OF SCIENCE MARKET MODEL

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1. Introduction

Financial markets are complex systems, which mean more than complicated systems. Although there is no universally accepted definition of 'complexity', most people would agree that any candidate complex system should have most or all of the following ingredients:

- Feedback. The nature of the feedback can change with time – for example, becoming positive one moment and negative the next – and may also change in magnitude and importance. It may operate at the macroscopic level or microscopic, or both. The presence of feedback implies that on some level, buried in the details of the dynamics, the system is remembering the past and responding to it, albeit in a non-trivial way.
- Non-Stationarity. We cannot assume that the dynamical or statistical properties observed in the system's past, will remain unchanged in the future.
- Many interacting agents. The system contains many components or participants known as agents, which interact in possibly time-dependent ways. Their individual behaviour will respond to the feedback of information, which is possibly limited, from the system as a whole and/or from other agents. Since this agents may effectively be competing to win, it is unlikely that there any such thing as typical agent.
- Adaptation. An agent can adapt its behaviour in the hope of improving its performance.

- Evolution. The entire multi-agent population evolves, driven by an ecology of agents who interact and adapt under the influence of feedback. The system typically remains far from equilibrium and can exhibit extreme behaviour.
- Single realization. The system under study is a single realization, implying that the standard techniques whereby averages over time are equated to averages over ensembles, may not work.
- Open system. The system is coupled to the environment, hence it is hard to distinguish between endogenous (ie outside) and endogenous (ie internal, self generated) effects.

We will see that our model mixes uses all of these ingredients to capture the financial market complexity and give a better understanding of real life behaviour and stylized facts.

In my opinion market nature is neither deterministic, nor chaotic and nor completely random, fractals and has very special scaling properties (power laws), markets are a mixture of these different behaviours. We will that mixing all the ingredients we are able to mimic real markets complex behaviour.

Introducing the New Kind of Science Market Model

New Kind of Science Artificial Market Model is an artificial market model using simple agents pursuing a small, enumerated set of trading strategies.

Daily model prices are based on the buy, sell and hold decisions of a population of heterogeneous agents, representing active traders. Each type of trader decides the day's trade (buy, sell or hold) according an strategy rule incorporating external news and two indicators based on the market price history - a "fundamental" or price level indicator, and a "technical" or recent trend indicator. Each of these indicators, including the news, can give strong or weak signals in either direction.

The strategy determines the recommended trade from the indicators; the strength of the signals determines the desired trade amount or volume. There is considerable possible variation in these simple strategies - 6561 (3 to the 8th power) distinct strategies are possible with these rules. In addition, the profit and loss (P&L) and current stock position of each strategy currently in use are tracked throughout, and traders are more likely to trade if ahead in P&L terms, and if reversing a current stock position (long or short) rather than adding to it.

There is some stochastic component to trade timing, and to the external news. Prices are a straightforward result of the day's desired trades - the price adjusts to reflect the order imbalance, and all desired trades then fill at the new price. The model can also, optionally, incorporate "herding" changes in the strategies in use, traders switch from the least successful strategies in use to better ones.

The output of the model is the evolution of the price of a theoretical asset, along with positions and P&Ls experienced by the different strategies used. We study how the market behaves with different mixes and weights of traders, using different strategies, as well as sensitivity to different inputs.

Moreover, we added features to the model sequentially and can speak to the question which features in the resulting behaviour depend on which internal features of the model. We looked in the price evolution for stylized facts seen in real world financial markets. Real financial markets are complex systems, showing non-stationary in the probability distributions especially changing volatility through time, non-trivial scaling

and serial correlation, and non-gaussian distributions of returns including high and variable kurtosis and fat tails. We were able to find all of these signals in our model.
NKS Artificial Market Model give us a basic set of stylized facts observed in many real markets:

- Fat - tailed PDF of price changes, with non trivial scaling properties
- Slow decay of the autocorrelation of absolute value of price changes
- Volatility clustering
- Fast decay of the autocorrelation of price - changes

We show that a combination of very simple trading rules, a mixture of different traders, with feedback and adaptation effects and some randomness through news and stochastic order arrival to the markets give the real market's very complicated behaviour.
Before introducing the model in Chapters 6 and 7, we start in Chapter 2 with a description of the theory and simple models, we describe in Chapter 3 the stylized facts observed in financial markets and we then describe in Chapter 4, three models Cont-Bouchaud, Bak et al. and Lux-Marchesi and a brief historical analysis of the artificial markets in Chapter 5.
In Chapter 6 we introduce a simple market equation, and we analyse the behaviour of these equation using different values and behaviours of the parameters. In chapter 7 we describe, define and analyse our model and on chapter 8 we discuss extensions and further work.

2. Theory and simple models

Agent-based modelling has become a popular methodology in social sciences, particularly in economics. Here we focus on the agent-based modelling of financial markets. The very idea of describing markets with models of interacting agents (traders, investors) does not fit well with the classical financial theory that is based on the notions of efficient markets and rational investors. However, it has become obvious that investors are neither perfectly rational nor have homogeneous expectations of the market trends. Agent-based modelling proves to be a flexible framework for a realistic description of the investor adaptation and decision-making process.

The paradigm of agent-based modelling applied to financial markets implies that trader actions determine price. This concept is similar to that of statistical physics within which the thermodynamic (macroscopic) properties of the medium are described via molecular interactions.

A noted expansion of the microscopic modelling methodology into social systems is the minority game (see Zhang and Challet). Its development was inspired by the famous El Farol's bar problem.

This problem considers a number of patrons N willing to attend a bar with a number of seats Ns. It is assumed that Ns < N and every patron prefers to stay at home if he expects that the number of people attending the bar will exceed Ns. There is no communication among patrons and they make decisions using only information on past attendance and different predictors (e.g., attendance today is the same as yesterday, or is some average of past attendance).

The minority game is a simple binary choice problem in which players have to choose between two sides, and those on the minority side win. Similarly to the El Farol's bar problem, in the minority game there is no communication among players and only a given set of forecasting strategies defines player decisions. The minority game is an interesting stylized model that may have some financial implications. But we shall focus further on the models derived specifically for describing financial markets.

In the known literature, early work on the agent-based modelling of financial markets can be traced back to 1980. In this paper, Beja and Goldman considered two major trading strategies, value investing and trend following. In particular, they showed that system equilibrium may become unstable when the number of trend followers grows.

Since then, many agent-based models of financial markets have been developed, we will review some of them in the following chapters.

We divide these models into two major groups. In the first group, agents make decisions based on their own predictions of future prices and adapt their beliefs using different predictor functions of past returns. The principal feature of this group is that price is derived from the supply-demand equilibrium. Therefore, we call this group the adaptive equilibrium models. In the other group, the assumption of the equilibrium price is not employed. Instead, price is assumed to be a dynamic variable determined via its empirical relation to the excess demand. We call this group the nonequilibrium price models. In the following two sections, we discuss two instructive examples for both groups of models, respectively. Finally, Section 2.3 describes a non-equilibrium price model that is derived exclusively in terms of observable variables

2.1 Adaptive Equilibrium models

In this group of models (Brock-Hommes, Lebaron-Arthur, Levy and Solomon , Chiarella- He) agents can invest either in the risk free asset (bond) or in the risky asset (e.g., a stock market index). The risk-free asset is assumed to have an infinite supply and a constant interest rate. Agents attempt to maximize their wealth by using some risk aversion criterion. Predictions of future return are adapted using past returns. The solution to the wealth maximization problem yields the investor demand for the risky asset. This demand in turn determines the asset price in equilibrium. Let us formalize these assumptions using the notations from Chiarella-He. The return on the risky asset at time t is defined as

$$\rho_t = (p_t - p_{t-1} + y_t)/p_{t-1} \quad (2.1.1)$$

where pt and yt are (ex-dividend) price and dividend of one share of the risky asset, respectively. Wealth dynamics of agent i is given by

$$W_{i,t+1} = R(1-\pi_{i,t})W_{i,t} + \pi_{i,t}W_{i,t}(1+\rho_{t+1}) = W_{i,t}\left[R + \pi_{i,t}(\rho_{t-1} - r)\right] \quad (2.1.2)$$

where r is the interest rate of the risk-free asset, $R = 1 + r$ and $\pi_{i,t}$, t is the proportion of wealth of agent i invested in the risky asset at time t. Every agent is assumed to be a taker of the risky asset at price that is established in the demand-supply equilibrium. Let us denote $E_{i,t}$ and $V_{i,t}$ the ‘‘beliefs’’ of trader i at time t about the conditional expectation of wealth and the conditional variance of wealth, respectively. It follows from (2.1.2) that

$$\begin{aligned} E_{i,t}\left[W_{i,t+1}\right] &= W_{i,t}\left[R + \pi_{i,t}(E_{i,t}\left[\rho_{t+1}\right] - r)\right], \quad (2.1.3) \\ V_{i,t}\left[W_{i,t+1}\right] &= \pi^2{}_{i,t}W^2{}_{i,t}V_{i,t}\left[\rho_{i,t+1}\right] \end{aligned}$$

Also, every agent i believes that return of the risky asset is normally distributed with mean $E_{i,t}\left[\rho_{t+1}\right]$ and variance $V_{i,t}\left[\rho_{t+1}\right]$. Agents choose the proportion $\pi_{i,t}$ of their wealth to invest in the risky asset, which maximizes the utility function U

$$\max_{\pi_{i,t}}\left\{E_{i,t}\left[U(W_{i,t+1})\right]\right\} \quad (2.1.4)$$

The utility function chosen in [9, 10] is

$$U(W_{i,t}) = \log(W_{i,t}) \quad (2.1.5)$$

Then demand $\pi_{i,t}$ that satisfies (2.1.4) equals

$$\pi_{i,t} = \frac{E_{i,t}[\rho_{t+1}] - r}{V_{i,t}[\rho_{t+1}]} \quad (2.1.6)$$

Another utility function used in the adaptive equilibrium models employs the so-called constant absolute risk aversion (CARA) function:

$$U(W_{i,t}) = E_{i,t}[W_{i,t+1}] - \frac{a}{2} V_{i,t}[W_{i,t+1}] \quad (2.1.7)$$

where a is the risk aversion constant. For the constant conditional variance $V_{i,t} = \sigma^2$, the CARA function yields the demand

$$\pi_{i,t} = \frac{E_{i,t}[\rho_{t+1}] - r}{a\sigma^2} \quad (2.1.8)$$

The number of shares of the risky asset that corresponds to demand $\pi_{i,t}$ equals

$$N_{i,t} = \pi_{i,t} W_{i,t} / p_t \quad (2.1.9)$$

Since the total number of shares assumed to be fixed $(\sum_i N_{i,t} = N = const)$, the market-clearing price equals

$$p_t = \frac{1}{N} \sum_i \pi_{i,t} W_{i,t} \quad (2.1.10)$$

The adaptive equilibrium model described so far does not contradict the classical asset pricing theory. The new concept in this model is the heterogeneous beliefs. In its general form:

$$\begin{aligned} E_{i,t}[\rho_{t+1}] &= f_i(\rho_{t-1}, \ldots., \rho_{t-L_i}), \quad (2.1.11) \\ V_{i,t}[\rho_{t+1}] &= g_i(\rho_{t-1}, \ldots., \rho_{t-L_i}) \end{aligned}$$

The deterministic functions f_i and g_i depend on past returns with lags up to Li and may vary for different agents.

While variance is usually assumed to be constant $(g_i = \sigma^2)$, several trading strategies f_i are discussed in the literature. First, there are fundamentalists who use analysis of the business fundamentals to make their forecasts on the risk premium δ_F

$$E_{F,t}[\rho_{t+1}] = r + \delta_F \quad (2.1.12)$$

In simple models, the risk premium $\delta_F > 0$ is a constant but it can be a function of time and/or variance in the general case. Another major strategy is momentum trading (traders who use it are often called chartists). Momentum traders use history of past returns to make their forecasts. Namely, their strategy can be described as

$$E_{M,t}[\rho_{t+1}] = r + \delta_M + \sum_{k=1}^{L} a_k \rho_{t-k} \quad (2.1.13)$$

where $\delta_M > 0$ is the constant component of the momentum risk premium and $a_k > 0$ are the weights of past returns ρ_{t-k}. Finally, contrarians employ the strategy that is formally similar to the momentum strategy

$$E_{C,t}[\rho_{t+1}] = r + \delta_C + \sum_{k=1}^{L} b_k \rho_{t-k} \quad (2.1.14)$$

with the principal difference that all b_K are negative. This implies that contrarians expect the market to turn around (e.g., from bull market to bear market).

An important feature of adaptive equilibrium models is that agents are able to analyze performance of different strategies and choose the most efficient one. Since these strategies have limited accuracy, such adaptability is called bounded rationality.

In the limit of infinite number of agents, Brock and Hommes offer a discrete analog of the Gibbs probability distribution for the fraction
of traders with the strategy i :

$$n_{i,t} = \exp\left[\beta\left(\Phi_{i,t-1} - C_i\right)\right] / Z_t , Z_t = \sum_i \exp\left[\beta\left(\Phi_{i,t-1} - C_i\right)\right] \quad (2.1.15)$$

In (2.1.15), Ci $\geq$ 0 is the cost of the strategy i, the parameter b is called the intensity of choice, and $\Phi_{i,t}$ is the fitness function that characterizes the efficiency of strategy i. The natural choice for the fitness function is

$$\Phi_{i,t} = \gamma\Phi_{i,t-1} + \varphi_{i,t} , \varphi_{i,t} = \pi_{i,t}(W_{i,t} - W_{i,t-1}) / W_{i,t-1} \quad (2.1.16)$$

where $0 \leq \gamma \leq 1$ is the memory parameter that retains part of past performance in the current strategy.

Adaptive equilibrium models have been studied in several directions. Some work has focused on analytic analysis of simpler models.

In particular, the system stability and routes to chaos have been discussed in these models. In the meantime, extensive computational modelling has been performed in Levy-Solomon and particularly for the so-called Santa Fe artificial market, in which a significant number of trading strategies were implemented (LeBaron-Arthur-Palmer , 1999).

2.2 NON-EQUILIBRIUM PRICE MODELS

The concept of market clearing that is used in determining price of the risky asset in the adaptive equilibrium models does not accurately reflect the way real markets work. In fact, the number of shares involved in trading varies with time, and price is essentially a dynamic variable. A simple yet reasonable alternative to the price-clearing paradigm is the equation of price formation that is based on the empirical relation between price change and excess demand.

Different agent decision-making rules may be implemented within this approach. Here the elaborated model offered by Lux is described. In this model, two groups of agents, namely chartists and fundamentalists, are considered. Agents can compare the efficiency of different trading strategies and switch from one strategy to another.

Therefore, the numbers of chartists, $n_c(t)$, and fundamentalists, $n_f(t)$, vary with time while the total number of agents in the market N is assumed constant. The chartist group in turn is sub-divided into optimistic (bullish) and pessimistic (bearish) traders with the numbers

$$n_c(t) + n_f(t) = N, n_+(t) + n_-(t) = n_c(t) \quad (2.2.1)$$

Several aspects of trader behaviour are considered. First, the chartist decisions are affected by the peer opinion (so-called mimetic contagion). Secondly, traders change strategy while seeking optimal performance. Finally, traders may exit and enter markets. The bullish chartist dynamics is formalized in the following way:

$$dn_+ / dt = (n_- p_{+-} - n_+ p_{+-})(1 - n_f / N) + \quad \text{Mimetic Contagion}$$

$$n_f n_+ (p_{+f} - p_{f+}) / N + \quad \text{Changes of strategy} \quad (2.2.2)$$

$$(b - a) n_+ \quad \text{Markey entry and exit}$$

Here, $p_{\alpha\beta}$ denotes the probability of transition from group β to group α. Similarly, the bearish chartist dynamics is given by

$$dn_- / dt = (n_+ p_{-+} - n_- p_{+-})(1 - n_f / N) + \quad \text{Mimetic contagion}$$

$$n_f n_-(p_{-f} - p_{f-})/N + \text{ Changes of strategy } \quad (2.2.3)$$

$$(b-a)n_- \quad \text{Market entry and exit}$$

It is assumed that traders entering the market start with the chartist strategy. Therefore, constant total number of traders yields the relation $b = aN/n_c$. Equations (2.2.1)–(2.2.3) describe the dynamics of three trader groups (n_f, n_+, n_-) assuming that all transfer probabilities $p_{\alpha\beta}$ are determined. The change between the chartist bullish and bearish mood is given by

$$\begin{aligned} p_{-+} &= 1/p_{-+} = v_1 \exp(-U_1), \qquad (2.2.4) \\ U_1 &= \alpha_1(n_+ - n_-)/n_c + (\alpha_2/v_1)dP/dt \end{aligned}$$

where v_1, α_1 and α_2 are parameters and P is price. Conversion of fundamentalists into bullish chartists and back is described with

$$\begin{aligned} p_{+f} &= 1/p_{f+} = v_1 \exp(-U_{21}), \qquad (2.2.5) \\ U_{21} &= \alpha_3((r - v_2^{-1}dP/dt)/P - R - s\left|(P_f - P)/P\right|) \end{aligned}$$

where v_2 and α_3 are parameters, r is the stock dividend, R is the average revenue of economy, s is a discounting factor 0 < s < 1, and Pf is the fundamental price of the risky asset assumed to be an input parameter. Similarly, conversion of fundamentalists into bearish chartists and back is given by

$$\begin{aligned} p_{-f} &= 1/p_{f-} = v_2 \exp(-U_{22}), \qquad (2.2.6) \\ U_{22} &= \alpha_3(R - (r - v_2^{-1}dP/dt)/P - s\left|(P_f - P)/P\right|) \end{aligned}$$

Price P in (2.2.4)–(2.2.6) is a variable that still must be defined. Hence, an additional equation is needed in order to close the system (2.2.1)–(2.2.6). As it was noted previously, an empirical relation between the price change and the excess demand constitutes the specific of the non-equilibrium price models

$$dP / dt = \beta D_{ex} \quad (2.2.7)$$

In the Lux model , the excess demand equals

$$D_{ex} = t_c(n_+ - n_-) + \gamma n_f (P_f - P) \quad (2.2.8)$$

The first and second terms in the right-hand side of (2.2.8) are the excess demands of the chartists and fundamentalists, respectively; β , t_c and γ are parameters.

The system (2.2.1)–(2.2.8) has rich dynamic properties determined by its input parameters. The system solutions include stable equilibrium, periodic patterns, and chaotic attractors. Interestingly, the distributions of returns derived from the chaotic trajectories may have fat tails typical for empirical data. Particularly in Lux- Marchesi, the Lux model was modified to describe the arrival of news in the market, which affects the fundamental price. This process was modeled with the Gaussian random variable $\varepsilon(t)$ so that

$$\ln P_f(t) - \ln P_f(t-1) = \varepsilon(t) \quad (2.2.9)$$

the modelling results exhibited the power-law scaling and temporal volatility dependence in the price distributions.

2.3 THE OBSERVABLE VARIABLES MODEL

The models discussed so far are capable of reproducing important features of financial market dynamics. Yet, one may notice a degree of arbitrariness in this field. The number of different agent types and the rules of their transition and adaptation vary from one model to another. Also, little is known about optimal choice of the model parameters (see Lebaron and Wagner). As a result, many interesting properties, such as deterministic chaos, may be the model artifacts rather than reflections of the real world.

A parsimonious approach to choosing variables in the agent-based modelling of financial markets was offered in A.B Schmidt. Namely, it was suggested to derive agent-based models exclusively in terms of observable variables. Note that the notion of observable data in finance should be discerned from the notion of publicly available data. While the transaction prices in regulated markets are publicly available, the market microstructure is not (many times). Still, every event in the financial markets that affects the market microstructure (such as quote submission, quote cancellation, transactions, etc.) is recorded and stored for business and legal purposes. This information allows one to reconstruct the market microstructure at every moment. We define observable variables in finance as those that can be retrieved or calculated from the records of market events. Whether these records are publicly available at present is a secondary issue. More importantly, these data exist and can therefore potentially be used for calibrating and testing the theoretical models.

The numbers of agents of different types generally are not observable. Indeed, consider a market analog of ‘‘Maxwell’s Demon’’ who is able to instantly parse all market events. The Demon cannot discern ‘‘chartists’’ and ‘‘fundamentalists’’ in typical situations, such as when the current price, being lower than the fundamental price, is growing.

In this case, all traders buy rather than sell. Similarly, when the current price, being higher than the fundamental price, is falling, all traders sell rather than buy.

Only price, the total number of buyers, and the total number of sellers are always observable. Whether a trader becomes a buyer or seller can be defined by mixing different behaviour patterns in the trader decision-making rule. Let us describe a simple non-equilibrium price model derived along these lines.

We discern ‘‘buyers’’ (+) and ‘‘sellers’’ (-). Total number of traders is N

$$N_+(t) + N_-(t) = N \quad (2.3.1)$$

The scaled numbers of buyers, $n_+(t) = N_+(t)/N$, and sellers $n_-(t) = N_-(t)/N$, are described with equations

$$\begin{aligned} dn_+/dt &= v_{+-}n_- - v_{-+}n_+ \quad (2.3.2) \\ dn_-/dt &= v_{-+}n_+ - v_{+-}n_- \end{aligned}$$

The factors v_{+-} and v_{-+} characterize the probabilities for transfer from seller to buyer and back, respectively

$$v_{+-} = 1/v_{-+} = v\exp(U), U = \alpha p^{-1}dp/dt + \beta(1-p) \quad (2.3.3)$$

Price p(t) is given in units of its fundamental value. The first term in the utility function, U, characterizes the ‘‘chartist’’ behaviour while the second term describes the ‘‘fundamentalist’’ pattern. The factor n has the sense of the frequency of transitions between seller and buyer behaviour. Since $n_+(t) = 1 - n_-(t)$, the system (2.3.1)–(2.3.2) is reduced to the equation

$$dn_+/dt = v_{+-}(1-n_+) - v_{-+}n_+ \quad (2.3.4)$$

The price formation equation is assumed to have the following form

$$dp/dt = \gamma D_{ex} \quad (2.3.5)$$

where the excess demand, D_{ex}, is proportional to the excess number of buyers

$$D_{ex} = \delta(n_+ - n_-) = \delta(2n_+ - 1) \quad (2.3.6)$$

The model described above is defined with two observable variables, nþ(t) and p(t). In equilibrium, its solution is $n_+ = 0.5$ and p=1. The necessary stability condition for this model is

$$\alpha\delta\gamma\nu \leq 1 \quad (2.3.7)$$

The typical stable solution for this model (relaxation of the initially perturbed values of n_+ and p) is given in Figure 12.1. Lower values of α and γ suppress oscillations and facilitate relaxation of the initial perturbations. Thus, the rise of the ‘‘chartist’’ component in the utility function increases the price volatility. Numerical solutions with the values of a and g that slightly violate the condition (2.3.7) can lead to the limit cycle providing that the initial conditions are very close to the equilibrium values . Otherwise, violation of the condition (2.3.7) leads to system instability, which can be interpreted as a market crash.

The basic model (2.3.1)–(2.3.6) can be extended in several ways. First, the condition of the constant number of traders (2.3.1)

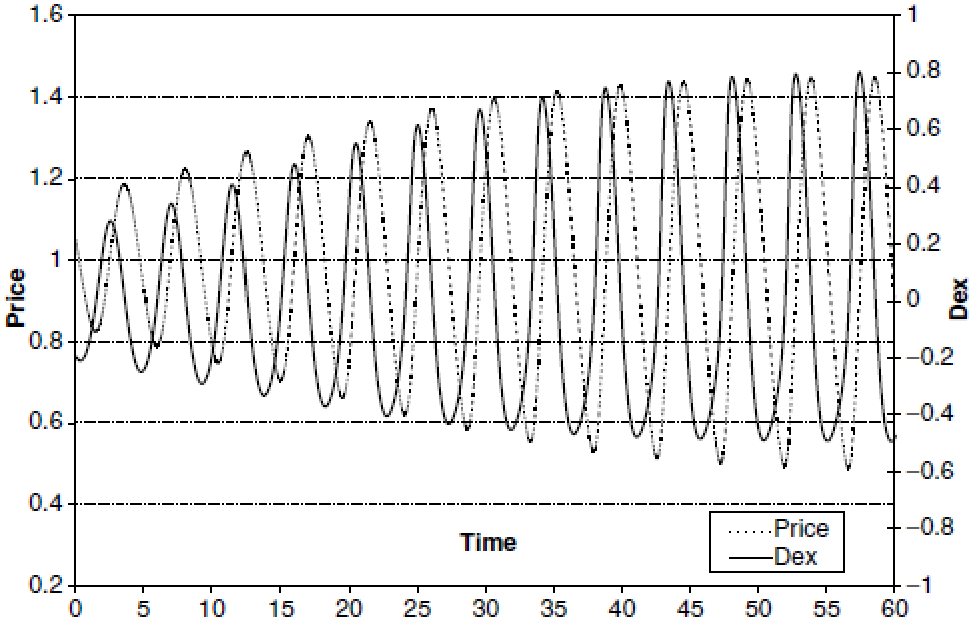


Figure 1.1 Dynamics of excess demand (Dex) and price for the model (12.4.5)–(12.4.7) with a $\alpha = \beta = \gamma = 1, n_+(0) = 0.4$ and $\mathrm{p}(0) = 1.05$

can be dropped. The system has three variables (n_+, n_-, p) and therefore may potentially describe deterministic chaos. Also, one can randomize the model by adding noise to the utility function (2.3.3) or to the price formation equation (2.3.5). Interestingly, the latter option may lead to a negative correlation between price and excess demand, which is not possible for the deterministic equation (2.3.5).

Market liquidity implies the presence of traders on both the bid/ask sides of the market. In emergent markets (e.g., new electronic auctions), this may be a matter of concern. To address this problem, the basic model (2.3.1)–(2.3.6) was expanded in the following way:

$$dn_+ / dt = v_{+-} n_- - v_{-+} n_+ + \sum R_{+i} + \rho_+ \quad (2.3.8)$$
$$dn_- / dt = v_{-+} n_+ - v_{+-} n_- + \sum R_{-i} + \rho_-$$

The functions $R_{\pm i}\left(i = 1,2,...M\right)$ and $\rho_{\pm}$ are the deterministic and stochastic rates of entering and exiting the market, respectively. Let us consider three deterministic effects that define the total number of traders. First, we assume that some traders stop trading immediately after completing a trade as they have limited resources and/or need some time for making new decisions

$$R_{+1} = R_{-1} = -bn_+ n_-, b > 0 \quad (2.3.9)$$

Also, we assume that some traders currently present in the market will enter the market again and will possibly bring in some ‘‘newcomers.’’

Therefore, the inflow of traders is proportional to the number of traders present in the market

$$R_{+2} = R_{-2} = a(n_+ + n_-), a > 0 \quad (2.3.10)$$

Lastly, we account for ‘‘unsatisfied’’ traders leaving the market. Namely, we assume that those traders who are not able to find the trading counterparts within a reasonable time exit the market

$$R_{+3} = \begin{cases} -c(n_+ - n_-) & \text{if } \mathrm{n}_+ > n_- \\ 0, & \text{if } \mathrm{n}_+ \le n_- \end{cases} \quad (2.3.11)$$

$$R_{+3} = \begin{cases} -c(n_- - n_+) & \text{if } \mathrm{n}_- > n_+ \\ 0, & \text{if } \mathrm{n}_- \le n_+ \end{cases} \quad (2.3.12)$$

We call the parameter c > 0 the ‘‘impatience’’ factor. Here, we neglect the price variation, so that $v_{+-} = v_{-+} = 0$. We also neglect the stochastic rates $\rho_{\pm}$ .Let us specify

$$n_+(0) - n_-(0) = \delta > 0 \quad (2.3.13)$$

Then equations (2.3.8) have the following form

$$dn_+ / dt = a(n_+ - n_-) + bn_+ n_- - c(n_+ - n_-) \quad (2.3.14)$$
$$dn_+ / dt = a(n_+ - n_-) + bn_+ n_-$$

The equation for the total number of traders $n = n_+ + n_-$ has the Riccati form:

$$dn / dt = 2an - 0.5bn^2 + 0.5b\delta^2 \exp(-2ct) - c\delta \exp(-ct) \quad (2.3.15)$$

This equation can be transformed into the Schrodinger equation with the Morse-type potential.

Equation (2.3.15) has the asymptotic solution

$$n_0 = 4a / b \quad (2.3.16)$$

Obviously, the higher the ‘‘impatience’’ factor, the deeper the minimum of n(t) will be. At sufficiently high ‘‘impatience’’ factor, the finite-difference solution to equation (2.3.15) falls to zero. This means that the market dies out due to trader impatience. However, the exact solution never reaches zero and always approaches the asymptotic value (2.3.16) after passing the minimum. This demonstrates the drawback of the continuous approach. Indeed, a non-zero number of traders that is lower than unity does not make sense. One way around this problem is to use a threshold, $n_{\min}$, such that

$$n \pm (t) = 0 \text{ if } n \pm (t) < n_{\min} \quad (2.3.17)$$

Still, further analysis shows that the discrete analog of the system (2.3.14) may be more adequate than the continuous model (Schmidt 2001).

2.4 The Minority Game

The Minority Game was introduced by Zhang and Challet (1997) as an extremely simplified model where the consequences of the agents heterogeneity can be studied in detail. Although rather remote from real financial markets, the structure of the model is extremely rich and allows us to study several relevant issues.

The equations

$$x_{\mathrm{t}} - \mathrm{x}_{\mathrm{t-1}} = \mathrm{D}_{\mathrm{t}^-}/\lambda \quad (2.4.1)$$

where $\mathrm{D}_{\mathrm{t}^-}$ represents the excess demand in the market just prior to time t , while time t represents the time when the new price p(t) is set. The scale parameter λ is the market depth.

$$g_R(\mathrm{t}+1) = -\,\mathrm{a}_{\mathrm{R}}^{\mu(\mathrm{t})}\chi(\mathrm{D}(\mathrm{t}+1))/\lambda - \mathrm{L}(\mathrm{t}+1) \quad (2.4.2)$$

$$\mathrm{D}_{\mathrm{t}^-} = \sum_{R=1}^{2^P} \mathrm{n}_{\mathrm{R}}(t) a_R^{\pi(t)} \quad (2.4.3)$$

Strategy reward structure of equation 2.4.2 rewards strategy books for suggesting the minority trading decision.

The ‘minority game’ consists, for each agent, to each at each time step between two possibilities, say +1,-1. An agent takes his decision based on the past history – say on the last M outcomes of the game. A strategy maps a string of M results into a decision. The number of strings is 2^M, to each of them can be associated +1 , -1. Therefore the number of strategies is 2^{2^M}.

Each agent is given from the start a certain number of strategies from which he can try to determine the best one. His bag of strategies is fixed in time – he cannot try new ones or slightly modify the ones he has in order to perform better. The only thing he can do is to try to rank the strategies according to the past performance. So he gives scores to his different strategies, say +1 every time a strategy gives the correct result, -1 otherwise. The crucial point here is that he assigns these scores to all his strategies depending on the outcome of the game, as if these strategies had been effectively played. Doing this neglects the fact that the outcome of the game is in fact affected by the strategy that the agent is actually playing at time t.

The parameters of this model are: the number of agents N, the number of story strings 2^M, and the number of strategies per agent. The last parameter does not really affect the results of the model, provided it is larger than one. In the limit of large N and M, the only relevant parameter is $\alpha = 2^M / N$. For $\alpha > \alpha_c$, where α_c can be computed exactly (and is found to be equal to 0.33740) the game is predictable in the sense that conditionally to given history h , the winning choice w is statistically biased towards +1, -1.

One can introduce predictability (Bouchaud 2003) q as:

$$q = \frac{1}{2^M} \sum_{h=1}^{2^M} \langle w | h \rangle^2 \quad (2.4.4)$$

Is non zero for $\alpha > \alpha_c$, and goes continuously to zero for $\alpha \to \alpha_c^+$, where the game becomes unpredictable by the agents (however agents with , say , longer memory $M' > M$ would still be able to extract information from the history).

Correspondingly, for $\alpha > \alpha_c$, the score of the strategies behave as a biased random walk, with a positive or negative drift. In this predictable phase some strategies perform systematically better than others. In the unpredictable (or efficient) phase $\alpha < \alpha_c$, all the strategies behave as random walks.

In particular, the time during which an agent keep playing a given strategy is related to the passage problem of a random walk, and is therefore a power-law variable with exponent $\mu = 1/2$, of infinite mean. When the game is extended to allow agents not to play if all their strategies have negative scores, one finds , as a function of α, an efficient active phase where a finite fraction of the agents play. In this phase, the duration of active periods of any given agent follows a power-law distribution with an exponent $\mu = 1/2$, producing non trivial temporal correlations in the volume of activity. The point $\alpha = \alpha_c$ is a critical point corresponding to a second order phase transition of the kind of statistical physics. The critical nature of the problem at and around this point leads to interesting properties, such as power-laws distributions and correlations, that may have some relation with the power-laws observed in financial time series.

Of course, the Minority game is too simple in my opinion to be compared to financial markets: there is no price and no exchange, and it is not clear to what extent the minority rule applies to financial markets.

There are several extensions of the Minority Game that reproduce, near the transition point, some of the stylized facts of financial time series, we talk about them in the next chapter. There is also a self-organizing effect that large α corresponds to small number of agents resulting in to some extent predictable time series, so new agents will join the game tuning down α to α_c.

3. Stylized Facts Observed in Financial Markets

The complex dynamics of financial markets can be characterized by some stylized facts of which the most important ones are: non-Gaussian distribution of return, short-term autocorrelation of return and long-term autocorrelation of volatility. Up to now, the traditional economic and financial theories cannot explain these satisfactorily. As complementary approach to the traditional ones, artificial market simulations have shown its strengths in studying the complex dynamics in economic and financial systems. In this work, we first describe the stylized facts by means of empirical data analysis.

We will study in the next chapter, three well-known artificial market models out of the many that have been developed so far: the Cont-Bouchaud model, the Lux-Marchesi model and the model by Bak et al. We have performed numerical experiments with these models for a wide range of parameters. These numerical studies elaborated that all three models can generate non-Gaussian distribution of return, but only two of them (the Lux-Marchesi model and the model by Bak et al) can confirm the volatility clustering phenomenon. In addition, from these case studies, we have obtained some insights with respect to the common mechanisms that account for these two stylized facts. Based on this, we have proposed two alternative ways to improve the Cont-Bouchaud model. It is demonstrated that both proposed methodologies can generate volatility clustering

The dynamics of financial markets is highly complex. Up to now, the main stream financial theories still lack a satisfactory explanation of its origin. The complexity of the dynamics can be characterized by some stylized facts, e.g., non-Gaussian (fat-tailed) distribution of return, relatively fast decay of the autocorrelation of return, and long term autocorrelation of volatility.

Traditional analysis of the financial time series (based on statistical techniques) fails to give a definite answer for the presence of these stylized facts. This is interesting but not surprising. Financial markets are composed of many heterogeneous traders and different types of financial assets. The complexity of the dynamics of financial markets is not only because of their great number of elements, their complex structure with respect to how the elements are located and linked to each other, but also due to the collective behaviour of the interacting elements under the influence of massive external factors. Without considering the highly non-linear interaction at this microscopic level, it is hard to describe properly the aggregative dynamics of financial markets. Thus, at least as a complementary approach in economics and finance, we should deal with the dynamics at the micro level. Recently, the technique of agent based modelling, which has been successfully applied in natural sciences, has been used by researchers in economics and finance. With this technique, we start with modelling the market, the behaviour of agents and the interactions between them. Then we can simulate the dynamics of interest and study the complexity of the dynamics from bottom up. Many artificial market models have been developed in the recent years. The technique has already shown its advantages. Many microscopic models are able to confirm (some of) the main stylized facts.

There are still some problems in this field. For instance, even some well-known models, e.g., the Cont-Bouchaud model, can not generate some important stylized facts. Moreover, although many models can generate the main stylized facts, they differ from each other in fundamental aspects in how to model the market and its dynamics. This implies that a common realization of the mechanisms that accounts for the complex dynamics of financial markets has not yet been achieved in this field.

For the last decades, some theories, techniques and tools from different disciplines, e.g. economics, mathematics, physics and computer science, etc., have been applied to investigate and analyze the dynamics of financial markets. Especially, thanks to the advanced computational technology, massive amounts of high frequency (tick-by-tick) financial data obtained at different frequencies became available and highly accessible. The analysis of these financial time series greatly contributed to a better understanding of the complex dynamics. Obviously, a thorough study of empirical financial data is a good starting point of the modelling of financial markets. For this sake, in this chapter, we will give a brief description of the data available in financial markets and, in particular, we will discuss main stylized features observed in the financial time series.

The main stylized facts are, namely: Non-Gaussian distribution of returns, short term correlation of return, and long term autocorrelation of volatility related to volatility clustering.

Non-Gaussian Distribution of Returns

The statistical study of return of financial assets can be traced back to the early years of the 20th century, in which financial return was regarded as a random process. Even until now, some notable economic models are based on the assumption that relative changes of price follow a Gaussian distribution. One example is the Black-Scholes model for option pricing. In 1963, Mandelbrot pointed out that a normal distribution for modelling the asset returns is not valid in general. This is due to the heavy-tailed character of the distribution of financial returns. For this reason he suggested that financial returns should be described by the Levy stable distribution. Since then, the non-Gaussian character of distributions of financial return has been studied intensively. In figure 3.1, we display the time series of S&P 500's daily return together with the time series of a variable that follows a Gaussian distribution. Compared to the Gaussian time series, the empirical financial time series exhibit some very large changes and clusters of large variability. For example, there is a large stroke around the end of 1987 which denotes the crash on the Black Monday, October 19, 1987, when the S&P 500 Index lost 20.5% or the current market situation of October 2008 in which the index lost 24% in 24 days ! There are also some turmoils around the end of 2001, when a 15% price drop occurred within a few weeks after September 11, 2001 an the current 2008 turmoil.

The probability density function (PDF) of return is characterized by a higher and narrower peak and heavier tails compared to a Gaussian distribution. Figure 3.2 presents the histogram of the daily returns of S&P 500 from January 1970 to September 2005. Noticed that we are using a logarithmic scale for the

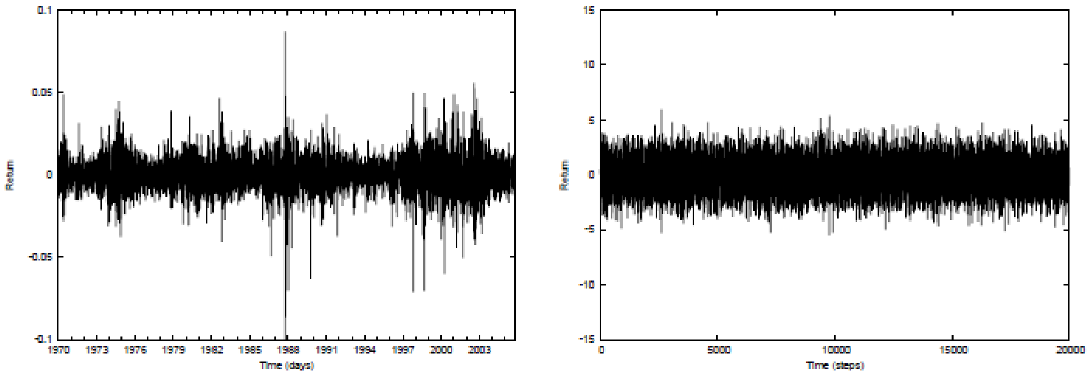


Figure 3.1: Comparison between the return of S&P 500 (left) and a Gaussian process (right).

y axis. Similar result can also be found in Figure 3.3, which shows histogram of returns of S&P500 on tick scale.

One way to measure the normality of a distribution is its Kurtosis, ·, which is defined as:

$$\kappa = \frac{\mu 4}{\sigma^4} \quad (3.1)$$

where μ4 is the fourth central moment and σ is the standard deviation of the distribution.

A data set with a high kurtosis value will have a distinct peak near the mean, declines rather rapidly around the mean, and then decays slowly leading to heavy tails. The kurtosis of a normal distribution is 3. Kurtosis less than 3 indicates a distribution that is flat at the mean and decays fast, while kurtosis larger than 3 indicates a distribution with a sharp peak and heavy tails.

In Table 3.1, the kurtosis of the log-returns of S&P 500 and DJIA5 of different time series are presented. The kurtosis of S&P 500 tick-by-tick data and daily data are 26.96 and 5.95, respectively, consistent with the return distributions shown in Figure 2.5 and

Figure 2.6. It can be noticed that tick and daily data usually give rise to kurtosis values larger than 3, implying that the tails of the corresponding distribution of return decay much slower than that of a normal distribution. Especially, the kurtosis of the returns of tick data is much greater than that of daily returns, deviating more from the Gaussian distribution.

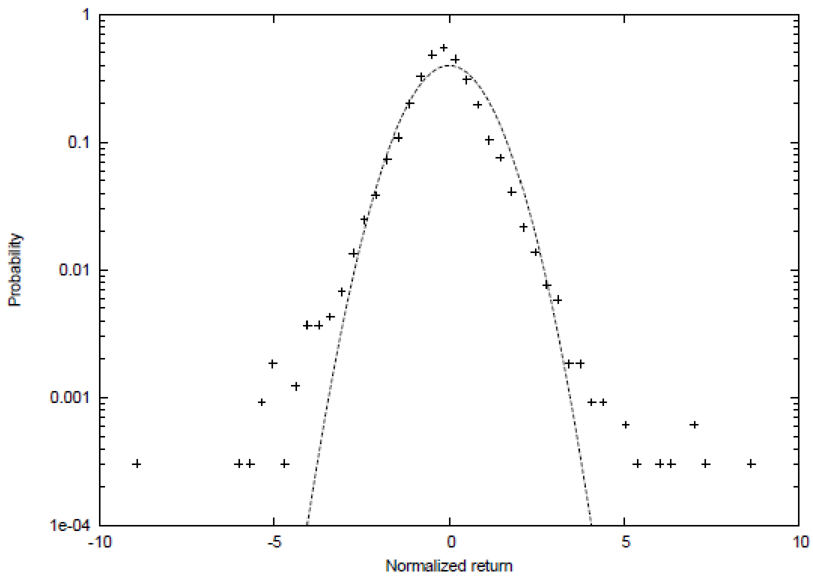


Figure 3.2: PDF of daily return of S&P 500 from January 1970 to September 2005 (the crosses) together with a Gaussian PDF (dotted line) for comparison.

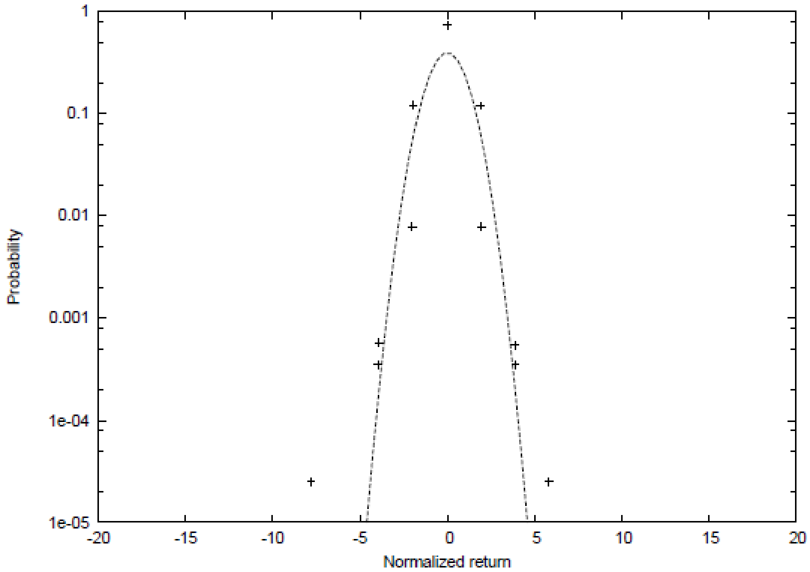


Figure 3.3: PDF of return on tick scale of S&P 500 data on September 1st, 2004 (the crosses), together with a Gaussian PDF (dotted line) for comparison.

Table 3.1: Kurtosis of empirical time series of return and the Kurtosis of the time series of a variable that follows a normal distribution.

Data	Kurtosis
S&P 500 tick data	26.96
S&P 500 daily data	5.95
DJIA tick data	20.1056
DJIA daily data	5.7348

Short-term Autocorrelation of Return

Traditionally, people thought that the prices or returns in liquid markets do not have time dependence. It has been taken as a strong support to the ”efficient market hypothesis (EMH)”. It is easy to understand that if there are correlations among price changes, there will be certain statistical arbitrage had attract arbitrageurs. Those statistical arbitrage will eventually eliminate the correlations, except on very small time scales that represents the amount of time the market takes to react on the arrived information.

Temporal independence of returns can be quantified by analyzing its autocorrelation function (ACF):

$$c(\tau) = \frac{\sum_{t=1}^{N-\tau}(Y_t - \overline{Y})(Y_{t-\tau} - \overline{Y})}{\sum_{t=1}^{N}(Y_t - \overline{Y})^2} \qquad (3.2)$$

where N denotes the total number of time steps, Y_t the return at time t, and $\overline{Y}$ the mean of the returns.

The results of the autocorrelation function of the S&P 500 daily return and tick return are shown in Figure 3.3. For comparison, in these figures and all plots that follow, we include corresponding results obtained for a series of independent normal distributed variables.

The results indicate that the autocorrelation of daily returns quickly drops to the noise range. However, obvious autocorrelations can be observed in the tick returns for the time lag from 0 to 25 ticks (about a few minutes). This suggests that financial returns have very short time ’memory’.

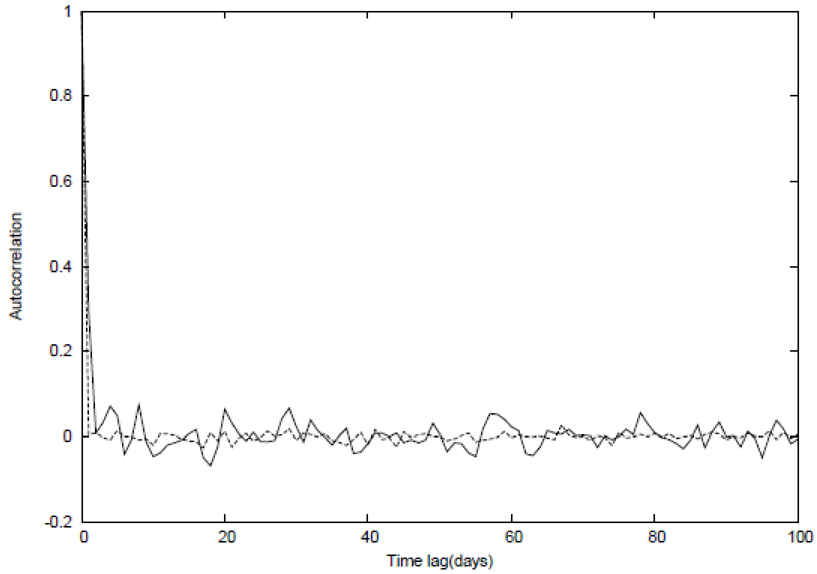


Figure 3.4: The autocorrelation function of S&P 500 daily returns

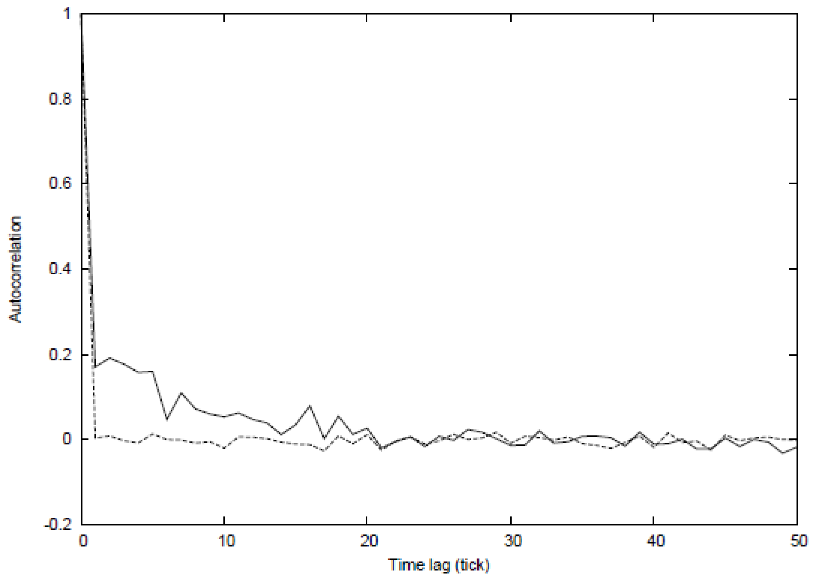


Figure 3.5: The autocorrelation function of S&P 500 tick returns

Long-term Autocorrelation of Volatility

Short-term autocorrelation in returns does not imply the independence of returns. The latter requires that any non-linear function of return is without

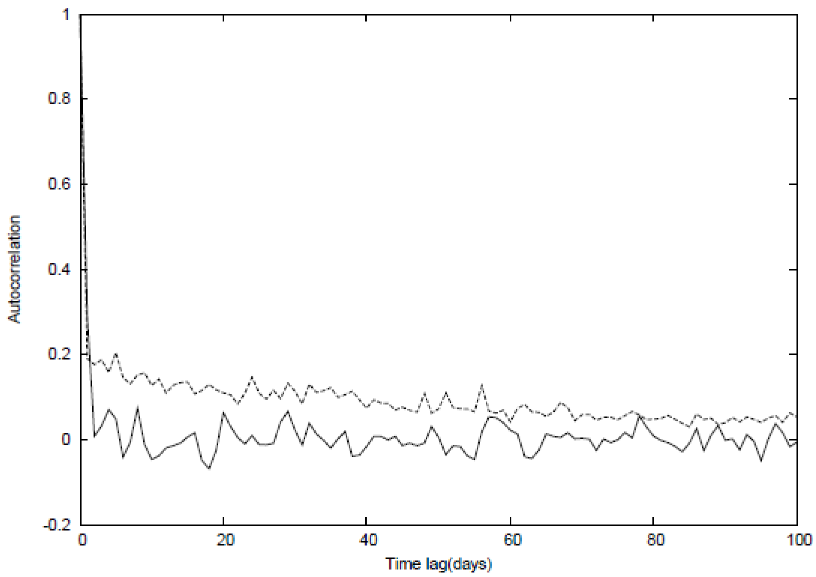


Figure 3.6: Autocorrelation functions of the daily return and the absolute daily return of S&P 500. It shows that absolute values of the daily returns are with long-term temporal correlations.

autocorrelation. Now, let us consider a non-linear function of return, i.e. volatility, which in principle is the standard deviation of the price changes during an appropriate period of time.

There are several reasons for considering the statistical properties of volatility:

• Volatility can be directly related to the amount of information arriving in the market;

• Volatility can be directly used in the modelling of stochastic process governing the price changes;

• From a practical point of view, volatility is a key indicator for measuring risk of financial investments.

Among the alternative methods to estimate volatility, here we only adopt the absolute return to measure the volatility.

The ACF of S&P 500 absolute daily returns is shown in Figure 3.6, together with the ACF of the daily returns. The time range of the data is from January 1970 to September 2004. Long positive autocorrelation of absolute returns can be observed in the plot. In contrast, both the autocorrelations of a set of Gaussian data and that of their absolute values, which are shown in Figure 3.7 (the solid line and dotted line, respectively), immediately drop to the noise.

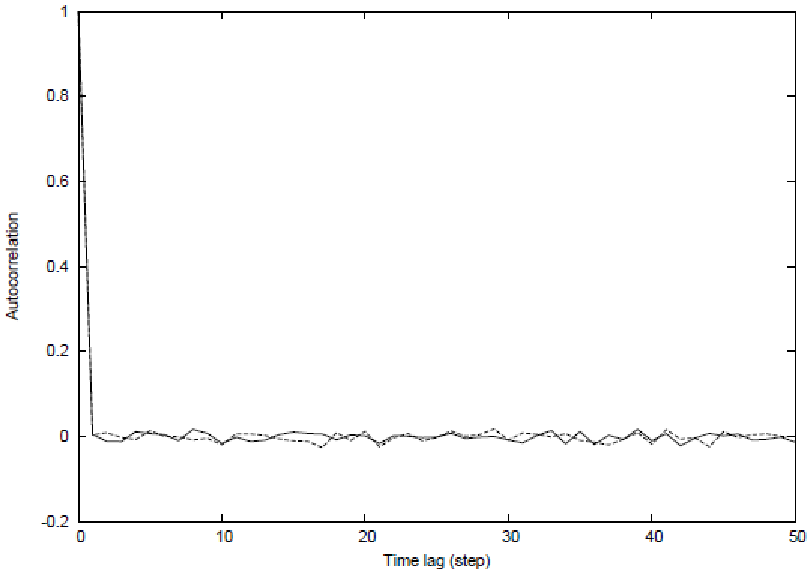


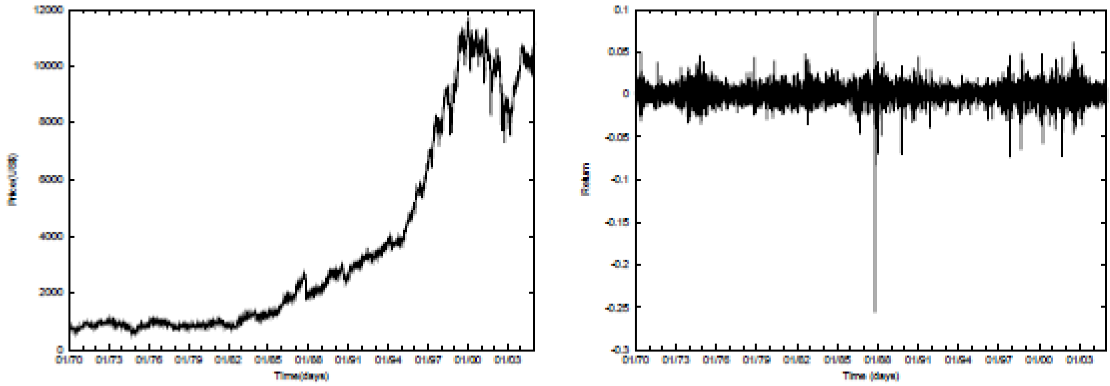
Figure 3.7: Autocorrelation functions of the values and absolute values of a set data following a Gaussian distribution. It shows that no temporal correlations exist in both the values and the absolute values.

range. It implies that there is no temporal correlation in both the data and their absolute values.

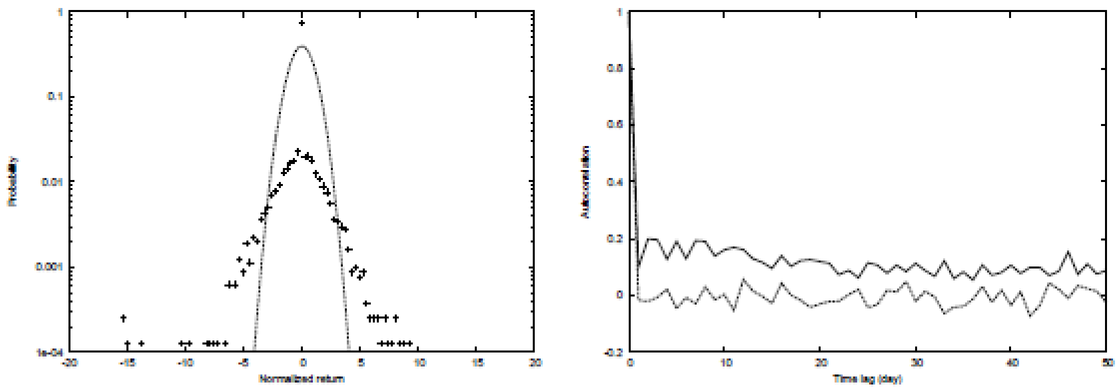
The absence of linear correlations and presence of long-time non-linear correlations in returns confirms that the signs of returns are weakly correlated over time, while the long time correlations exist in the amplitudes of returns.

The long term autocorrelation of volatility is directly related to a characteristic phenomenon — volatility clustering, i.e. ”periods of quiescence and turbulence tend to cluster together”. Recalling the return distribution plotted in Figure 2.3, it clearly shows some high returns and that they tend to group together.

Similar stylized facts can be found in different financial time series. Here we further investigate some data of Dow Jones Industrial Average (DJIA) and the exchange rate of Japanese Yen / US Dollar in foreign currency exchange markets.



Price (Left) , Return (Right)



Probability Density Function (Left) , Autocorrelation of the return and absolute return (Right).

Figure 3.8 shows the results of DJIA daily data, which were recorded from January 1970 to December 2004. The stylized facts observed in exchange rate of Japanese Yen/ US Dollar in the same period show results similar to that of S&P 500, with respect to the stylized facts.

As discussed above, the stylized facts of financial time series can not be effectively explained by traditional economic and financial theories. In order to identify the origin of these stylized features, we resort to artificial markets.

4. Three Artificial market models : Cont-Bouchaud, Bak et al. and Lux-Marchesi

Let’s discuss now in depth three important models: the Cont-Bouchaud model [34], the model by P. Bak et al. [33] and the model by T. Lux et al. [96] [98]. We will investigate how these models represent real financial markets and agent’s behaviour, whether they confirm the main stylized facts and what their merits and disadvantages are.

4.1 The Cont-Bouchaud model

Model Description

The market

According to the Cont-Bouchaud model, the market is represented by a pool of N agents whose locations are not important. Instead of grouping traders into a few certain types, as other artificial market models do, it adopts only homogeneous agents.

The behaviour of agents

The agents face three alternatives at each time period: to buy a unit of a financial asset, to sell a unit of asset, or not to trade. Agents tend to form (binary) links between each other with probability c/N. Here c is a connectivity parameter, which represents the willingness of agents to align their actions. The resulting market structure is then described by a random graph consisting of N vertices. The connected components correspond to groups of investors. In Figure 4.1, a simple example is shown. These groups decide, independently from other groups, whether to buy, to sell, or not to trade. According to theoretical analysis, when $N \to \infty$ and $c \to 1$, the cluster size distribution follows a power law:

$$P(W)W \underset{\to}{\sim} \infty \frac{A}{W^{5/2}} \quad (4.1)$$

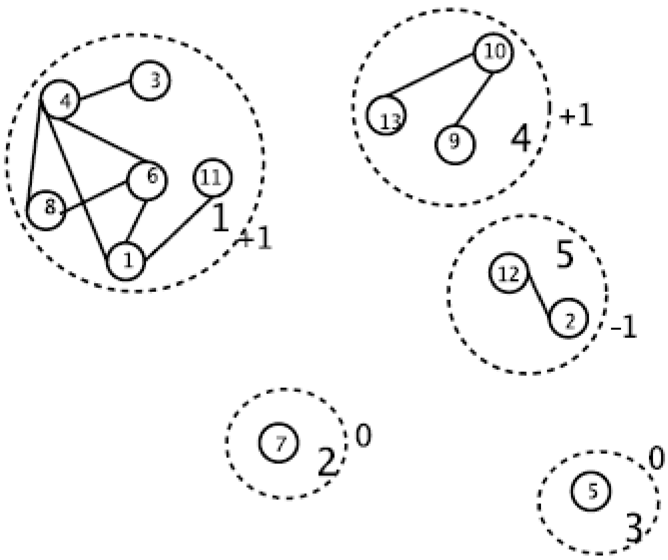


Figure 4.1: An example of a pool of traders consisting of 5 subgroups. 0,−1, 1 denote the behaviour of the different groups, namely remaining inactive, selling and buying, respectively.

where W is cluster size, A is a constant. When c is close to and smaller than 1 (0 < 1 − c << 1), the cluster size distribution is cut off by an exponential tail:

$$P(W)W \underset{\sim}{\rightarrow} \infty \frac{A}{W^{5/2}} \exp(-\frac{(1-c)W}{W_0} \quad (4.2)$$

where W_0 are constants. For the critical value c = 1, each agent tends to establish a link with at least another agent.

The demand of a cluster α is denoted as $\phi_\alpha(t)$ which is equal to 1 for buying, −1 for selling, and 0 for not trading. The variables Á are independent with a symmetric distribution: $\mathrm{P}(\phi\alpha_\alpha = +1) = \mathrm{P}(\phi\alpha_\alpha = -1) = \mathrm{a}, \mathrm{P}(\phi\alpha_\alpha = 0) = 1 - 2\mathrm{a}$ .The demand of a certain group of traders is proportional to its size.

Price formation

Price change is proportional to excess demand,

$$\Delta p = p(t+1) - p(t) = \frac{1}{\lambda} \sum_{\alpha=1}^{k} W_\alpha \phi_\alpha(t) \quad (4.3)$$

where W_α is the size of cluster $\alpha, \phi_\alpha(t)$ is the individual demand of agents belonging to that cluster, k is the number of clusters, and λ is a so-called market depth parameter which measures the sensitivity of price to changes in excess demand.

In the original paper about the Cont-Bouchaud model a rigorous analytical derivation is given for the cluster size distribution and the distribution of price changes. However, volatility clustering is not considered in the paper.

Furthermore, to the best of our knowledge, no analytical results are known regarding volatility clustering. Therefore, we implemented this model and carried out various numerical experiments. In our implementation we adopted a recursive labelling algorithm to identify clusters.

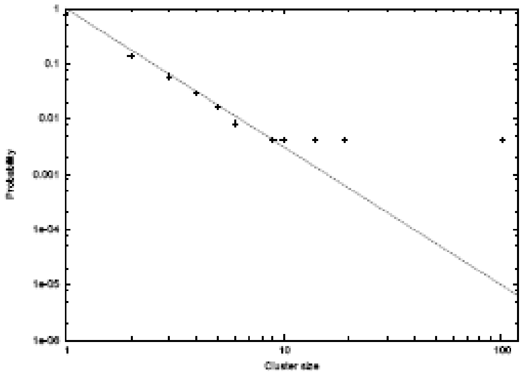
Simulation Results

From the model description above, it is clear that the cluster formation process is crucial for the price dynamics. From theoretical analysis, it is expected that the cluster size distribution will follow a power law under certain conditions.

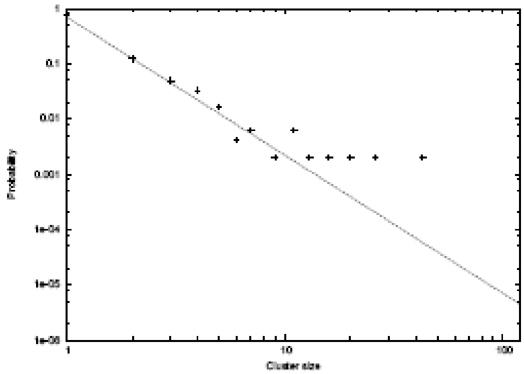
Figure 4.2 and Figure 4.3 show the cluster size distribution for different values of N when c = 1 and c = 0.8, respectively. In both experiments the activity parameter is set to 0.001. The dashed line denotes the analytical curve (with A = 1 and W0 = 0.8) and the dots represent the estimated probabilities from our simulations. It is clear that the estimated cluster size distribution follows a power law for small values of the cluster size. The range for which the power law is valid becomes wider as the number of agents is increased. The discrepancy between theory and simulations for large values of the cluster size is due to the finite number of agents that are used in the simulations. It should be noted that the analytical expressions are only valid in the limit of infinite number of agents.

Price and price change

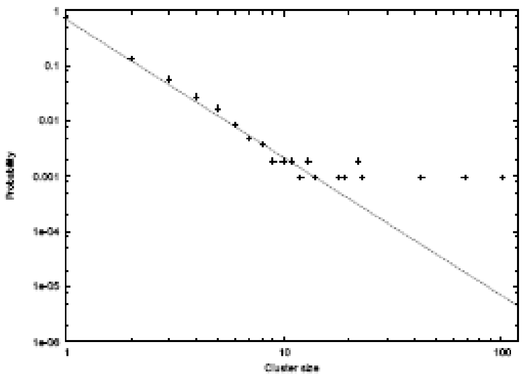
Figure 4.4 shows the evolution of the price and the time series of price change. In Figure 4.4(a), there are some large ’jumps’ in the price, which correspond to the big ’strokes’ in Figure 4.4(b) that reflect large (positive or negative) returns. Due to these large strokes, the probability distribution of price change does not follow a Gaussian distribution. This will be discussed in more detail in the next section.



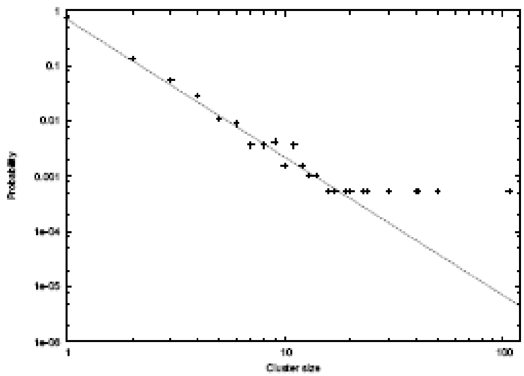
(a) N=500.



(b) N=1000.



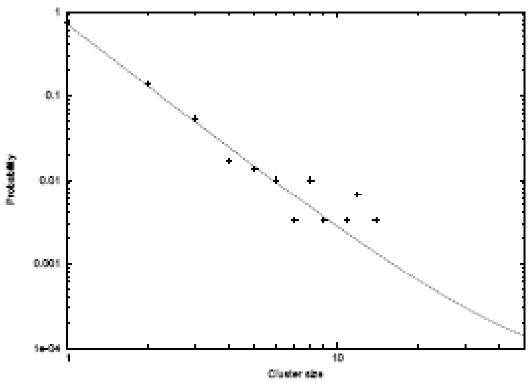
(c) N=2000.



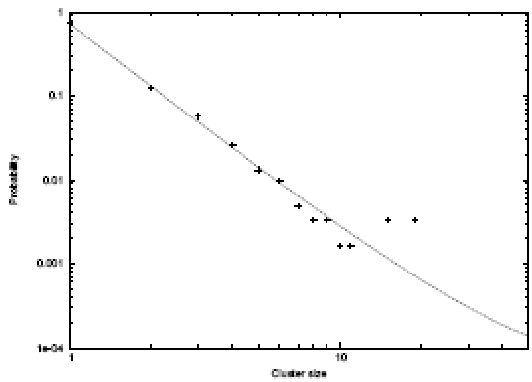
(d) N=4000.

Figure 4.2: The cluster size distribution for different values of N, when c = 1.

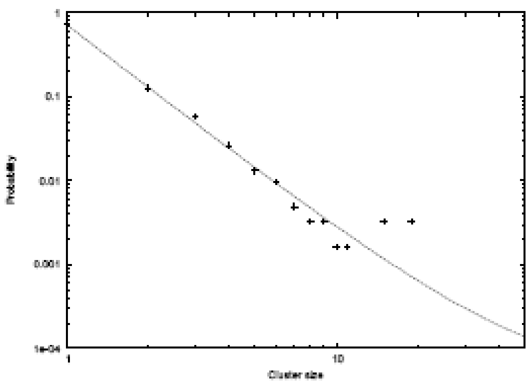
The scales of both axes are logarithmic.



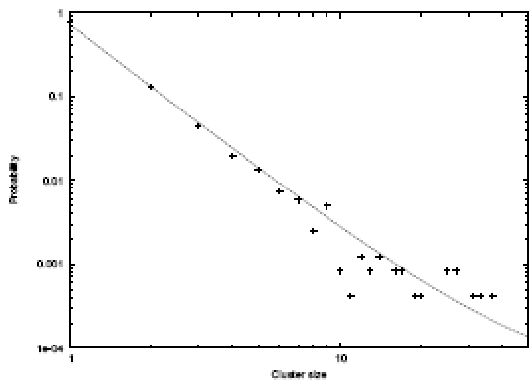
(a) N=500.



(b) N=1000.



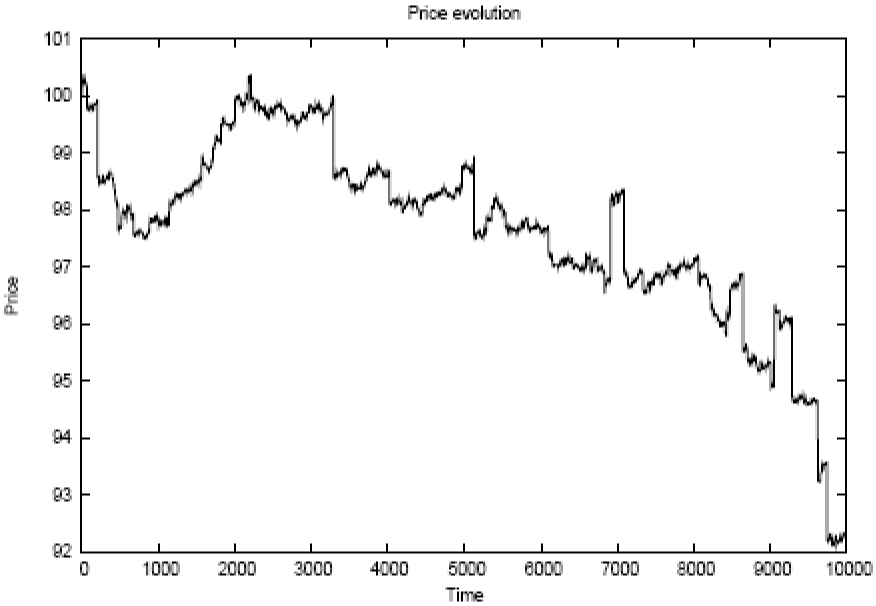
(c) N=2000.



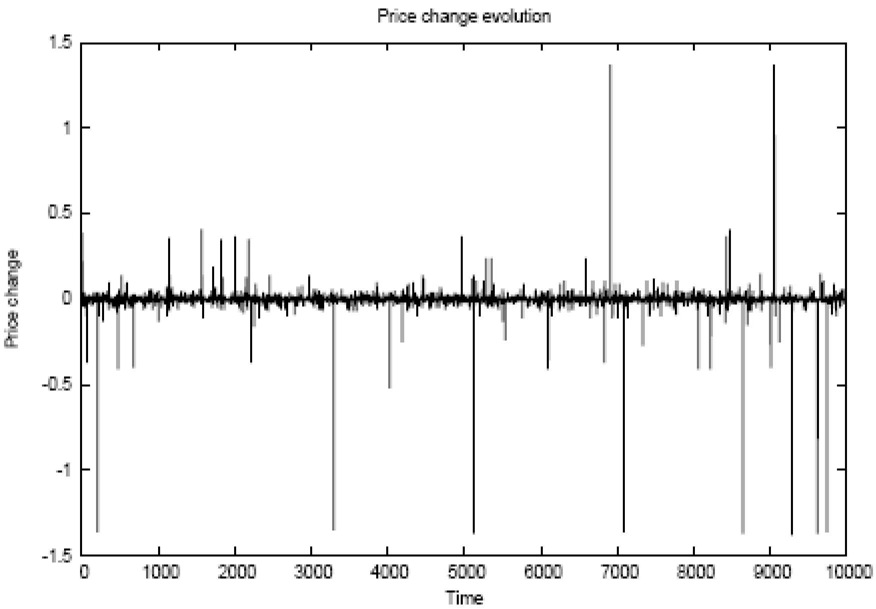
(d) N=4000.

Figure 4.3: The cluster size distribution for different values of N, when c = 0.8.

The scales of the axes are logarithmic.



(a) Time evolution of the price.



(b) Time evolution of Price change.

Figure 4.4: The evolution of the price and time series of price changes over a time period of 10000 time steps. Here, N = 1000, c = 0.9 and a = 0.0005.

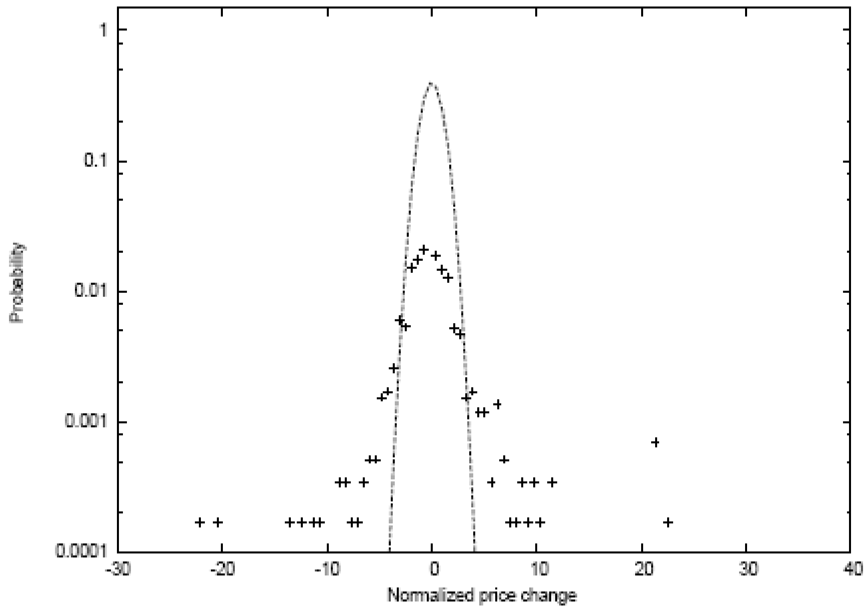


Figure 4.5: Probability distribution of normalized price changes compared with a Gaussian PDF. Here, c = 0.9, N = 1000 and a = 0.005. The scale of the y axis is logarithmic.

Probability distribution of price changes

Figure 4.5 confirms our expectation:"fat tails" can be clearly observed in the histogram of price changes. From the model description above, we can conclude that the large (positive or negative) returns may correspond to the actions of some very large groups. As shown in Figure 4.6, when we decrease the value of c slowly down to 0, meaning that agents are becoming more and more independent, the probability distribution of price change converges to a Gaussian distribution.

We have also performed some investigations with respect to the parameter a. This parameter denotes the activity of agents, and is defined as the probability that agents buy or sell. When $a << \frac{1}{2}$ some traders do not participate in trading, while $a \cong \frac{1}{2}$

indicates the situation that almost all traders are buying or selling. Figure 4.7 shows the probability distributions of price change for different values of a. It can be seen that the smaller the activity is, the fatter the corresponding distribution will be. When $a \cong \frac{1}{2}$, the distribution is close to a Gaussian distribution. This can be understood as follows: when nearly all of agents are trading, there are no sharp changes in price. Generally these sharp changes are caused by heterogeneities in the demand of the different groups.

This is of course connected with the cluster size distribution.

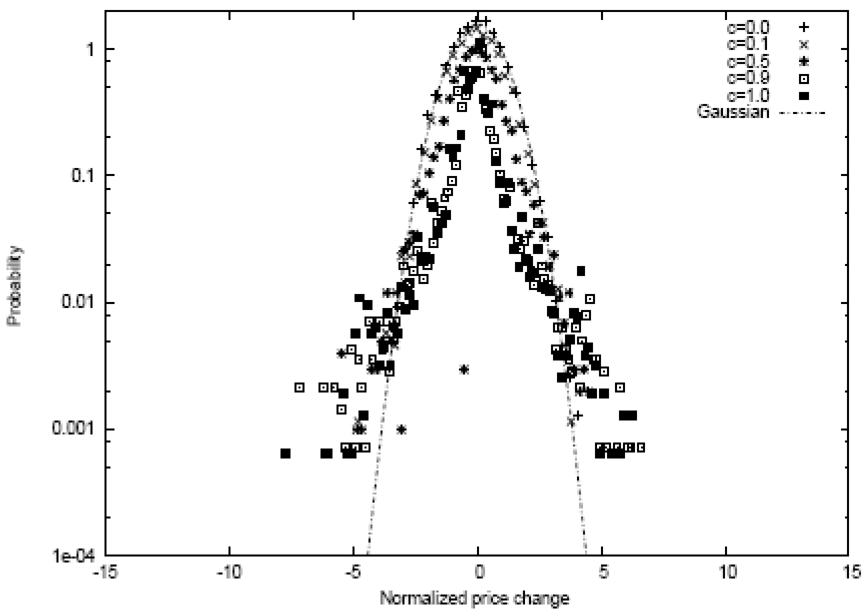


Figure 4.6: The Probability distribution of normalized price changes for different values of c, compared with a Gaussian distribution. Here, N = 1000 and a = 0.005. For $c \to 0$, a progressive convergence to a Gaussian distribution is observed.

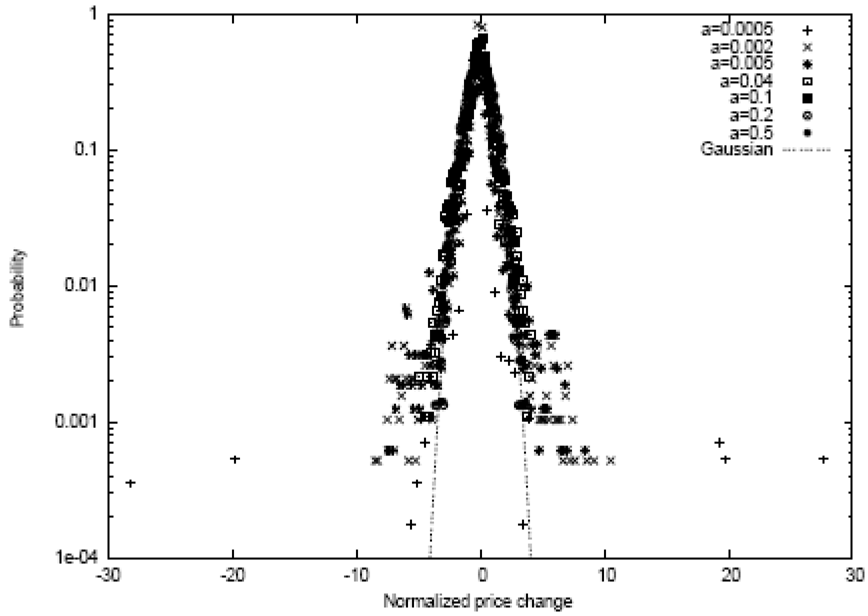
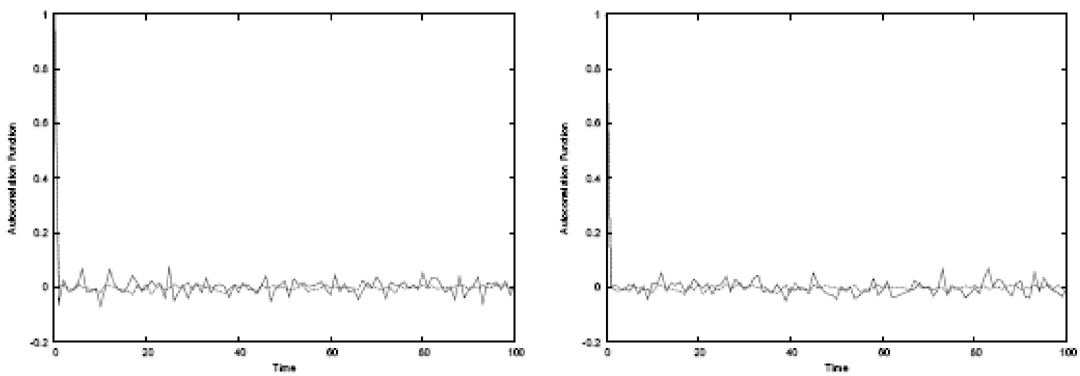


Figure 4.7: The Probability distribution of normalized price changes for different values of the activity parameter a, compared with a Gaussian distribution. Here, N = 1000 and c = 0.9. For $a \to \frac{1}{2}$, a progressive convergence to a Gaussian distribution is observed.



(a) ACF of price changes.

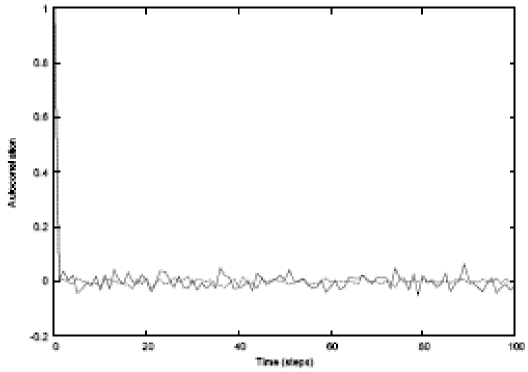
(b) ACF of absolute price changes.

Figure 4.8: Plots of the autocorrelation Function (ACF) of the raw and absolute price changes. Here, c = 0.9 and N = 1000.

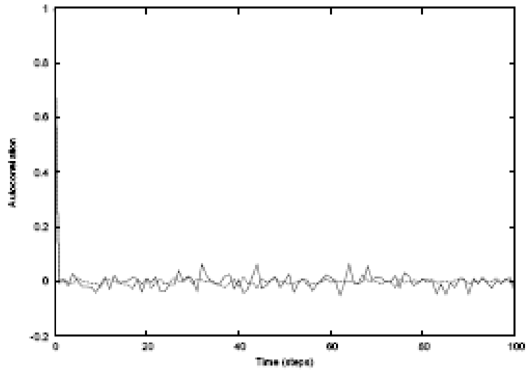
Autocorrelation of price changes and volatility

Next, we consider the autocorrelation of price change and volatility. Similar to our empirical data analysis (see Chapter 2), we will study the ACF of the price change (return) and its volatility. As a comparison, in all figures showing autocorrelation functions, the corresponding ACF of a series of independent normally distributed variables will be included.

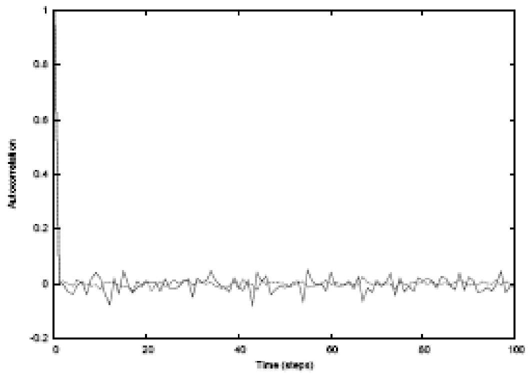
Figure 4.8 shows that, when we set the value of c close to 1, the autocorrelation of the price change drops quickly to the noise range. When c is set to other values, we get similar results (see Figure 4.9). The absence of memory in price change and in its absolute value is the consequence of the trading behaviour of agents: At each time step, if the agents are going to trade, their demands (either +1 or −1) are randomly determined. Thus there is no dependence between the demands at any two successive time steps.



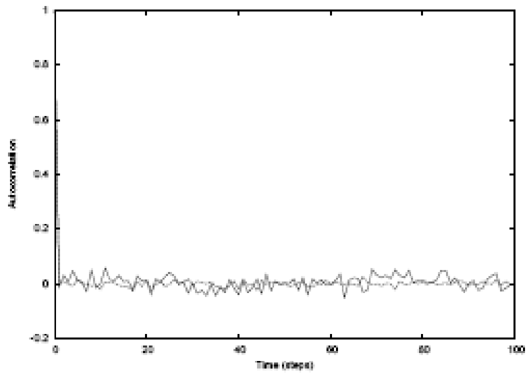
(a) ACF of price changes (c=0.5).



(b) ACF of absolute price changes (c=0.5).



(c) ACF of price changes (c=0.8).



(d) ACF of absolute price changes (c=0.8).

Figure 4.9: Plots of the autocorrelation Function (ACF) of the raw and absolute price changes according to different c value. Here, N = 1000.

Discussion

From the discussion above we learn that, according to the Cont-Bouchaud model, (fat-tailed) non-Gaussian distribution is directly caused by heterogeneities in the demand of the different groups and the herding behaviour of agents within a group. These heterogeneities are directly connected with the cluster size distribution and the activity parameter. One obvious advantage of this model is that, due to its simplicity, the distribution of return can be solved theoretically. Unfortunately, numerical experiments show that the Cont- Bouchaud model is not able to produce an important stylized fact, namely long term autocorrelation of volatility. During this project, we initiated a subproject with the objective to study the behaviour of this model when using a small world network instead of a random graph. The results were similar to what we have reported in this chapter, and more importantly, this modification does not explain volatility clustering.

4.2 The Model by Bak et.al.

Model Description

The market

According to this model, the market contains of N/2 stock shares of only one type. There are totally N agents in the market who are classified into two groups: fundamental value traders and noise traders. Each agent can own at most one share. So at each time step, he can either be a potential buyer or a potential seller of the stock. Each share owner advertises a price at which he is willing to sell his share, and each buyer advertises a price that he is willing to pay for a share. Locations and geographical relations of agents are not considered in this model, i.e. interaction can happen between any two agents in the market.

The behaviour of agents

According to the model, the biding and asking prices of any fundamental trader and the biding or asking price of any noise trader is initialized in a certain range of values. Fundamental traders choose their prices in order to optimize their utility functions, based on the expected dividends of the company that issued the stock and their risk aversion degrees. Their biding or asking prices will remain the same throughout the simulation, if they do not take participation in trading. As will be discussed below, the price updating mechanism of the noise traders is more versatile. The whole simulation process is composed of a set

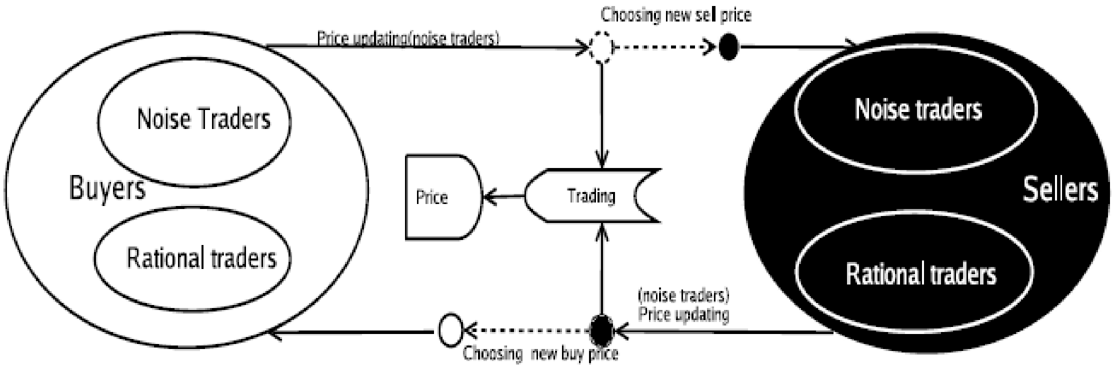


Figure 4.10: Diagram of a trading cycle in the Bak model. Note that if a transaction takes place the status of the trader (buyer or seller) will be switched.

of repetitive trading cycles as shown in Figure 4.10. In each trading cycle, an agent is chosen randomly to perform his operation in the market. Each trading cycle goes through the following three main stages:

1. Before a transaction: The selected agent will update his biding/asking price according to the situation of market and his particular strategy (unless he is a fundamentalist).
2. During a transaction: If there are trading opportunities, he will trade at the most favourite price for him.
3. After a transaction: both traders will exchange their buyer/seller status and choose new biding/asking prices. The behaviour of a fundamental trader is relatively simple,

while that of a noise trader, may be quite involved. The latter is defined in for the different trading stages during a cycle as follows:

1. In the stage before trading, the price can be updated in two alternative ways:

(a) the price is changed randomly with equal probability in either the downward or the upward direction. This is similar to the diffusion process in physics. The traders who behave in this way are called ”independent noise traders”.

(b) The price drifts to the current market price with a probability that is higher than 0.5.

2. In the stage before trading, how strong (in terms of unit) the diffusion or drifting is. There are also two possibilities:

(a) The strength is one unit.

(b) The strength is proportional to the actual recent variations of market prices, a feature called ’volatility feedback’.

3. In the stage after trading, how the new initial offering price is set after becoming a new seller (buyer). There are two possibilities here:

(a) The new asking (bidding) price is determined by randomly choosing a price between the current price and the maximum price (between zero and the transaction price).

(b) Or by randomly picking a seller (buyer) in the market and copying his price. To summarize, there are eight ways in which the trading strategy of a noise trader can be modelled.

Price formation

The market price is set to the actual transaction price after each transaction. This is the lowest selling price when a buyer is chosen in the market, or the highest buying price when he is a seller.

Simulation Results

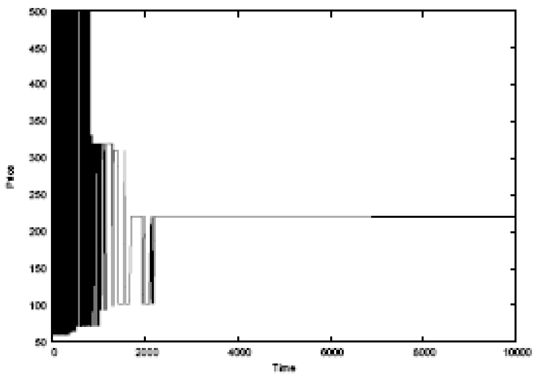
In this section we will present the results obtained in our simulations using different price update mechanism for the noise traders and other parameter settings. We will first show, in the first subsection, the time evolution of price and return. In the next subsection, we will study the distribution of return and the relevant autocorrelation functions following a similar approach as our empirical data analysis as presented in before.

Price and return

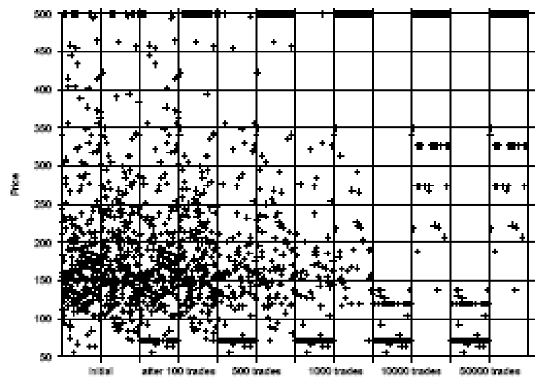
Let's first suppose that all the agents in the market are rational agents. In Figure 4.11(a) the price evolution corresponding to this case is shown. The initial prices of the fundamental traders are between 50 and 500. It is clear that the price converges to a certain value (around 220) after about 2000 steps. The reason for this is that after a period of trading, no further trade will occur since all the sellers hold the price beyond the market price (220) and all the buyers are waiting to trade at prices below the market price. Figure 4.11(b) gives the snapshots of prices hold by all the agents at different time steps. It confirms that after certain time steps, the buyers and sellers phase separate into non-overlapping regions. In this case, return quickly goes to zero. Clearly this is not realistic.

An interesting question is, how will the market price and return evolve if there are only noise traders? We first consider the situation when all noise traders are independent. Figure 4.12 illustrates the simulation results when we use N = 500 independent noise traders with initial prices lower than pmax =

500. Subfigure 4.12(a) shows the price variation, 4.12(b) shows the snapshots of the selling and buying prices of the traders at different time steps, and 4.12(c) depicts the time series of return. We observe that the behaviour of the price and return are not similar to characteristics seen in empirical data. However,



(a)



(b)

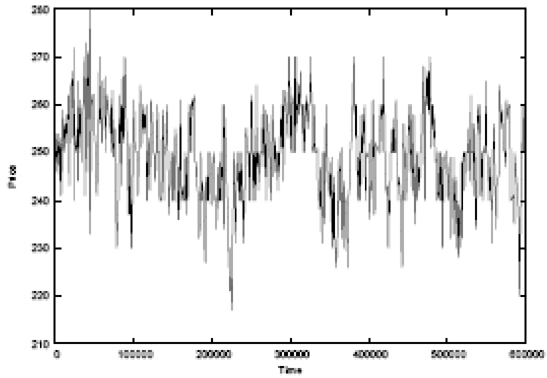
Figure 4.11: Price evolution when all the agents in market are rational agents. (a) Price evolution. (b) Prices of the agents at selected time steps. Here, N = 500, and data for 6 time instances are shown.

some big strokes can be observed in the plot of return which suggests that the distribution of return will not be Gaussian. This will be discussed in more detail in the next section.

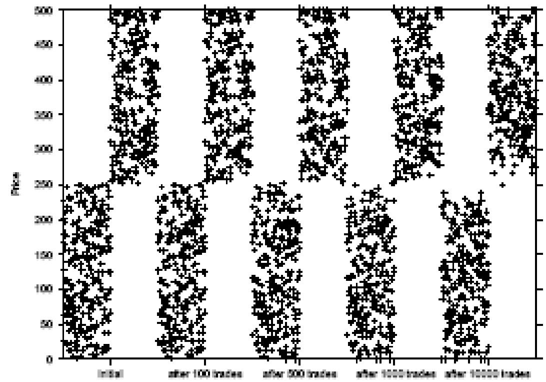
In the next experiment the noise traders are allowed to adjust their holding price towards the current market price. A drift coefficient D = 0.05 is chosen to determine how strongly the traders want to follow the market price. Time series of the price and return corresponding to this case are shown in Figure 4.13(a) and (b), respectively. We see that

there are relatively more big strokes in Figure 4.13(b) compared to Figure 4.12(c). This implies that this type of herding behaviour can generate larger variation in return.
Next, another type of herding behaviour is added into simulations, namely the so-called ”urn” model. After each transaction the agent who had just executed the transaction will randomly pick a buyer or seller in the market and copy his price.

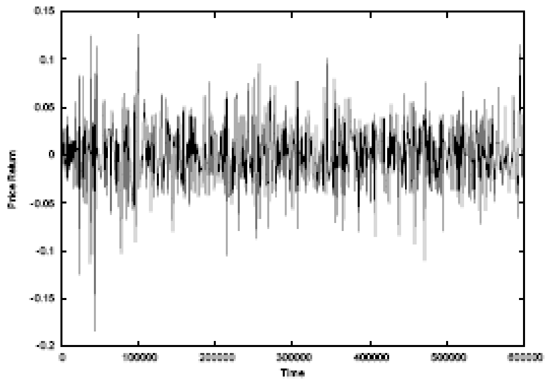
Figure 4.14 shows the simulated prices and returns obtained with the urn model when using N = 200 traders. The buying price of each agent pb(j) is initialized randomly within the range from 1900 to 2000, and the selling price ps(j) within the range from 2001 to 2100. The highest price is set to pmax = 4000. It is clear that the price changes more dramatically compared to the previous models. Moreover, it is qualitatively similar to that observed in real markets, e.g. that of S&P 500 (see Figure 2.2). The plot of return does not look similar to that of real data. However, there are still some big strokes which might lead to a non-Gaussian distribution. In addition, it seems that the number of agents does not influence the results very much. The results obtained using 1000 agents are shown in Figure 4.15. It is clear that these are similar to the results obtained with 200 agents. An important feature that we have not seen so far is the volatility clustering phenomenon.
When ’volatility feedback’ is added in the ”urn” model the results are more
Figure 4.12: (a) Price evolution with only independent noise traders. (b) Snapshots of the prices hold by traders at different time steps. (c) Time series of return. Here, N = 500, Pmax = 500; all agents in the market are independent noise traders.



(a)



(b)

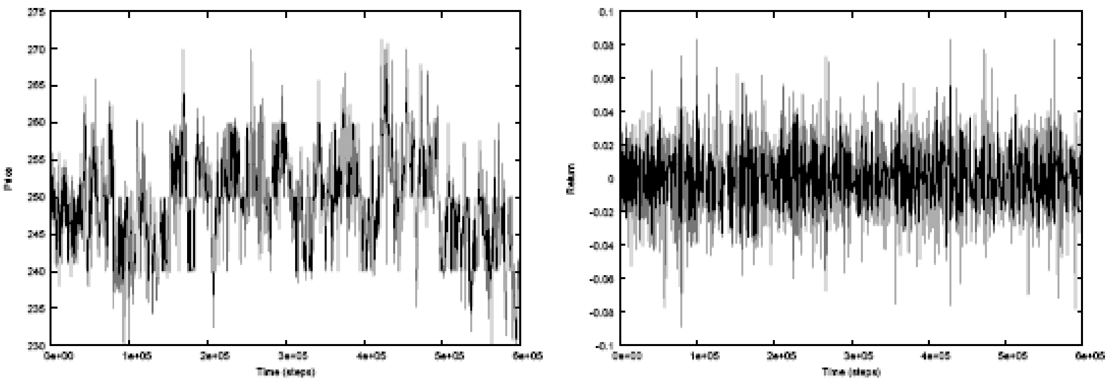


(c)

Figure 4.12: (a) Price evolution with only independent noise traders. (b) Snapshots of the prices hold by traders at different time steps. (c) Time series of return. Here, N = 500, Pmax = 500; all agents in the market are independent noise traders.

realistic. We have carried out experiments using N = 200 and N = 1000 traders, respectively. Historical prices of 400 time steps are used for calculating the average volatility. By examining the results of both experiments as shown in Figure 4.16(a) and (b), and Figure 4.16 (c) and (d), we conclude that the price evolution obtained with this model is quite similar to that observed in real markets. More importantly, volatility clustering is clearly observed. In our final simulation with this model, we considered a market consisting of a mixture of rational traders and noise traders. More specifically, we were interested in the effect of the fraction of rational traders (or noise traders) on the time series of price and return. Figure 4.17 shows the simulation results corresponding to different fractions of noise traders: 2%, 20% and 50%. The rational traders' bid and ask prices are initialized in the range from 1960 to 2060. The total number of agents is N = 1000.

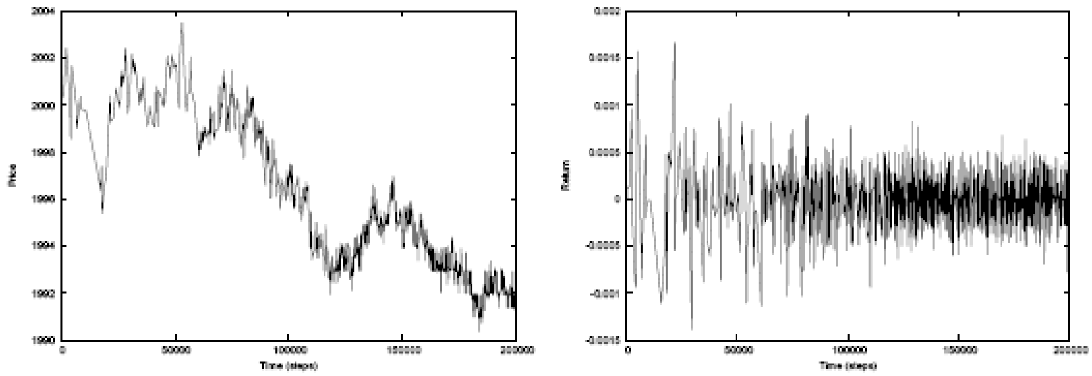
When there are only 2% rational traders, a few extremely large prices (bubbles)



(a) (b)

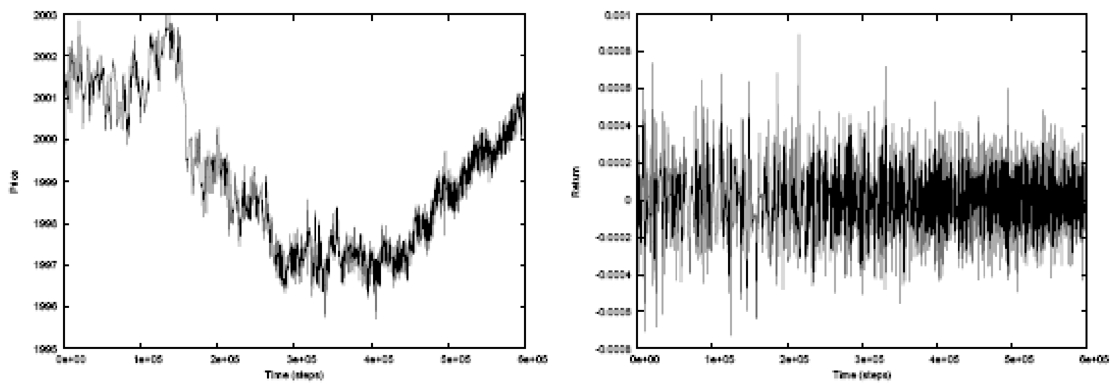
Figure 4.13: (a) Price evolution when the noise traders drift their biding /asking prices toward the current market price. (b) Price returns. Here, N = 500, initial prices of traders are set in the range between 0 and 500.

occur (see Subfigure 4.17(a) and 4.17(b)). When the relative number of rational traders is increased to 20% (see Subfigure 4.17(c) and 4.17(d)), the prices vary within a small range around the fundamental value. This range becomes even smaller when the fraction is further increased to 50%, as shown in Subfigure 4.17(e) and 4.17(f). It is clear that as fraction of rational traders is increased, the market becomes more stable. An overwhelming fraction of noise traders will induce anomalous large price changes. According to this model a behaviour similar to empirical data can only be obtained if a small fraction of rational traders is used.



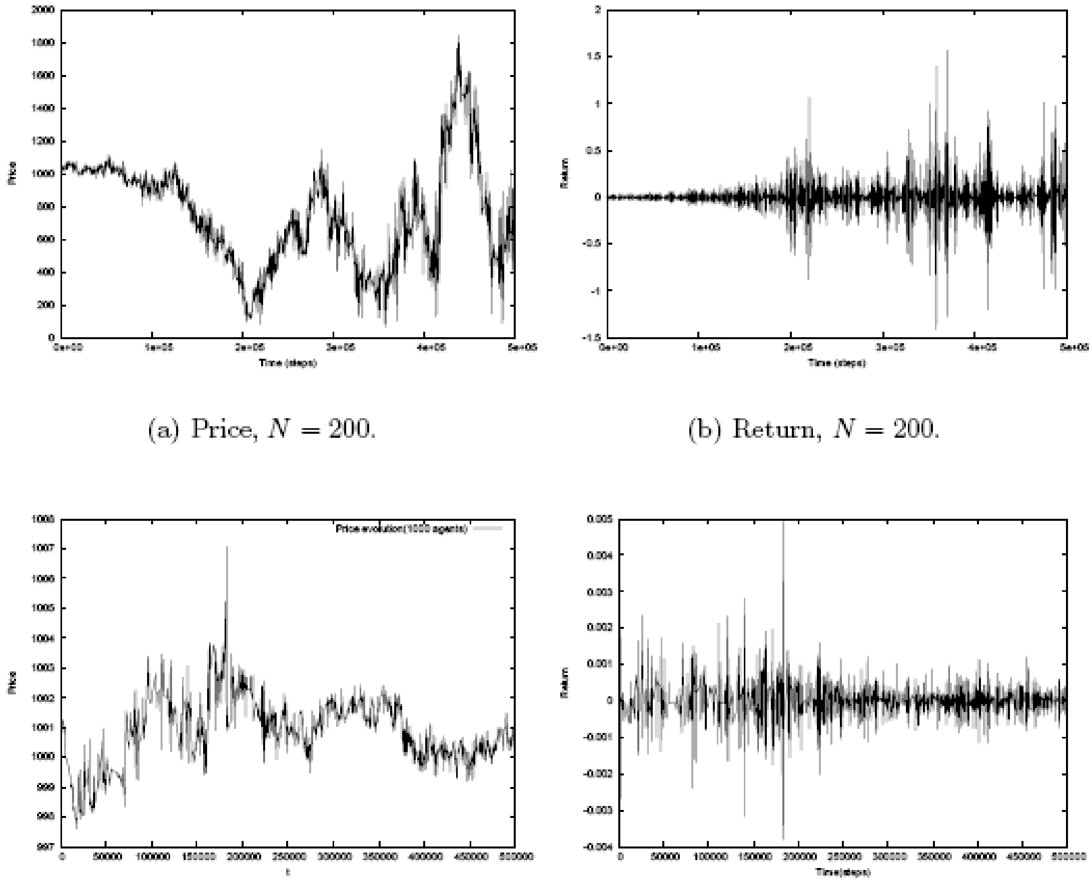
(a) (b)

Figure 4.14: (a) Price evolution of the urn model. (b) Time series of return of the urn model. Here, N = 200.



(a) (b)

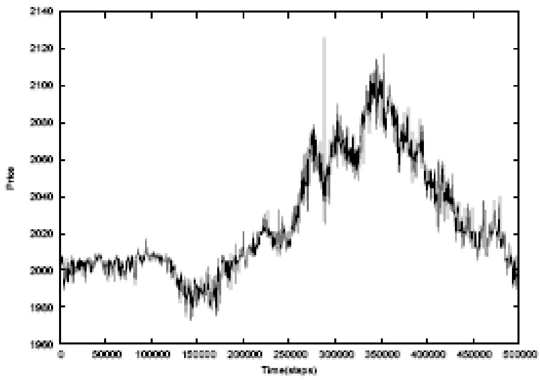
Figure 4.15: (a) Price evolution of the urn model. (b) Time series of return of the urn model. Here, N = 1000.



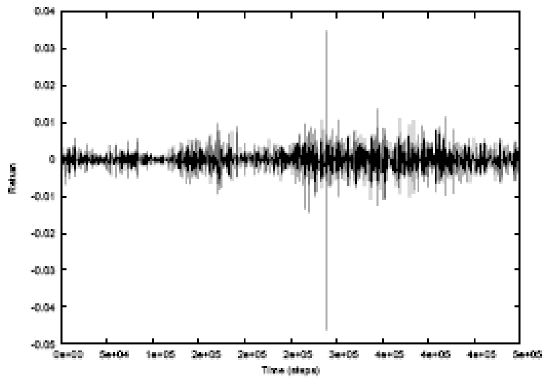
(a) Price, $N = 200$.　(b) Return, $N = 200$.

(c) Price, $N = 1000$.　(d) Return, $N = 1000$.

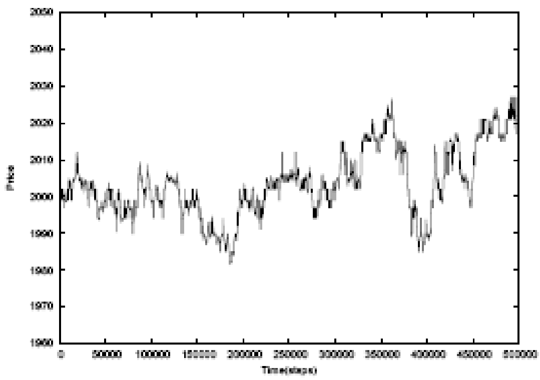
Figure 4.16: Price variations and time series of return, when volatility feedback is added into the urn model. Here, the number of agents are: (a) and (b), N = 200; (c) and (d), N = 1000.



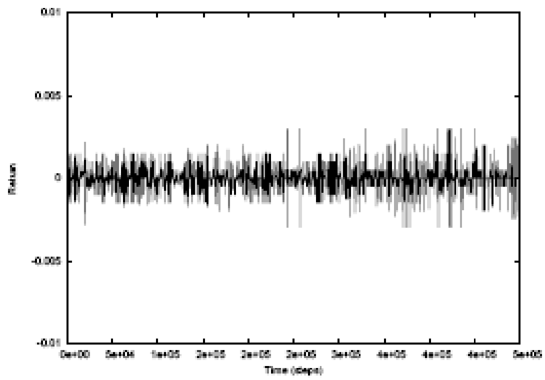
(a) Price, 2% rational traders.



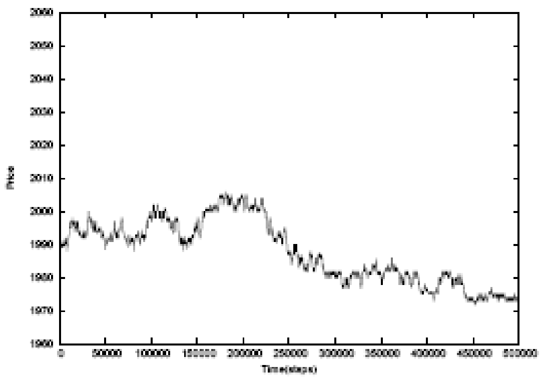
(b) Return, 2% rational traders.



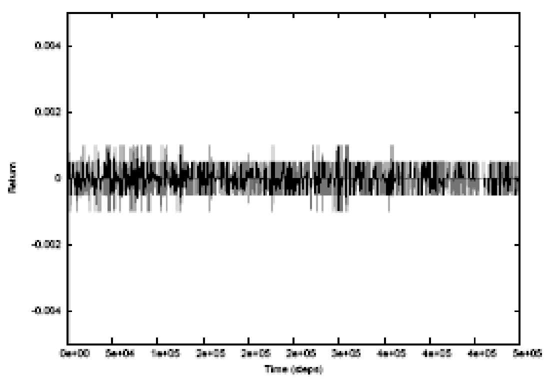
(c) Price, 20% rational traders.



(d) Return, 20% rational traders.



(e) Price, 50% rational traders.



(f) Return, 50% rational traders.

Figure 4.17: Evolution of price and return when the market consists of a mixture of noise and rational traders. Results corresponding to different fractions of noise traders

are displayed. (a) and (b): 2%. (c) and (d): 20%. (e) and (f): 50%. Here, the rational traders' asking and biding prices are initialized in the range from 1960 to 2060. The prices hold by noise traders are initialized within the range from 1800 to 2200. The number of agents is N = 1000.

Type of trades	**Kurtosis**
100% noise traders (independent)	3.6143
100% noise traders (drifting to market price)	4.7434
100% noise traders (urn)	6.0847
100% noise traders (urn with volatility feedback)	248.2751
Mix type 2% rational traders (urn with volatility feedback)	209.7075
Mix type 20% rational traders (urn with volatility feedback)	104.08
Mix type 50% rational traders (urn with volatility feedback)	12.23

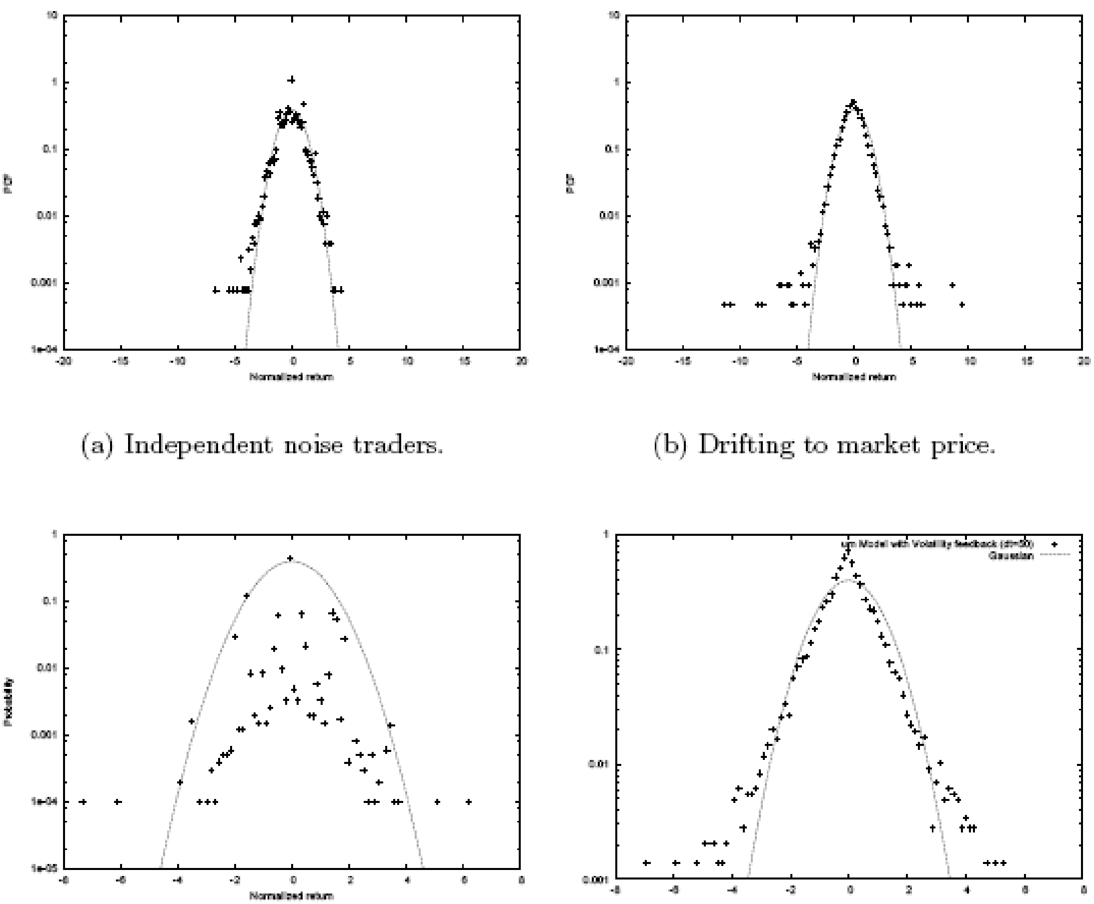
Table 4.1: Kurtosis of the time series obtained using different fractions of rational traders and different models for the behaviour of the noise traders.

Probability distribution of return

In the previous subsection we have presented the time series of price and return for different strategies employed by the noise traders and market scenarios. Here we will study the probability density function of return and the autocorrelation function of return and volatility for the corresponding cases. When a market consisting of only rational traders is considered, we have seen that the price converges quickly to a constant level. Obviously the return is simply zero for most of the time.

Figure 4.18 shows the PDFs of the normalized return when considering a market with only noise trades. We consider four different strategies for the price updating, namely a random price adjustment, a price drifting towards the current market price, a strategy following the 'urn' model, and a strategy following the 'urn' model with volatility feedback. As the model is refined, we clearly see that the PDF becomes non-Gaussian.

In the previous chapter we have seen that the Cont-Bouchaud model ascribes the 'fat-tailed' distribution to the "herding behaviour". From the experiments with Bak's model, we have found that 'fat-tailed' distribution of return can be generated when herding is included as is the case in the 'urn' model. The PDF then is clearly non-Gaussian. However, when a model is used with based on a drift towards the market price, fat tails are also observed. We believe that due to the drifting of the price towards the market price, an indirect herding mechanism is introduced. It should be noted that the current market price is the price at which a trade was executed in a previous time step. This of course is a price that a seller or buyer was holding. By drifting towards this price the behaviour of that specific trader is mimicked to some extend. When only diffusion is considered, the distribution does not deviate significantly from a Gaussian distribution. In Table 4.1 the kurtosis of the time series of return are presented for the different cases. When a herding mechanism is present the Kurtosis is significantly greater then 3. As expected when volatility feedback is included, the kurtosis is much higher compared to the cases without volatility feedback. If in addition, rational traders are included, the kurtosis decreases. This is expected because the rational traders tend to stabilize the market. We further investigate the impact of the volatility feedback. Figure 4.19 shows the PDFs of return when using various lengths for the time window that is used to calculate the average volatility. It is evident that the longer the time window is, the stronger the PDF deviates from the Gaussian distribution. This can be understood by looking at the time series of the return (see Figure 4.15). It is clear that as the time window is increased to 3000 time units, the more volatile the behaviour within the time window is.



(a) Independent noise traders.

(b) Drifting to market price.

(c) According to the "urn" model.

(d) According to the "urn" model with volatility feedback.

Figure 4.18: Probability density function corresponding to the following cases: (a) Model with independent noise traders only. (b) Model with noise traders who drift their price towards the current market price. (c) Model with noise traders who follow the 'urn' model. (d) Model with noise traders who follow the 'urn' model with volatility feedback. Here, N = 1000, and a market with only noise traders is considered.

Normalized price changes

Probability distributions of normalized price changes

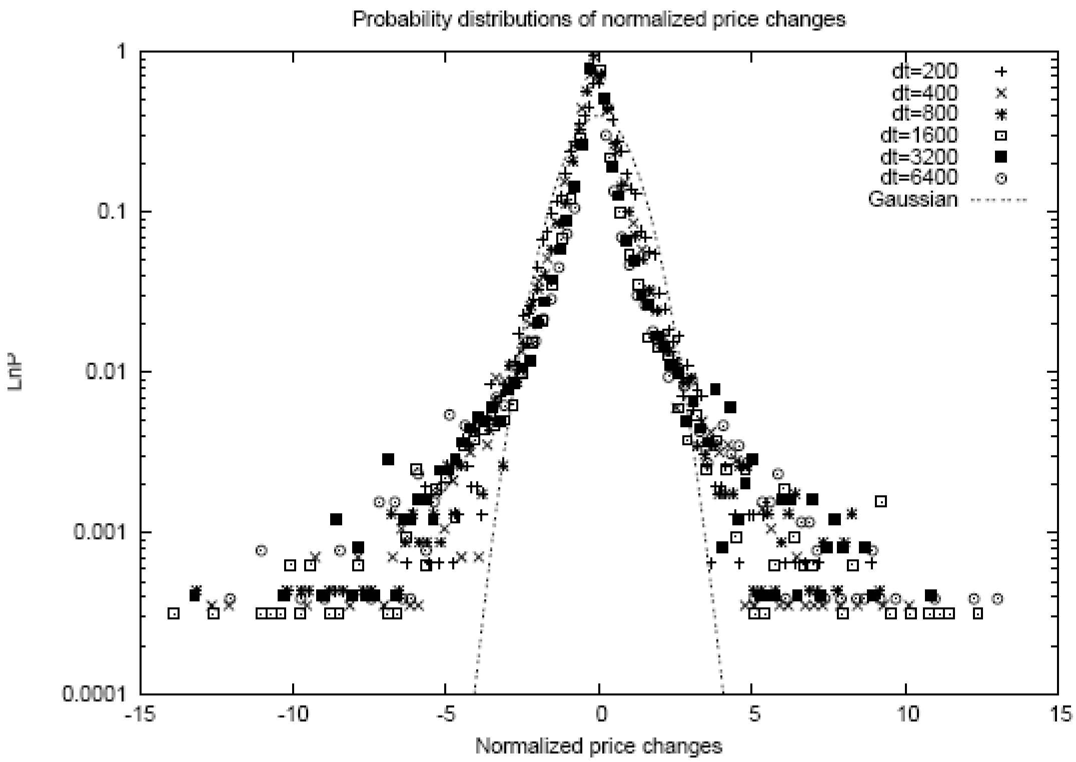


Figure 4.19: Return distributions of the cases with volatility feedback added into the model. Different lengths of memory time are used here.

Autocorrelation of return and volatility

In the remainder of this section we will focus on the autocorrelation of return and absolute return. In Figure 4.20 the ACF of return and that of absolute return are shown for the different cases considered above. We see that the ACF decays quickly to the noise range when a model is used with only independent noise traders or with noise traders using a drifting mechanism or with noise traders mimicking the ’urn’ behaviour. When volatility feedback is introduced into the model a long term decay of the ACF of absolute return is observed. This is expected because in that case the volatility clustering effect was clearly observed in the time series of return.

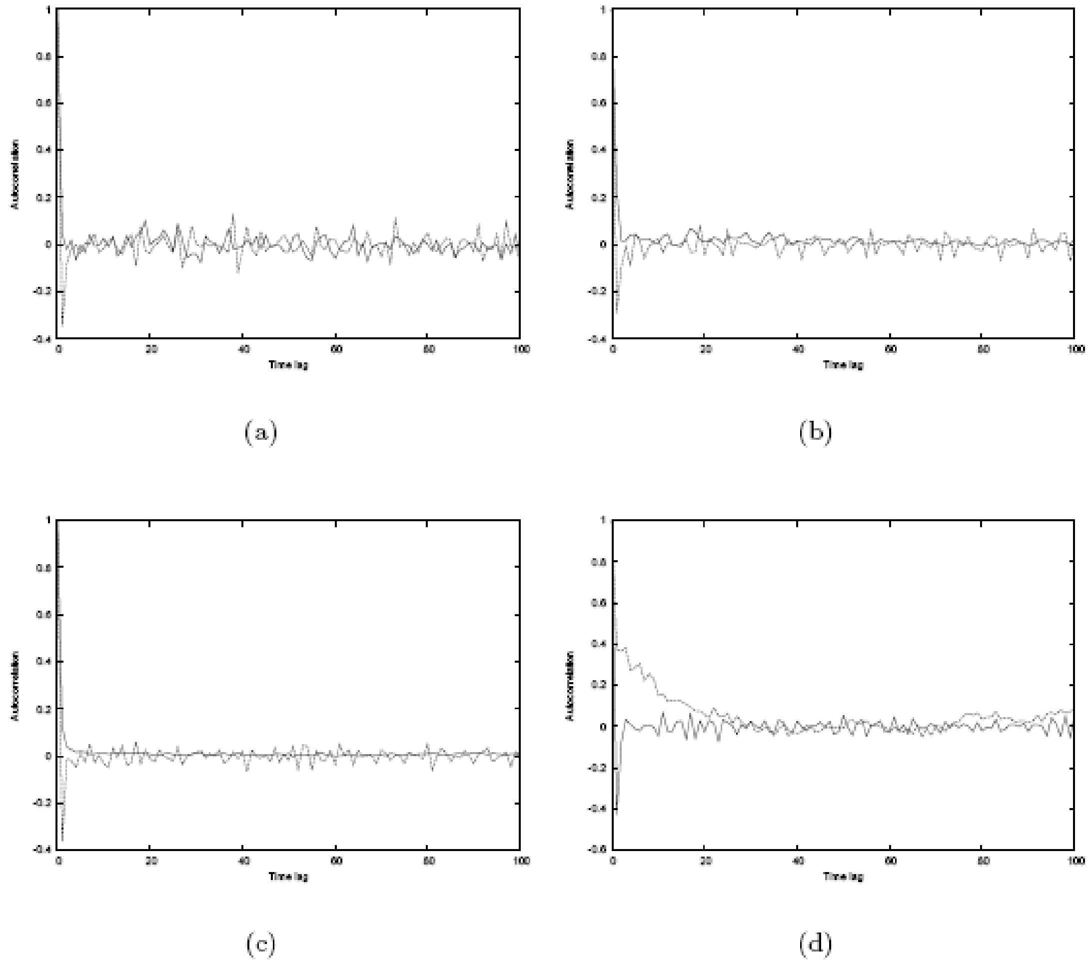


Figure 4.20: ACF of raw and absolute return obtained using Bak's model with different types of traders. (a) Independent noise traders. (b) Noise traders with drifting behaviour. (c) Noise traders with 'urn' behaviour. (d) Noise traders with 'urn' behaviour and volatility feedback.

Finally, in Figure 4.21, the ACF of return and that of absolute return are shown when using different fractions of rational traders in a model where the noise traders follow the 'urn' behaviour with volatility feedback. In Figure 4.21(a), we see that the autocorrelations of return quickly drop to the noise range for all cases. Figure 4.21(b) shows that the corresponding autocorrelations of volatility have different lengths of memory. The top curve is the ACF of absolute return corresponding to a fraction of 2%, the middle one corresponding to a fraction of 20% and the lowest one corresponding to a fraction of 50%. This implies that the noise traders tend to amplify the effect of volatility clustering.

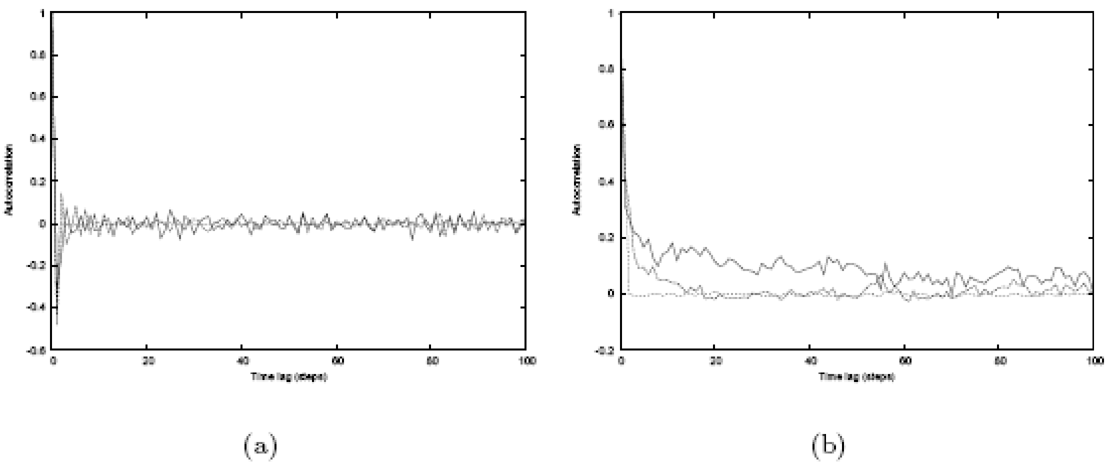
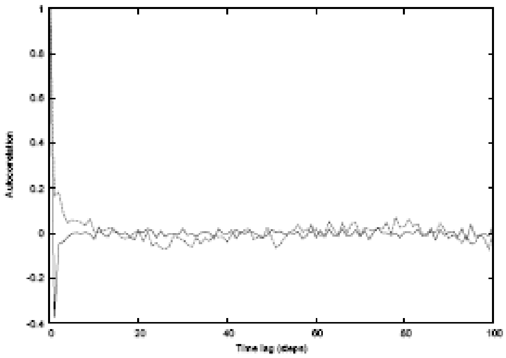


Figure 4.21: (a). ACF of return when the market consists of a mixture of rational and noise traders. (b) The corresponding ACF of absolute return.

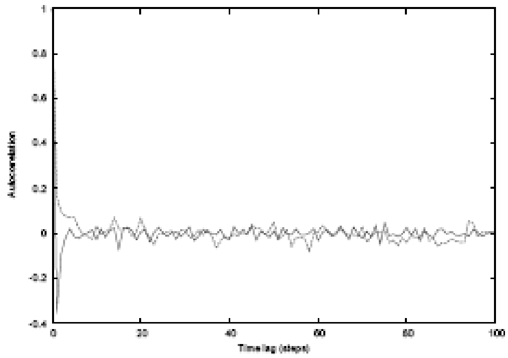
Similar to what we have done above (see Figure 4.19), we show the ACFs when different time windows are used to calculate the average volatility. Here, the parameter setting is the same as that used for Figure 4.18. The results are depicted in Figure 4.22. It is shown that for increasing length of the time window, the ACF of absolute return decays slower.

Discussion

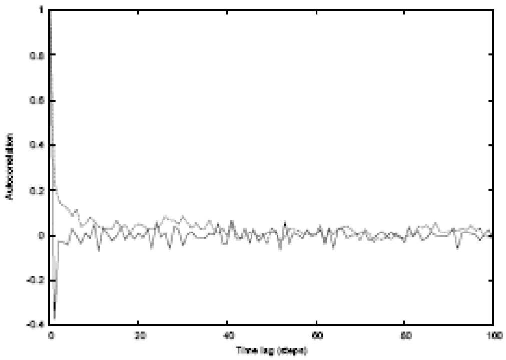
This model is based on rational and noise traders. The strength of the imitating behaviour depends on the fraction of rational traders and on the model that is used for the price updating of the noise traders. If this fraction is high the market will be more stable. Conversely, if the fraction of noise traders is high, large price variations might occur. While the 'diffusion' behaviour of noise trader can barely generate non-Gaussian distributions of return, it is the behaviour of imitation or herding that accounts for distinct 'fat-tailed' distributions. Volatility feedback is responsible for the phenomenon of volatility clustering: the historic price does influence the current or future return.



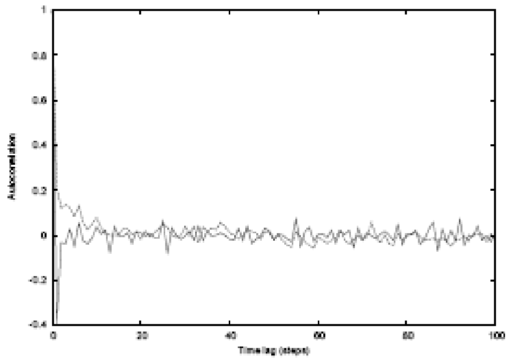
(a) dt=200.



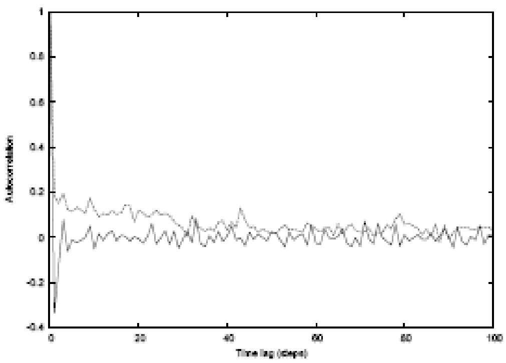
(b) dt=400.



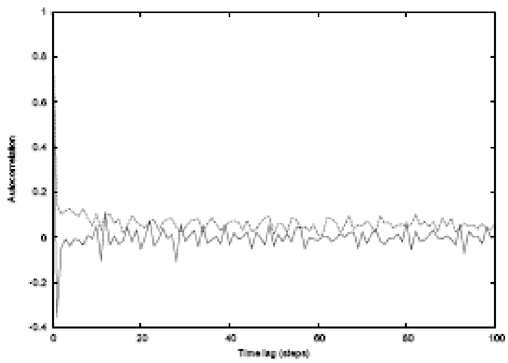
(c) dt=800.



(d) dt=1600.



(e) dt=3200.



(f) dt=6400.

Figure 4.22: ACF of return and absolute return for the cases when volatility feedback is added into the model. Different lengths of the time window are used here for comparison. Here, 1000 noise traders are used.

4.3 The Lux-Marchesi Model

Model description

The market

The Lux-Marchesi model uses two types of agents: Fundamentalists and chartists. Chartists are further divided into optimists and pessimists. Optimists will always buy units of asset, while pessimists will always sell units of asset. Fundamentalists look at the fundamental value of the stock (pf), and buy (sell) if the actual market price (p) is below (above) the fundamental value. Contrarily, chartists look at the present price trend on the market and the behaviour of other agents, rather than the fundamental value.

Agent behaviour

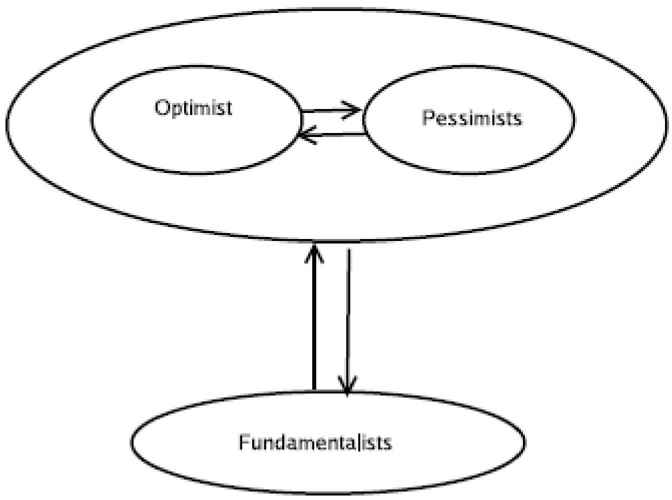


Figure 4.23: Diagram of the Lux-Marchesi model.

The key feature of the Lux-Marchesi model is the movement of agents among different groups or equivalently, the switching of traders to different trading strategies. Figure 4.23 illustrates the switching scheme which consists of following possibilities:

1. Movements between optimists and pessimists.

Switches of noise traders between the optimistic strategy and the pessimistic strategy are dependent on the combined effect of the majority opinion of all noise traders which is referred as the flow and the prevailing price trend which is called chart. The former is represented by opinion index x, which is defined as

$$x = \frac{n_+ - n_-}{n_c} \quad (4.3.1)$$

where n_+ is the number of optimistic chartists, n_- is the number of pessimists and n_c is the number of chartists.

The latter is expressed as $\frac{\dot{p}}{v_1}$, here p is the price, . $\dot{p} = \frac{dp}{dt}$ (the price change), v_1 is a parameter that denotes the frequency of revaluation of opinion.

The transition probabilities of chartists are described as follows:

$$\pi_{+-} = v_1(\frac{n_c}{N}\exp(U_1)), \quad (4.3.2)$$

$$\pi_{-+} = v_1(\frac{n_c}{N}\exp(-U_1)), \quad (4.3.3)$$

$$U_1 = \alpha_1 x + \alpha_2 \frac{\dot{p}}{v_1}$$

where N is the total number of traders; α_1 and α_2 are, respectively, the parameters measuring the importance that the individuals place on the majority opinion and the actual price trend in forming expectations about future price changes; π_{+-} is the probability of pessimists switching to optimists , and π_{-+} is the probability of optimists switching to pessimists. This equation implies that the possibility of changing to another strategy becomes higher when the actual movement of price is in a direction which is in contradiction to a trader's own expectation.

2. Movements between fundamentalists and chartists

Movements between noise traders and fundamentalists are triggered by the comparison of the excess profits from their respective strategies. Generally, traders will move to the more successful group. Specifically, transition probabilities relies on pay-off differential, the net gain from one investment after the deduction of the gain that would have received from other investment. The gains of the traders of various types are shown here.

(a) The gain expected by fundamentalists is

$$s\left|\frac{p_f - p}{p}\right|,$$

where pf is the fundamental value, p is the price, s < 1 is a discount factor because a gain will only be realized after the price reverses to the fundamental value.

(b) The gain of optimists is

$$(r + \dot{p}) / p,$$

here r is the nominal dividend of the asset.

(c) The gain of pessimists is R, the average real return from other investments.

Then, by considering the opportunity costs, the pay-off differential of the strategy of optimists is

$$(r + \frac{\dot{p}}{v_2}) / p - R - s\left|\frac{p_f - p}{p}\right|),$$

and that of the pessimists is

$$(R-(r+\frac{\dot{p}}{v_2})/p-s\left|\frac{p_f-p}{p}\right|),$$

where v_2 is a parameter for the frequency of this type of transition.

Consequently, the transition probability from fundamentalists to optimists π_{+f} and vice versa π_{f+} from fundamentalists to pessimists π_{-f} and vice versa π_{f-}, respectively, are described by the following set of equations:

$$\pi_{+f}=v_2(\frac{n_+}{N}\exp(U_{2,1})), \qquad (4.3.4)$$

$$\pi_{f+}=v_2(\frac{n_f}{N}\exp(-U_{2,1}), \qquad (4.3.5)$$

$$\pi_{-f}=v_2(\frac{n_-}{N}\exp(U_{2,2})), \qquad (4.3.6)$$

$$\pi_{f-}=v_2(\frac{n_f}{N}\exp(-U_{2,2})), \qquad (4.3.7)$$

$$U_{2,1}=\alpha_3(r+\frac{\dot{p}}{v_2})/p-R-s\left|\frac{p_f-p}{p}\right|),$$

$$U_{2,2}=\alpha_3(R-(r+\frac{\dot{p}}{v_2})/p-s\left|\frac{p_f-p}{p}\right|)$$

where α_3 is a measure of the pressure exerted by profit differentials.

Price formation

The prices are adjusted by an auctioneer (market maker) in balancing demand and supply. The aggregate excess demand is $\mathrm{ED}=(\mathrm{ED})_{\mathrm{c}}+(\mathrm{ED})_{\mathrm{f}}$, where

$(ED)_c$ and $(ED)_f$ are the excess demands of chartists and fundamentalists, respectively. Since all chartists either buy or sell the same number t_c units, $(ED)_c = (n_+ + n_-)t_c$. Depending on the deviation of the price from the fundamental value, reaction strength ° and number of fundamentalists nf , the excess demand of fundamentalists is given by $ED_f = n_f \gamma(p_f -)$. Assuming that there are additional liquidity traders in the market whose excess demand is stochastic or that the value of aggregate excess demand is perceived with some imprecision by the auctioneer, the model further adds a (small) noise term μ that follows a Gaussian distribution with mean 0.0 and standard deviation $\sigma = 0.05$. Then the transition probabilities for an increase or decrease of the market price by a fixed amount $\delta p = \pm 0.01$ is expressed as

$$\begin{aligned} \pi_{\uparrow P} &= \max[0, \beta(ED + \mu)], \\ \pi_{\downarrow P} &= -\min[0, \beta(ED + \mu)] \end{aligned} \qquad (4.3.8)$$

where β is a parameter for the reaction speed of the auctioneer.

Another stochastic factor can also be embedded into the fundamental price, by assuming that changes of the (log) fundamental value follows a Wiener process:

$\ln(p_{f,t}) = \ln(p_{f,t-1}) + \varepsilon_t \Delta t$ with $\varepsilon_t \approx N(0, \sigma_\varepsilon)$.In our experiments, we assume σ_ε = 0.005 .

Simulation Results

Figure 4.24 shows the time evolution of price, return and the fraction of chartists z, when the μ is adopted as the only source of noise.

The parameter set is:

$N = 500, v_1 = 3, v_2 = 2, \beta = 6, T_c (\equiv Nt_c) = 10, T_f\ (\equiv N\gamma) = 5, \alpha_1 = 0.6,$
$\alpha_2 = 0.2, \alpha_3 = 0.5, p_f = 10, r = 0.0004, s = 0.75.$

When we only take the uncertainty in the fundamental value, similar results are obtained as shown in Figure 4.25. Here, 80% of the traders in the market are fundamentalists, and 20% are chartists. The parameters setting is the same as the ones used above.

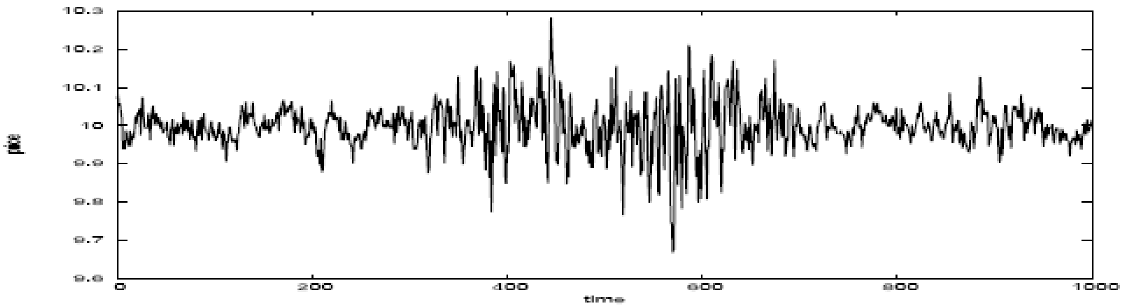
The simulation results when we adopt both types of noise simultaneously is displayed in Figure 4.26. Same parameters are again used.

Price and return

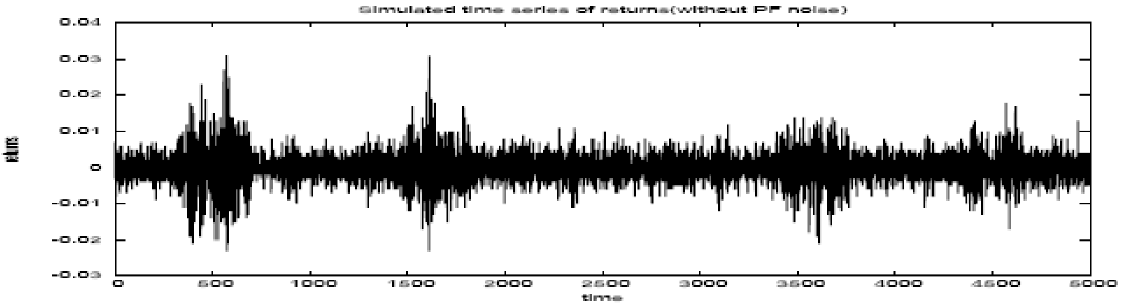
In the plots of price in both figures, we notice that some very large events happen and they tend to occur together. Correspondingly, some very large returns and clusters of larger returns are clearly shown in the plots of return. In addition, we see that fluctuations of return are remarkably correlated with the changes of the fraction of chartists — z. This correspondence provides a clue to the cause of volatility clustering: If the market is with a high fraction of fundamentalists, the market will move strongly towards the fundamental equilibrium. When the market is under the dominance of the chartists' practice, deviations from the fundamental equilibrium become self-reinforcing and the system can no longer maintain its local stability.

In the case that both noises are present at the same time, we see that the plots of price and return are more similar to the ones shown in Figure 4.25.

Therefore, it seems that the noise in the fundamental value has a stronger impact on the dynamics than the noise in the process of price adjustment, provided that they have the same standard deviation.



(a)



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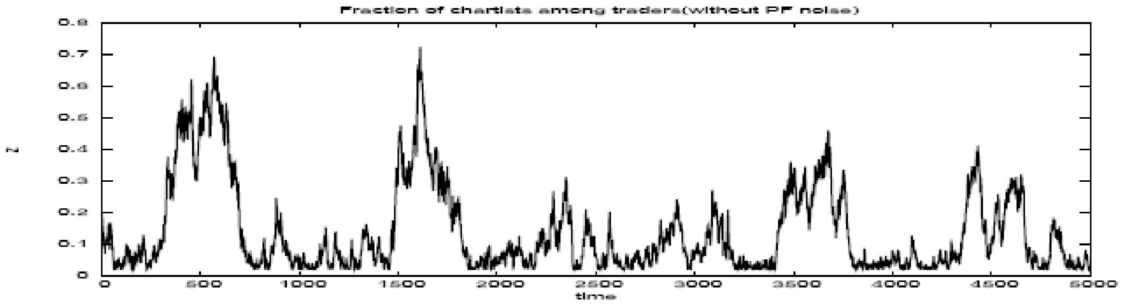
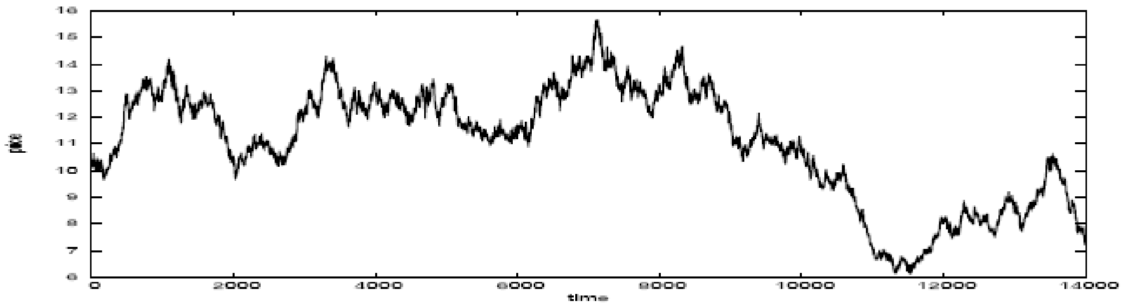
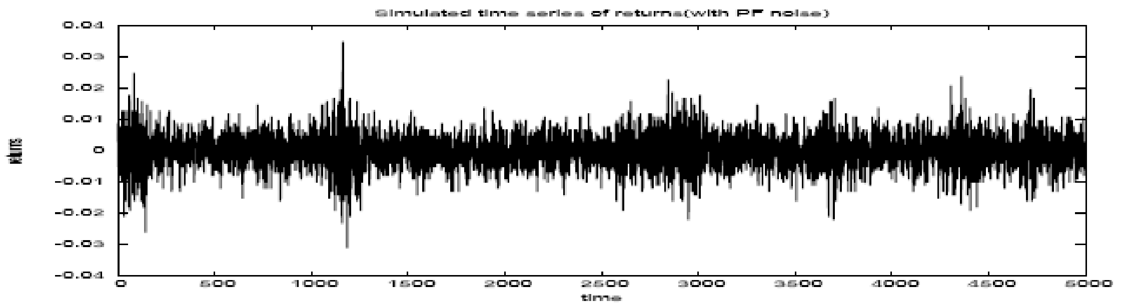


(c)

Figure 4.24: Simulation results of the Lux-Marchesi model when μ is adopted as the only source of noise. (a) Price evolution. (b) Time series of log-return.(c) Evolution of the fraction of chartists.



(a)



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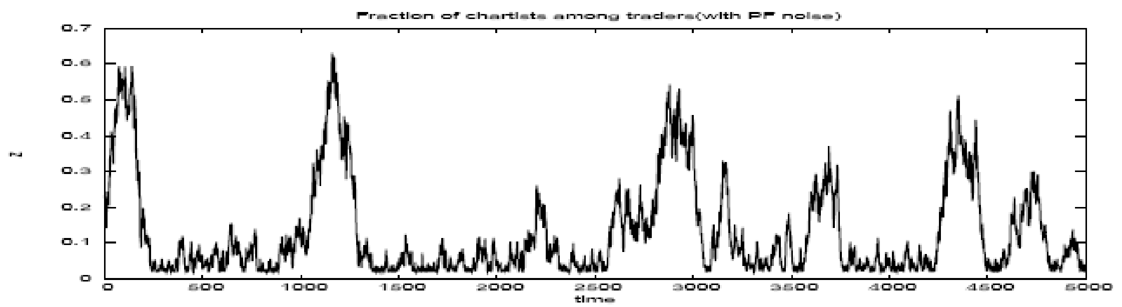
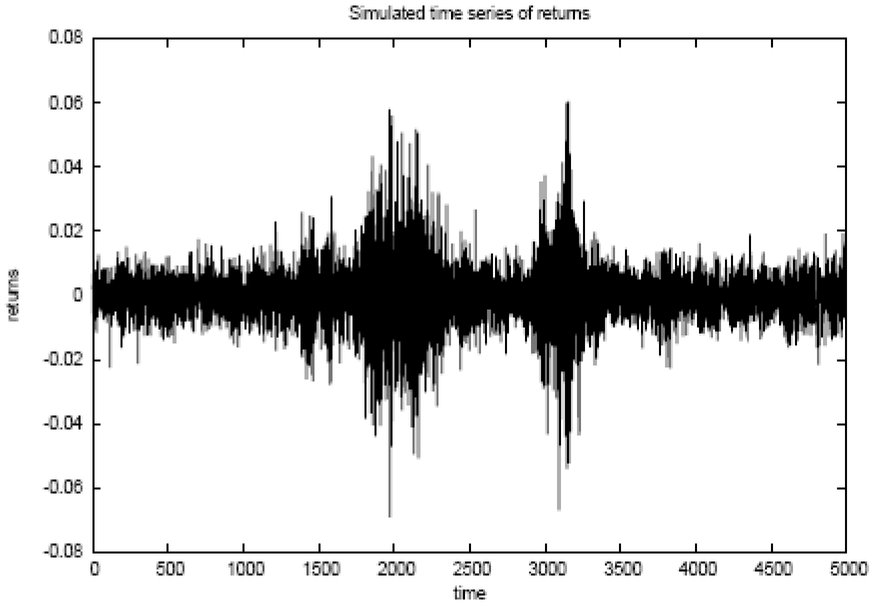
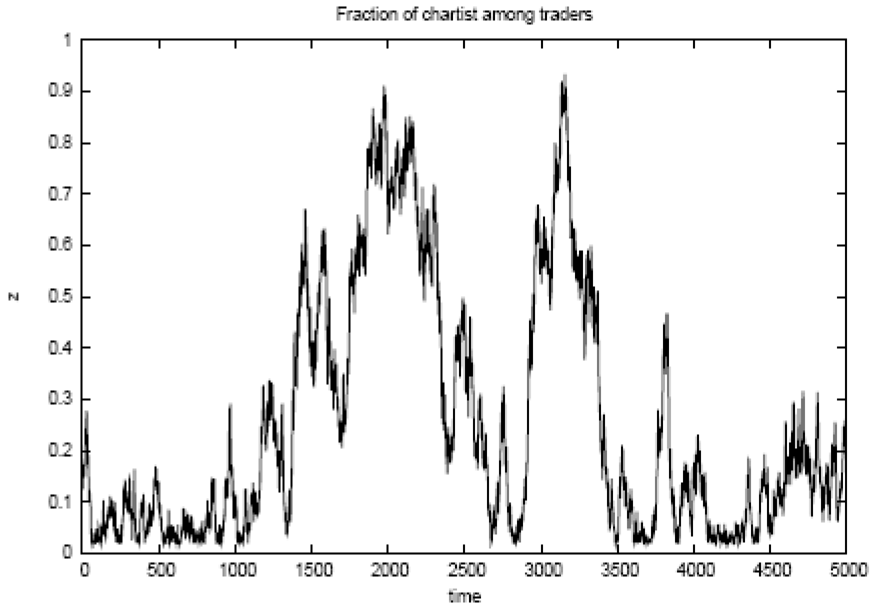


(c)

Figure 4.25: Simulation results of the Lux-Marchesi model when we only take the uncertainty in the fundamental value. (a) Price evolution. (b) Time series of log-return. (c) Evolution of the fraction chartists.



(a)



(b)

Figure 4.26: (a) simulated time series of log price returns, with fundamental price and excess demand noise of μ . (b). simulated time series of fraction of chartists. z = nc/N. With noise of fundamental price and noise of μ in price adjustment.

Probability distribution of returns

Figure 4.27 presents the PDF of return for the three situations with regard to varying level of noise. Figure 4.27(a), 4.27(b) and 4.27(c) are, respectively, the PDFs corresponding to the case when only the noise term μ is used, only the noise in the fundamental value is adopted, and both noises are chosen.

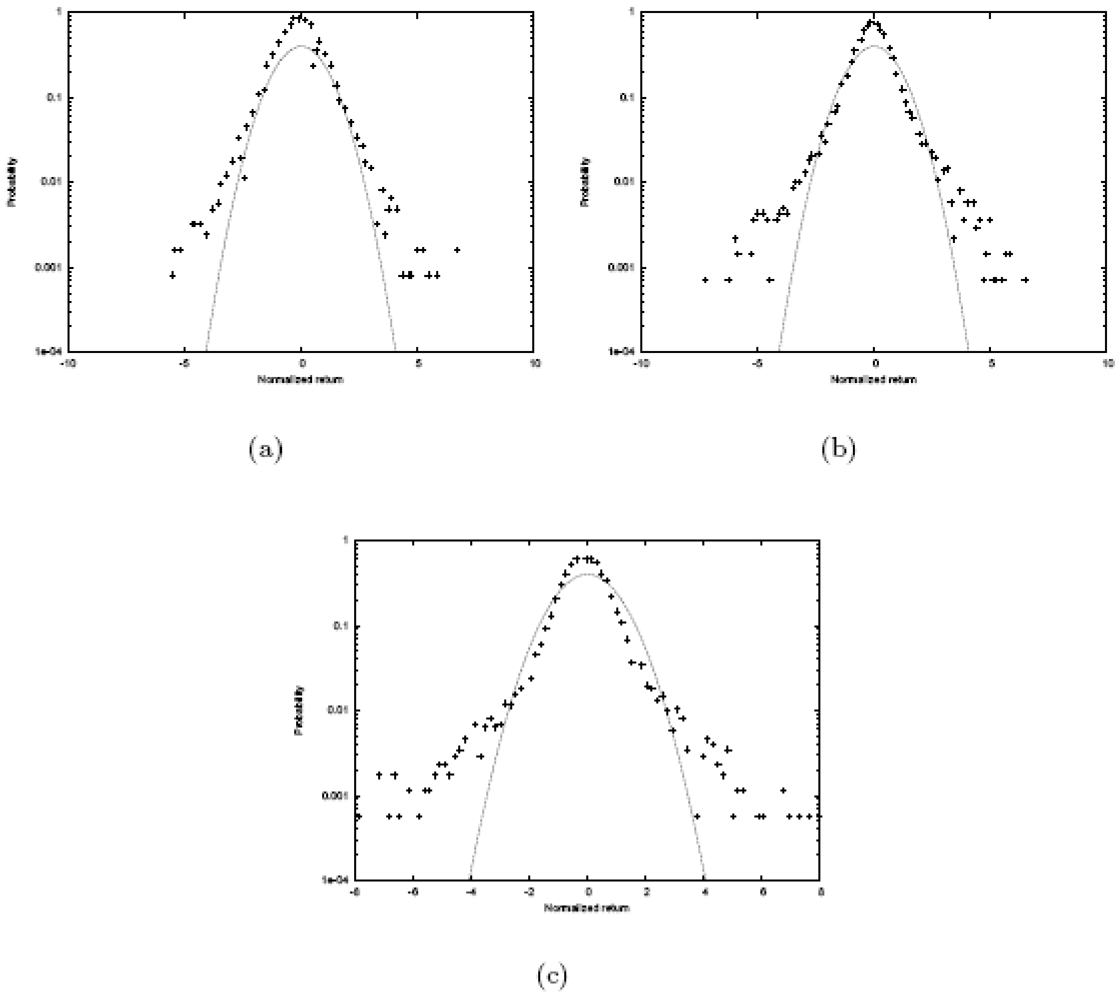


Figure 4.27: (a) PDF of return of the case that μ is the only noise. (b) PDF of return of the case that only the source of uncertainty in the fundamental price is adopted. (c) PDF when both types of noise are included.

We further study the properties of the probability density function of return at different levels of time aggregation, i.e. the values of τ in $\ln(p_{f,t}) = \ln(p_{f,t-\tau}) + \varepsilon_t \Delta t$. The corresponding PDFs are shown in Figure 4.28. This plot demonstrates that less value of τ leads to fatter tails; increasing the value of τ give rise to PDFs close to the Gaussian distribution. This confirms the observation on empirical data discussed in Chapter 2 that lower frequency (larger τ) financial data exhibits more features close to Gaussian distribution

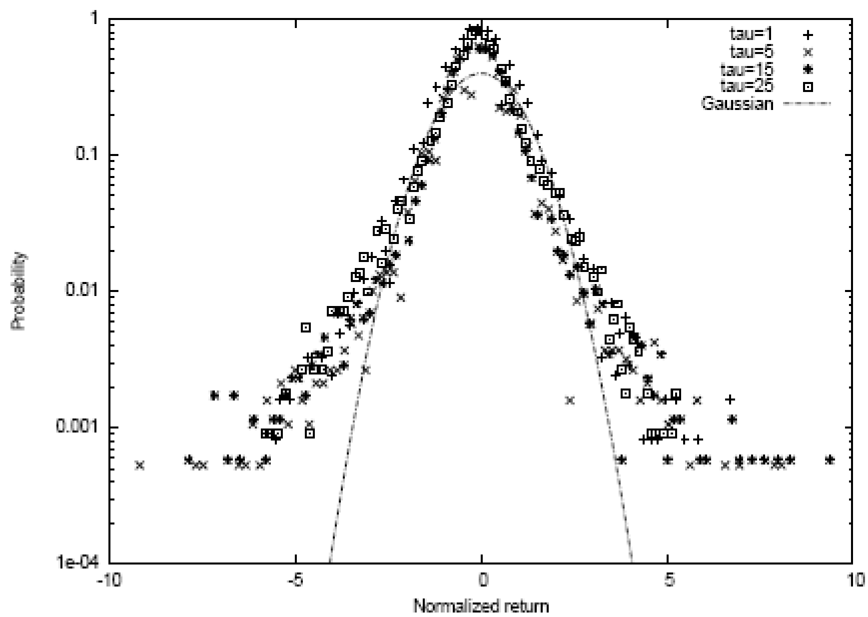
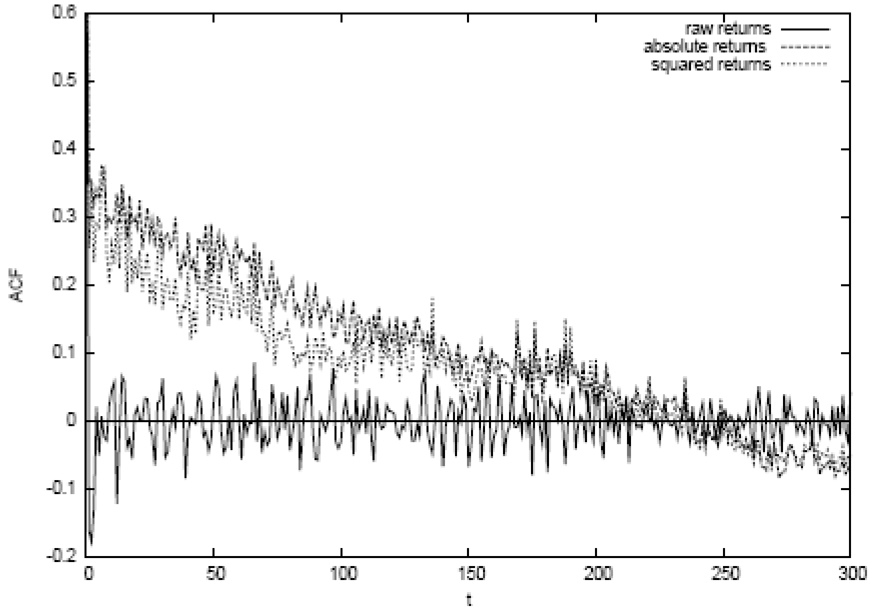


Figure 4.28: PDFs at different level of time aggregation.

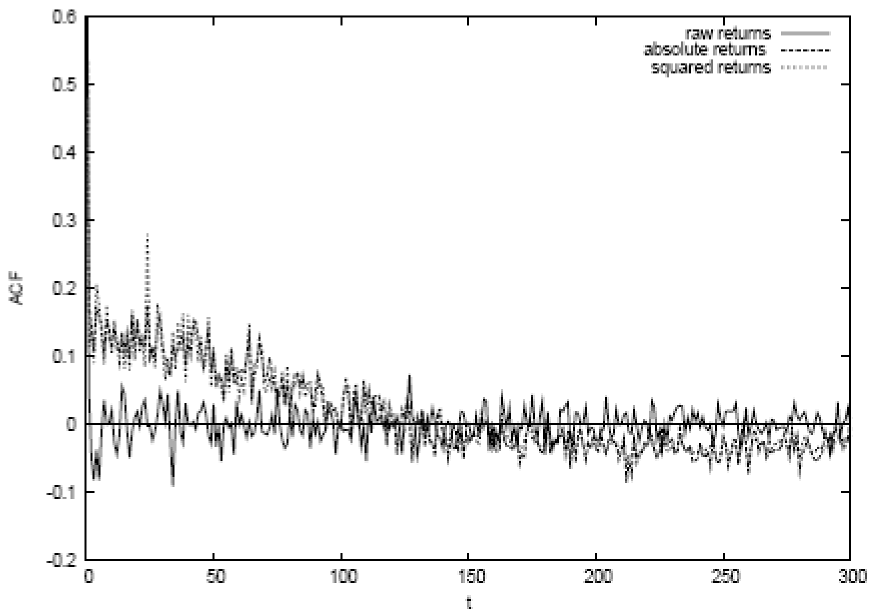
Autocorrelation of returns

To measure the effect of volatility clustering, we calculate and plot the autocorrelations of the simulated return, squared return and absolute return. All the situations discussed above are investigated. Figure 4.29 shows the corresponding ACF plots for the case with fundamental noise and the case without fundamental price noise, respectively.

Both plots show that autocorrelations of return fluctuate around zero, indicating lack of memory in return. However, squared and absolute returns (volatility) possess autocorrelations for long time. In addition, when the noise of fundamental price is present, the autocorrelations of squared and absolute returns decay more slowly that the other case.



(a)



(b)

Figure 4.29: (a) The ACF of return (the lowest one), squared return (the middle one) and absolute return. Fundamental price is fixed to 10. (b) The ACF of return (the lowest

one), squared return (the middle one) and absolute return. Changes of fundamental price follow a Gaussian distribution with mean 0 and deviation 0.0005.

Discussion

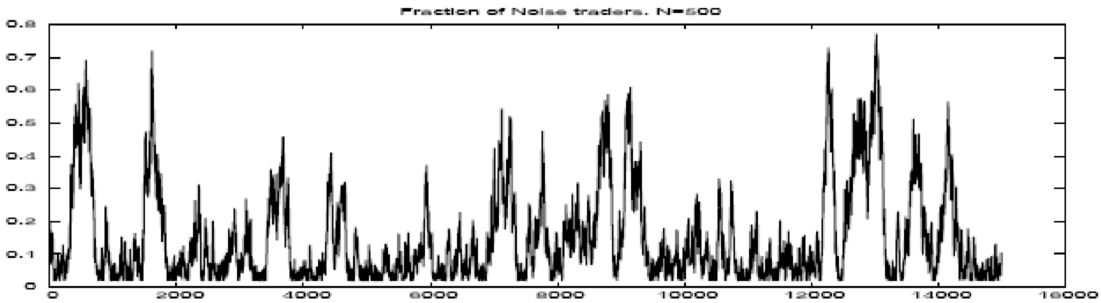
This model is suitable for simulating low frequency market dynamics, considering that traders need time to compare the performances of different strategies and make decision to switch among them. The movement of the traders among the different groups is the key of the dynamics. The periods of high fractions of chartists correspond to the periods of large clusters of high return.

The Lux-Marchesi model reproduces most of the main stylized facts including the ”fat tails” distribution of return, and volatility clustering. However, this model does not apply to a market with large number of traders. We have carried out a series of experiments concerning the influence of number of agents.

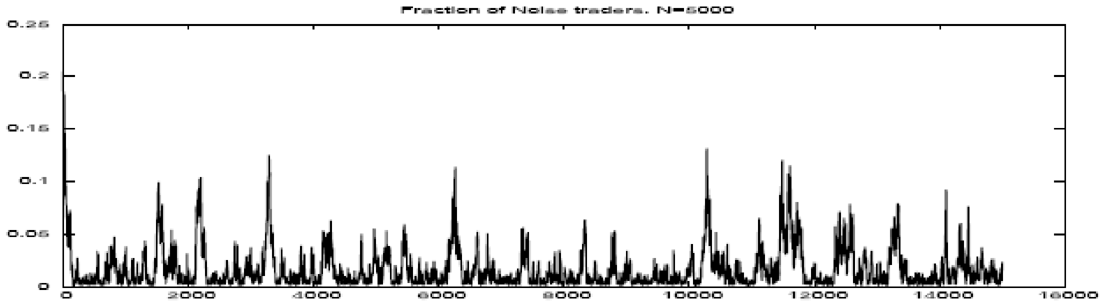
In the experiments the number of agents is varying from 500 to 100000. Figure 4.30 shows the simulation results of z when the number of agents is set to 500, 5000, 50000 and 100000, respectively. For each experiment, the fraction of chartist among all agents is initially set to 20%. For small N, for example 500 or 5000, the returns show strong fluctuations and clustering of volatility. As N becomes larger, it shows that z quickly falls to zero with little fluctuations.

Consequently, the returns exhibit much weaker or no volatility clustering, the price is closer to the fundamental price. Thus, large number of agents may significantly change the dynamics generated by the Lux-Marchesi model. These results are consisted with that of the authors.

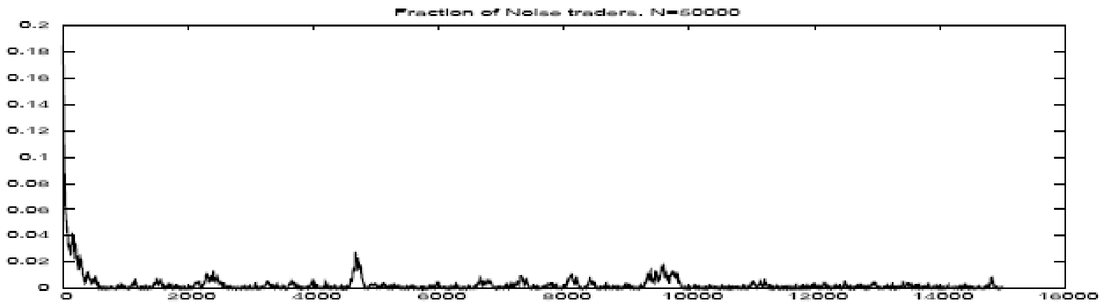
Another drawback of the Lux-Marchesi model is that the simulation time required is much longer than that required by the other two models. Our simulation program was implemented in JAVA. One simulation took about several hours on a machine with 2.80 GHZ Pentium 4 and 512 MB memory, when using 1000 traders. One of the reasons for this is that the Lux-Marchesi model has more types of trader (three) than the Bak model (two) and the Cont-Bouchaud model (one). Moreover the rules are more complicated than those of the other two models. Another important reason is that at every time step, every agent needs to compare his profitability with other agents and a new probability of switching needs to be determined, while in the Bak model in every time step, there are only two agents in the transaction, and only those two agents' behaviour are considered. Moreover, in the Cont-Bouchaud model, the random graph structure is formed only once in the beginning, the price calculation is calculated at a group level whose sizes are not changed through the simulation.


Fraction of Noise traders. N=500

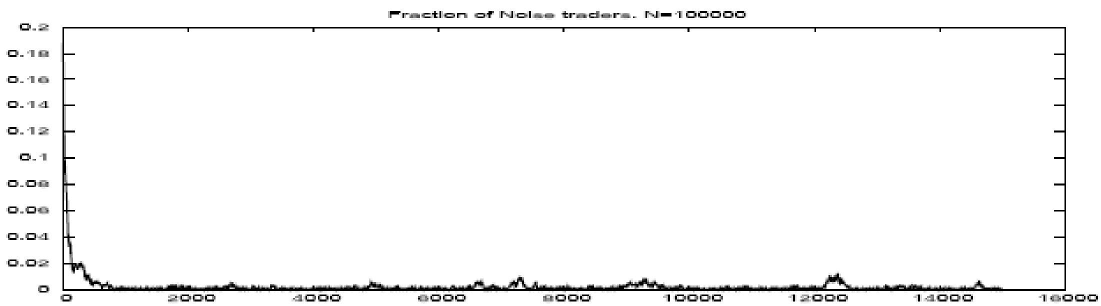
(a)


Fraction of Noise traders. N=5000

(b)


Fraction of Noise traders. N=50000

(c)


Fraction of Noise traders. N=100000

(d)

Figure 4.30: Evolution of fraction of chartists of different cases in terms of total number of traders. (a). N = 500, (b). N = 5000, (c). N = 50000 and (d). N = 100000. The scale of vertical axes in the various plots are :(a) 0 − 0.8,(b) 0 − 0.2, (c)0 − 0.06 and (d) 0 − 0.04. The scale of the horizontal axis is 0 − 150000.

5. Historical analysis of Artificial Market models – A view from statistical physics perspective

Most models proposed in the behaviourally orientated econophysics literature attempt to model this speculative interaction within certain prototypes of financial markets. Behavioural models of speculative markets have been among the first publications inspired by statistical physics.

First wave models

The earliest example is Takayasu et al. (1992) who in a continuous double auction, let agents' bid and ask prices change according to given rules and studies the statistical properties of the resulting price process. A similar approach has been pursued by Sato and Takayasu (1998) and Bak, Paczuski and Shubik

(1997). Independently, Levy, Levy and Solomon (1994, 1995, 2000) have developed a multi-agent model which, at first view, looks relatively conventional: agents possess a well-defined utility function which they attempt to maximize by choosing an appropriate portfolio of stocks and bonds. They assume particular expectation formation schemes (expected future returns are assumed to be identical to the mean value of past returns over a certain time horizon), short-selling and credit constraint as well as idiosyncratic stochastic shocks to individuals' demand for shares. Under these conditions, the market switches between periodic booms and crashes whose frequency depends on agents' time horizon. Although this model and its extension produces spectacular price paths, its statistical properties are not really inline with the empirical findings outlined above, nor are those of the other early models. Somewhat ironically, the early econophysics papers on financial market dynamics have been in fact similarly ignorant of the stylized facts like most of the traditional finance literature.

Second wave models

The second wave of models was more directly influenced by the empirical literature and had the declared aim of providing candidate explanations for the observed scaling laws. Mostly, they performed simulations of 'artificial' financial markets with agents obeying a set of plausible behavioural rules and demonstrated that pseudo-empirical analysis of the generated time series yields results close to empirical findings. To our knowledge, the model by Lux and Marchesi (1999, 2000) has been the first which generated both the (approximate) cubic law of large returns and temporal dependence of volatility (with realistic estimated decay parameter) as emergent properties of their market model. This model had its roots in earlier attempts at introducing heterogeneity into stochastic models of speculative markets. It had drawn some of its inspiration from Kirman's (1993) model of information transmission among ants which in Kirman (1991) had already been used as a model of interpersonal influences in a foreign exchange market. While Kirman's model had been based on pair-wise interaction, Lux and Marchesi had a mean-field approach in which an agent's opinion was influenced by the average opinion of all other traders.

Using statistical physics methodology, it could be shown that a simple version of this model was capable of generating bubbles with over- and undervaluation of an asset as a reflection of the emergence of a majority opinion among the pool of traders. Similarly, periodic oscillations and crashes could be explained by the breakdown of such majorities and the change of "market sentiment" (Lux, 1995). A detailed analysis of second moments can be found in Lux (1997) where the explanatory power of stochastic multi-agent models for time-variation of volatility has been pointed out (Ramsey, 1996, had already emphasized the applicability of statistical physics methods for deriving macroscopic laws for second movements from microscopic behavioural assumptions). The group dynamics of these early interaction models have been enriched in the simulation studies by Lux and Marchesi (1999, 2000) and Chen, Lux and Marchesi (2001) by allowing agents to switch between a chartist and fundamentalist strategy in response to differences in profitability in both strategies. Interpersonal influences enter via chartists' attempts to trace out information from both ongoing price changes as well as the observed 'mood' of other traders.

Assuming that the market is continuously hit by news on fundamental factors, one could investigate in how far price changes would mimic incoming information (the traditional view of the efficient market paradigm), or would be disconnected from fundamentals. The answer to this question turned out to have two different aspects: first, the speculative market on average kept close track of the development of the fundamentals. All new information was incorporated into prices relatively quickly as otherwise fundamentalist traders would have accepted high bets on reversals towards the fundamental value. Second, upon closer inspection, the output (price changes) differed quite significantly from the input (fundamental information) in that price changes were always characterized by the scaling laws of the equations even if the fundamental news were modelled as a white noise process without these features. Hence; the market was never entirely decoupled from the real sphere (the information), but in processing this information it developed the ubiquitous scaling laws as emergent properties of a system from the distributed activity of its independent subunits.

This result could also be explained to some extent via an analysis of approximate differential equations for the mean values of state variables derived from the mean-field approach (Lux, 1998; Lux and Marchesi, 2000). In particular, one finds that, in a steady state, the composition of the population is indeterminate. The reason is that in a stationary environment, the price has to equal its fundamental value and no price changes are expected by agents. In such a situation, neither chartists nor fundamentalist would have an advantage over the other group as neither mispricing of the asset nor any discernible price trend exists. In the vicinity of such a steady state, movements between groups would, then, only be governed by stochastic factors which would lead to a random walk in strategy space. However, in this model (and in many related models), the composition of the population determines stability or instability of the steady state. Quite plausible (and in line with a large body of literature in behavioural finance), a dominance of chartists with their reinforcement of price changes will be destabilizing.

Via bifurcation analysis one can identify a threshold value for the number of chartists at which the system becomes unstable. The random population dynamics will lead to excursions into the unstable region from time to time which leads to an onset of severe fluctuations. The ensuing deviations of prices from fundamental value, however, causes profit differentials in favour of the fundamentalist traders so that their group increases and the market moves back to the stable subset of the strategy space.

The joint dynamics of the population composition and the market price has a close resemblance to empirical records. With the above mechanism of intermittent switching between stability and (temporal) instability the model not only has interesting emergent properties, but it also can be characterized by another key term of complexity theory: criticality. Via its stochastic component, the system approaches a critical state where it temporally looses stability and the ensuing out-of-equilibrium dynamics give rise to stabilizing forces. One might note that "self-organizing criticality" would not be an appropriate characterization as the model does not have any systematic tendency towards the critical state. The trigger here is a purely stochastic dynamics without any preferred direction. Lux and Marchesi argue that irrespective of the details of the model, indeterminateness of the population composition might be a rather general phenomenon in a broad range of related models (because of the absence of profitability of any trading strategy in any steady state).

Together with dependency of stability on the population composition, the intermittent dynamics outlined above should, therefore, prevail in a broad class of microscopic interaction models. Support for this argument is offered by Lux and Schornstein (2005) who investigate a multi-agent model with a very different structure. Adopting the seminal Kareken and Wallace (1983) approach to exchange rate determination, they consider a foreign exchange market embedded into a general equilibrium model with two countries and overlapping generations of agents. In this setting, agents have to decide simultaneously on their consumption and savings together with the composition of their portfolio (domestic vs. foreign assets). Following Arifovic (1996), agents are assumed to be endowed with artificial intelligence (genetic algorithms) which leads to an evolutionary learning dynamics in which agents try to improve their consumption and investment choices over time.

This setting also features a crucial indeterminacy of strategies: in a steady state, the exchange rate remains constant so that holdings of domestic and foreign assets would have the same return (assuming that returns are only due to price changes). Hence, the portfolio composition would be irrelevant as long as exchange rates remain constant (in steady state) and any configuration of the GA for the portfolio part would have the same pay-off.

However, out-of-equilibrium, different portfolio weights might well lead to different performance as an agent might profit or loose from exchange rate movements depending on the fraction of foreign or domestic assets in her portfolio. The resulting dynamics again shares the scaling laws of empirical data. Similarly, Giardina et al. (2000) allow for more general strategies than Lux and Marchesi (1999) but also found a random walk in strategy space to be at the heart of emergent realistic properties.

A related branch of models with critical behaviour has been launched by Cont and Bouchaud (2000). Their set-up also focuses on interaction of agents. However, they adapt the framework of percolation models in which agents are situated on a lattice with periodic boundary conditions. In percolation models, each site of a lattice might initially be occupied (with a certain probability p) or empty (with probability 1-p). Clusters are groups of occupied neighbouring sites (various definitions of neighbourhood could be applied). In Cont and Bouchaud, occupied sites are traders, and trading decisions (buying or selling a fixed number of assets or remaining inactive) are synchronized within clusters. Whether a cluster buys or sells or does not trade at all, is determined by random draws. Given the trading decisions of all agents, the market price is simply driven by the difference between overall demand and supply. Being modelled closely along the lines of applications of percolation models in statistical physics, it follows immediately that returns (price changes) will inherit the scaling laws established for the cluster size distribution. Therefore, if the probability for connection of lattice sites, say q, is close to the so-called percolation threshold qc (the critical value above which an infinite cluster will appear), the distribution will follow a power law. As detailed in Cont and Bouchaud, the power-law index for returns at the percolation threshold will be 1.5, some way apart from the "cubic law". As has been shown in subsequent literature, finite-size effects and variations of parameters could generate alternative power laws, but a cubic law would only emerge under particular model designs (Stauffer and Penna, 1998). Autocorrelation of volatility is entirely absent in these models, but could be introduced by sluggish changes of cluster configurations over time (Stauffer et al., 1999). If clusters dissolve or amalgamate after transactions, more realistic features could be obtained (Equiluz and Zimmerman, 2000) As another interesting addition, Focardi et al. (2002) consider latent connections which only become relevant in times of crises.

Alternative lattice types have been explored by Iori (2002) who considers an Ising type model with interactions between nearest neighbours. This approach appears to generate more robust outcomes and seems not to suffer from the percolation models essential need to fine tune the parameter values at criticality for obtaining power laws. As long as no self-organizing principles are offered for the movements of the system towards the percolation threshold, the extreme sensitivity of percolation models with respect to parameter choices is certainly unsatisfactory. Though sweeping these systems back and forth through a critical state might be an interesting addition (explored by Stauffer and Sornette,1999), behavioural underpinnings for such dynamics are hard to imagine.

6. Simple market model equation – From Dynamical systems to stochastic calculus and mixtures

Now we turn to define from scratch a more plausible market model. As a starting point we define some toy equations that theoretically we can prove, have "good" behaviour.

Equation of the Price change

$$\delta x = \frac{1}{\lambda}\sum_i \varphi_i \equiv \frac{\Phi}{\lambda} \quad (6.1)$$

- δx is the price increment
- λ is a measure of the market depth
- φ_i each operator i may buy , sell or hold , values of -1,0,+1

Dynamical equation for the excess of demand

$$\Phi_{t+1} - \Phi_t = a\delta x_t - b\delta^2 x_t - a'\delta x_t - K(x_t - x_0) + \chi\xi_t \quad (6.2)$$

- a > 0 describes trend following effects that is how price increments amplifies the average propensity to buy and sell.
- b>0 describes risk aversion effects and encodes the fact that price drops are given more weight than increases: the term $-b\delta^2 x_t^2$ can be seen as giving to the parameter a dependence on δx_t , such that a increases when the price goes down. Alternatively δx_t^2 can be seen as a proxy for the short term volatility: an increase of

volatility should decrease the demand for the stock. The most important point is that a term proportional to δx_t^2 breaks the $\delta x_t^2 \rightarrow -\delta x_t^2$ symmetry.

- $a' > 0$ is a stabilizing term (which has an opposite to that of a) that arises from market clearing mechanisms: the very fact that the price moves clears a certain number of orders and reduces the imbalance. In fact, because the size of the queue in the order book is on average an increasing function of the distance from the current price, one expects non linear corrections to this market clearing argument, that will add terms of higher order in δx_t .
- K>0 is a mean reverting force that models the fact that the price x_t is too far above a reference fundamental price x_0 , sell orders will be generated and vice-versa.
- Finally, ξ is a random variable noise term representing random external news that induces more buy (sell) orders. The coefficient χ can be seen as the susceptibility to these news. A more refined version of the theory should allow this coefficient itself to depend on past values of δx : an increase of the volatility tends to make agents more nervous and feeds back onto itself, leading to volatility clustering.
- We won't be considering m effects and risk aversion b maybe not considered in our model. If we don't consider that we are left with the linear-non linear effects of the other parameters and the nature of the ξ random variable.

Langevin equation

$$u \equiv \frac{dx}{dt}$$
$$\frac{du}{dt} = -\alpha u - \beta u^2 - \gamma u^3 - \kappa(x - x_0) + \tilde{\xi}(t) \qquad (6.3)\text{-}(6.4)$$
$$\alpha = (a' - a)/\lambda, \beta = b/\lambda, \gamma = c/\lambda, \kappa = K/\lambda, \tilde{\xi}(t) = \chi\xi(t)/\lambda$$

Eliminating from the equations 6.1 and 6.2 and the continuous time limit we can derive the "langevin" equation for the price velocity $u \equiv \frac{dx}{dt}$. This equation has very interesting features and phenomenas.

Potential equation

$$V(u) = \kappa(x - x_0)u + \alpha\frac{u^2}{2} + \beta\frac{u^3}{3} + \gamma\frac{u^4}{4} + \ldots\ldots$$
$$\frac{du}{dt} = -\frac{\partial V}{\partial u} + \tilde{\xi}(t) \qquad (6.6)\text{-}(6.7)$$

Linear case

In the case where β and γ are absent, that means the linear case, what we have is an equation well known in physics, the forced harmonic oscillator with friction:

$$\frac{d^2x}{du^2} = -\alpha\frac{dx}{dt} - \kappa(x - x_0) + \tilde{\xi}(t) \quad (6.8)$$

If $\alpha > 0$ the above equation can be described as an Ornstein – Uhlenbeck process, price increment is given by:

$$\langle u(t)u(t+\tau)\rangle = \frac{D}{2\alpha}e^{-\alpha\tau} \quad (6.9)$$

Where D is the variance of $\tilde{\xi}(t)$. The volatility $\langle u(t)^2 \rangle = \frac{D}{2\alpha}\alpha\frac{\chi^2}{\lambda(a' - a)}$

In mathematics, the Ornstein–Uhlenbeck process, also known as colored noise, (named after Leonard Ornstein and George Eugene Uhlenbeck), also known as the mean-reverting process, is a stochastic process rt given by the following stochastic differential equation:

$$dr_t = -\theta(r_t - \mu)dt + \sigma dW_t \quad (6.10)$$

where θ, μ and σ are parameters and Wt denotes the Wiener process.

This equation is solved by variation of parameters. Apply Itō's lemma to the function $f(r_t, t) = r_t e^{\theta t}$ to get

$$df(r_t, t) = \theta r_t e^{\theta t} dt + e^{\theta t} dr_t = \mu\theta e^{\theta t} dt + \sigma e^{\theta t} dW_t \quad (6.11)$$

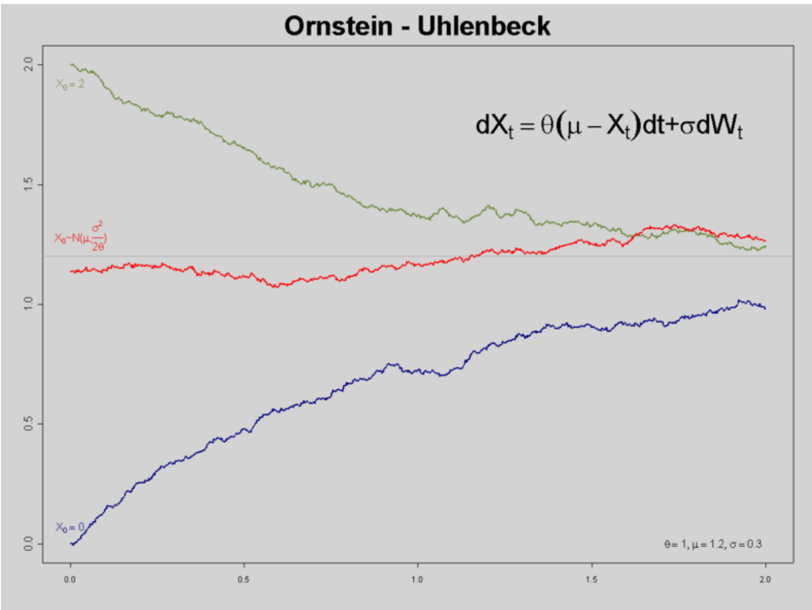


Figure 6.1 Orstein-Uhlenbeck process.

Bubbles and crashes

Suppose now that feedback effects are strong compared to stabilizing effects: $a > a'$ and b>0, such that $\alpha < 0$ and $\beta > 0$. Suppose that the initial price is $x \approx x_0$ and that

$\gamma = 0$ for simplicity. The above potential develops a minimum for a non zero value of u given by $u^* = -\alpha / \beta > 0$. The fictitious particle hovers around its minimum. This means that price fluctuate around a positive mean u^* that has no economic justification (recall that a true rise of future expectations, modelled by parameter m , is chosen to be zero here). This growth is entirely sustained because of trend following effects: this is a speculative bubble.

In a first approximation, the price or log-price grows linearly with time as $u^* t$. Interestingly, when t reaches a time t_c given, within this simple approximation, by $t_c = \alpha^2 / 4\beta\kappa u^* = -\alpha / 4\kappa$, the minimum of the potential ceases to exist and u flows to -$-\infty$: the bubble collapses and the price plummets. The price decay is again stopped by the mean reverting 'fundamental' term κ. Not surprisingly, the bubble lifetime t_c is long if trend following effects are strong ($|\alpha|$large) and or/if the mean reverting fundamental effect is weak (κ is small).

If trend following effects are not too strong, α is positive and $V(u)$ has a local minimum for u=0 , and a local minimum for $u^* = -\alpha / \beta < 0$, beyond to which the potential decrease $-\infty$. A potential barrier $V^* = \alpha u^{*2} / 6$ separates the (meta-stable) region around u=0 from the instable region $u < u^*$.

The nature of the motion of u in such a potential is the following: starting at u u =0 , $x = x_0$, the fictious particle oscillates in the the vicinity of u=0 , corresponding to a Brownian Like-Motion of the price itself.

However, occasionally, an 'activated' event (ie. driven by the noise term $\tilde{\xi}(t)$) brings the particle near u^*.

Once the potential barrier is crossed the fictious particle reaches $u = -\infty$ in finite time. (The stabilizing term γu^3 would actually prevent this divergence and create a secondary minimum for a negative value of u, corresponding to the crash regime). In financial terms, the regime where u oscillates around u=0 and where β can be neglected is the normal fluctuation regime.

This normal regime can however be interrupted by crashes , where the derivative of the price becomes very large and negative, due to risk aversion term b which desestabilizes the price by amplifying selling orders. The interesting point is that these two regimes can be clearly separated since the average time t^* needed for such crashes to occur is exponentially large in V^* / D (corresponding to the time needed to overcome the barrier V^*, and thus appear only very rarely.

A very long time scale is therefore naturally generated in this model. Note that in this line if thought, a crash occurs because an improbable succession of unfavourable events, and not due to a single large event in particular. However, volatility feedback effects mean that stronger and stronger fluctuations might precede the crash, each large fluctuation increasing D and making the crash more probable.

Now will turn our work to the modelling of a new Artificial Market Model we think may correspond to the observed facts and real behaviours of agents and we will analyse if market experiment paths and statistical facts correspond to market reality.

7. NKS artificial market model

7. 1 Equations and description

Our model describes the behaviour of a heterogeneous population of traders that at a given moment in time. Each type of trader decides the day's trade (buy, sell or hold) according an strategy rule incorporating external news and two indicators based on the market price history - a "fundamental" or price level indicator, and a "technical" or recent trend indicator. Each of these indicators, including the news, can give strong or weak signals in either direction.

The strategy determines the recommended trade from the indicators; the strength of the signals determines the desired trade amount or volume. There is considerable possible variation in these simple strategies - 6561 (3 to the 8th power) distinct strategies are possible with these rules. In addition, the profit and loss (P&L) and current stock position of each strategy currently in use are tracked throughout, and traders are more likely to trade if ahead in P&L terms, and if reversing a current stock position (long or short) rather than adding to it.

There is some stochastic component to trade timing, and to the external news. Prices are a straightforward result of the day's desired trades - the price adjusts to reflect the order imbalance, and all desired trades then fill at the new price. The model can also, optionally, incorporate "herding" changes in the strategies in use, traders switch from the least successful strategies in use to better ones.

The output of the model is the evolution of the price of a theoretical asset, along with positions and P&Ls experienced by the different strategies used. We study how the market behaves with different mixes and weights of traders, using different strategies, as well as sensitivity to different inputs.

Moreover, we added features to the model sequentially and can speak to the question which features in the resulting behaviour depend on which internal features of the model. We looked in the price evolution for stylized facts seen in real world financial markets. Real financial markets are complex systems, showing non-stationary in the probability distributions especially changing volatility through time, non-trivial scaling and serial correlation, and non-gaussian distributions of returns including high and variable kurtosis and fat tails. We were able to find all of these signals in our model.

We show that a combination of very simple trading rules, a mixture of different traders, with feedback and adaptation effects and some randomness through news and stochastic order arrival to the markets give the real market's very complicated behaviour.

As shown before with the theoretical equations it is possible to account for the stylized facts and market qualitative paths with the chosen parameters and trading strategies, memories of these indicators.

We have defined a much richer market model than the simple market model, given that market practice and effects are much more complicated than the previous description, introducing in the model real life indicators and real life memory effects, portfolio probabilistic limits, strategy switching, a random discrete model for the news, non-linearity in the response to the indicators and a rich universe of traders.

We cannot prove that this is the only configuration of these arbitrary equations we show that the stylized facts and market paths fit the market and we think that the conditions, trading strategies, and news effects we introduce are perfectly plausible given our experience as a financial practitioners and applied mathematicians.

The reader will see that we have chosen a model close to market reality and practice, and we won't be introducing more mathematical formalisms because the outputs and results are straightforward.

We use Mathematica 6.0 as a programming code and the calculations are fast and accurate with a simple Dell Latitude D510 Laptop.

Detailed Description of the Market

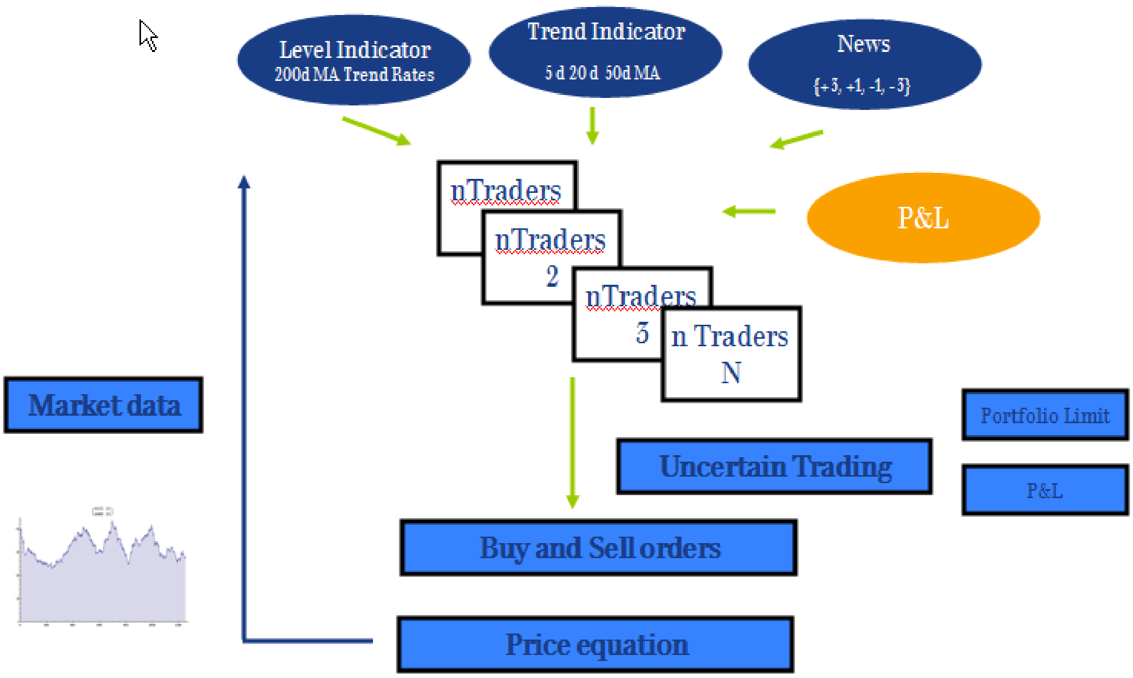


Figure 7.1.1 Graphic with the Framework

Price Equation

$$dx = (vol * Size[indicators]) \times \sum_{i=1}^{N} (BinominalDistribution[weight[i], Tradechance[perform]]) * Choice[i][indicators]$$

note = (1/lamba) market depth parameter of the prior market equation has become a (volparameter * Size[indicators]) factor, and has thereby become time-varying; the sum of trader decisions acts as in the prior model, although determining the trade decisions for each actor is more involved.

Compared to the prior market model equation, the market depth parameter (1/lamba) has been replaced by the product of a fixed vol parameter and a time - varying trade Size function.

Effectively, market depth is time - varying, through the second of these factors. The time - variation of market depth is a function of the same indicators, {level, trend, news} driving individual trade decisions.

Size = a trading-scale or market-depth function. The form is simple - the product of the clarity of the 3 indicators, with "very" = 2, normal = 1. Thus market depth increases when the news is very bad or very good, when the market is above or below all 3 reference levels, when the market is above or below all three lagged moving averages. When all indicators show strong signals, market depth is 8 times its minimum level, which occurs when none of the indicators are giving strong signals. This change is responsible for stochastic volatility in the resulting behaviour.

The sum of trade decisions is of the same general form as in the equation model - a sum of buy, hold or sell decision for each unit of market weight at that time-step. The individual trade decisions have been aggregated into weighted strategies, reflecting how they respond to the 3 indicators. The decision process is more complicated than in the market equation model, in several ways, but once arrived at those trade decisions affect the price - change in exactly the same way as in the equation model. (One might explore reducing those complexities to get a more easily solved form etc).

The trade decisions have a stochastic component influenced by current P&L and current stock position and a strategy recommendation component determined by the indicators and the specific strategy involved. Basically the strategy gives a trade direction, buy hold or sell, but some portion of traders will "Hold" anyway, thus "smearing out" timing of trade executions. Traders are more likely to trade to close existing positions than to add to them, and are more likely to trade when ahead than when behind, in overall profit and loss to date terms.

Details of the NKS AM

Overview

- external parameters = {volparameter, expectedtrendrate}
- trader parameters = {weight[i], Choice[i]}
- indicators = {level, trend, news}

Variable explanations - capitals denote functions, lower case denotes scalars

- weight[i] = scalar weight of the ith trading strategy (number of traders or amount of capital committed to that strategy)
- Choice[i] = ith trading strategy decision function
- TradeChance[performance] = {0.5, .25, 0.125}, default 0.5, halved if proposed trade direction is the same as existing stock position, halved if current P & L is negative
- A specific Choice function is any complete map from all possible triples of the sign of the 3 indicators to {1, 0, -1} < == > {buy, hold, sell}
- volparameter = scalar (constant) parameter of volatility

Indicators

- level, trend, news values are each in {+3, +1, -1, -3}
- level = relation of current price to several long term rates - this includes an expected trend rate parameter set externally.
- expectedtrendrate = a fundamental trader's expected long run average price change; price level thresholds are set at 1.5, 1, and 0.5 times that rate, projected, above initial baseline level of an initial 200 day MA. Thus if expectedtrendrate is 8%, the trend rates used for levels are 4%, 8%, and 12%. The level indicator is a strong positive signal if the price has increased less than 4% on average since the baseline period, and a strong negative signal if the price has increased more than 12% on average since the baseline period, etc.
- trend = relation of current price to several lag-length trends; trend is sign of moving average minus current price for lags of 5, 20, and 50 days. Thus trend is strongly positive if the price is above its 5, 20, and 50 day moving averages, strongly negative if below all 3 of those, etc.
- news = random new information, very good, good, bad, or very bad, externally specified. Standard case is random {1/8, 3/8, 3/8, 1/8} distribution of news, independent.

What is a full initial condition?

A full initial condition consists of a **trader set** and a price history. A **trader set** is a weighted collection of strategies.We also track a current stock position and running profit or loss (P & L) for each strategy.

These start out at 0 for all strategies.

traders = {{4060,2,{0},{0}}, {6192,8,{0},{0}}, {6480,10,{0},{0}}};

Because strategies can depend on past prices, a full initial condition also includes a history of prices before the run itself.

past = 180*N@TrendUpAboveAve

{9000.`,9540.`,9540.`,9360.`,9540.`,9000.`,8640.`,8640.`,8280.`,8280.`,8460.`,8820.`,8640.`,8280.`,8640.`,8100.`,7740.`,7920.`,7560.`,7920.`,8460.`,8280.`,8460.`,8820.`,8280.`,7740.`,7380.`,7560.`,8100.`,7560.`,7020.`,6480.`,6120.`,6120.`,6300.`,6120.`,5940.`,6120.`,6300.`,6660.`,7200.`,6660.`,7200.`,7020.`,7380.`,7560.`,8100.`,7920.`,7920.`,8100.`,8460.`,8100.`,8280.`,8640.`,9000.`,8460.`,8280.`,7920.`,8100.`,8280.`,8640.`,9000.`,9180.`,9360.`,9000.`,8820.`,9360.`,9900.`,10080.`,10080.`,9540.`,9360.`,9720.`,9540.`,9000.`,9000.`,8820.`,9000.`,9540.`,9360.`,9900.`,10440.`,9900.`,10080.`,10080.`,10440.`,10800.`,10440.`,10980.`,11160.`,11340.`,11880.`,12060.`,11880.`,12240.`,12600.`,12780.`,13320.`,13500.`,13320.`,13320.`}

What other parameters are there?

There is a single **volatility** parameter in the price change equation. For a given order imbalance, it determines the percentage price change that results before the trades are executed. It is a scalar, all time variation in volatility reflects variations in actual orders. The total weight of all strategies and the volatility parameter have similar effects; one can raise or lower measured standard deviations by adjusting either.

There is a single expected **growth rate** parameter used by the level indicator. The price is considered low as of a given date if it is below a projection of a 200 day moving average at the start, growing at the given rate.

The price is very low for the level indicator if it is below projected growth at half this rate and very high if above projected growth at one and a half times this rate. Note that this only effects the clarity of this indicator and with it, order sizes for recommended trades.

One can evolve the run for any number of **steps**. We have chosen our typical parameter values to reflect days, but for suitable values for other parameters one could model intraday trading or longer scales etc.

Potentially one can feed in an arbitrary sequences of **news**, each day's tick being very good, good, bad, or very bad. For our explorations here, we just used random news evenly distributed with probabilities {1/8, 3/8, 3/8, 1/8} in those categories, but these could be adjusted.

What data is returned?

Data returned is a pair, the first element consisting of the strategies and their ending stock position and P&L, the second element consisting of the price series generated by their trading decisions.

One can use a single Sow and Reap to collect all the individual trading decisions for each strategy, and the running stock and P&L positions at each time step, if those are desired for analysis.

All standard financial measures can be performed on the resulting price series - descriptive statistics, return histograms, moving averages of volatility, etc.

Example of a fully specified run

```
traders = {{4060, 2, {0}, {0}}, {6192, 8, {0}, {0}}, {6480, 10, {0}, {0}}};
past = 180*N@TrendUpAboveAve;
```

data = evolve[{traders, past}, 250, .001, 8];

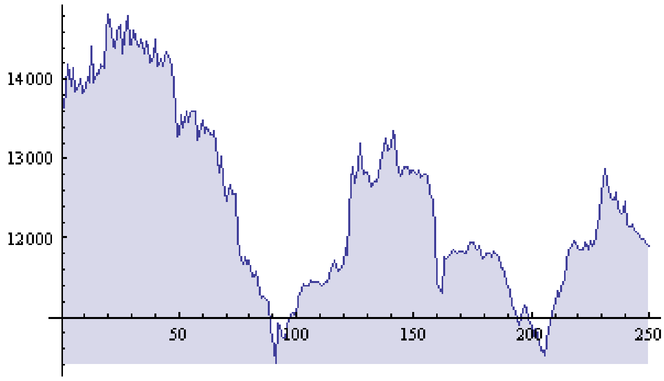


Figure7.1.1 NKS Price evolution

Changing strategies inside a single run

Although most of our analysis was conducted with the model described above using a static set of traders, we also coded a dynamic version in which strategy - weight moves from losing to winning strategies. Please contact me for the experiments with strategy switching.

7. 2 Experiments and outputs – Stylized facts and Behaviours

Evaluations

traders = {{6192, 12, {0}, {0}}, {4060, 3, {0}, {0}}, {6480, 6, {0}, {0}}};

withtrades = Reap[evolve[{traders, N@TrendUpBelowAve}, 1250, .002, 25]];

volume = Abs[Total[#]] & /@ withtrades[[-1, 1]];

volMA = Sqrt[251]* MovingAverage[Log[1 + periodchanges[#, 1] &@Last@withtrades[[1]]]^2, 20]^.5;

StandardDeviation[volMA] = 0.152888

volumeVolCo = Correlation[Drop[MovingAverage[volume, 20], 1], volMA]0.988863

returns = periodchanges[#, 1] &@withtrades[[1, 2]];

vols = (returns^2)^.5 / (1/251^.5);

Sqrt[251]*StandardDeviation@returns = 0.319495

Kurtosis@returns = 10.6088

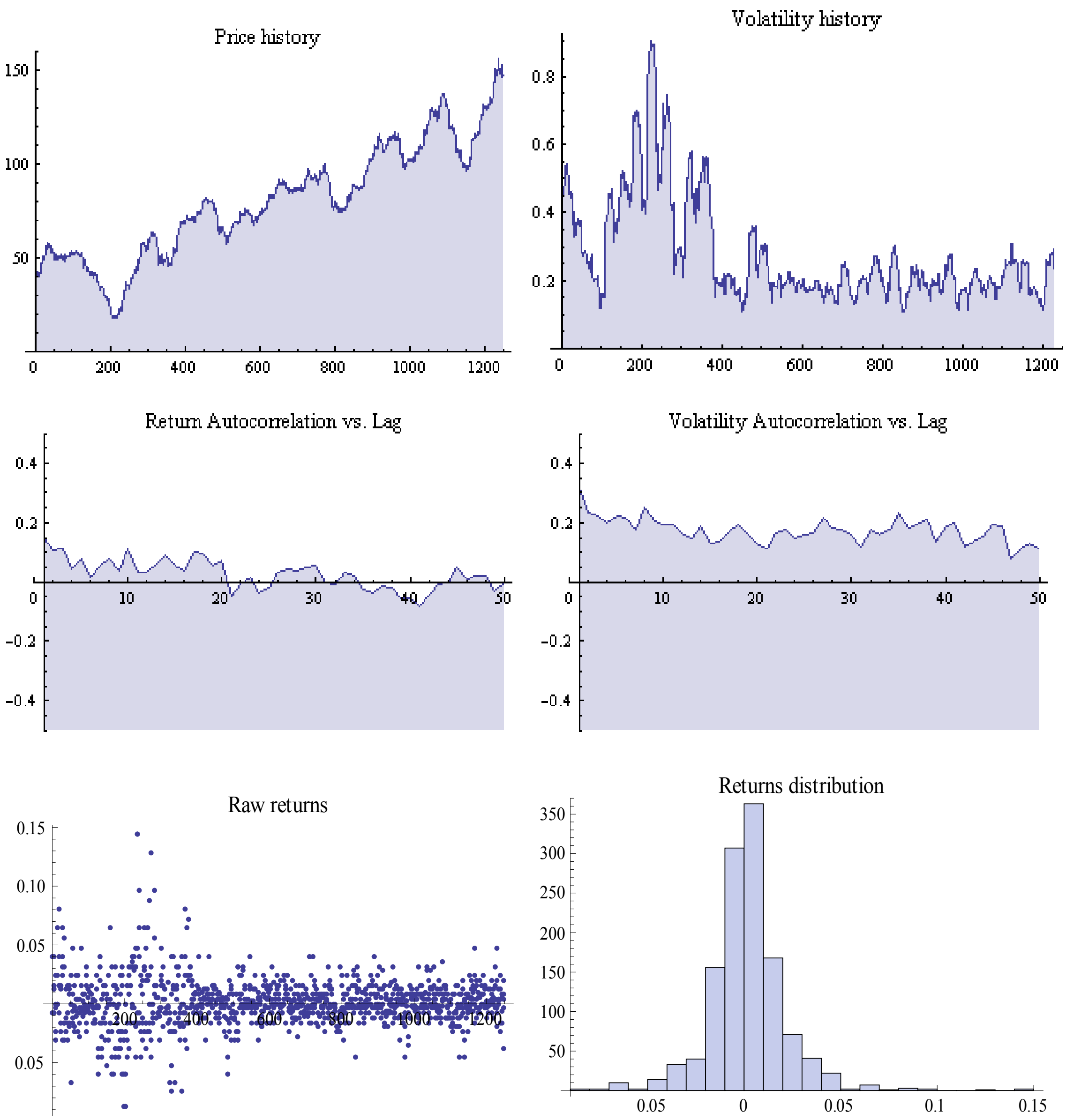


Figure 7.2.1 Prices (Up Left) , Volatilities (Up right) , Return Autocorrelation (Middle Left) , Volatility Autocorrelation (Middle Right) Returns (Down Left), Histogram (Down Right)

{mean annual return: 0.380617},
{standard deviation: 0.319495},

{returns kurtosis: 10.6088},
{volume-Vol correlation: 0.988863}

Multiple runs with the same strategy set

traders = {{6192,12,{0},{0}},{4060,3,{0},{0}},{6480,6,{0},{0}}};

data = Table[evolve[{traders, N@TrendUpBelowAve}, 750,.001,25],{5}];

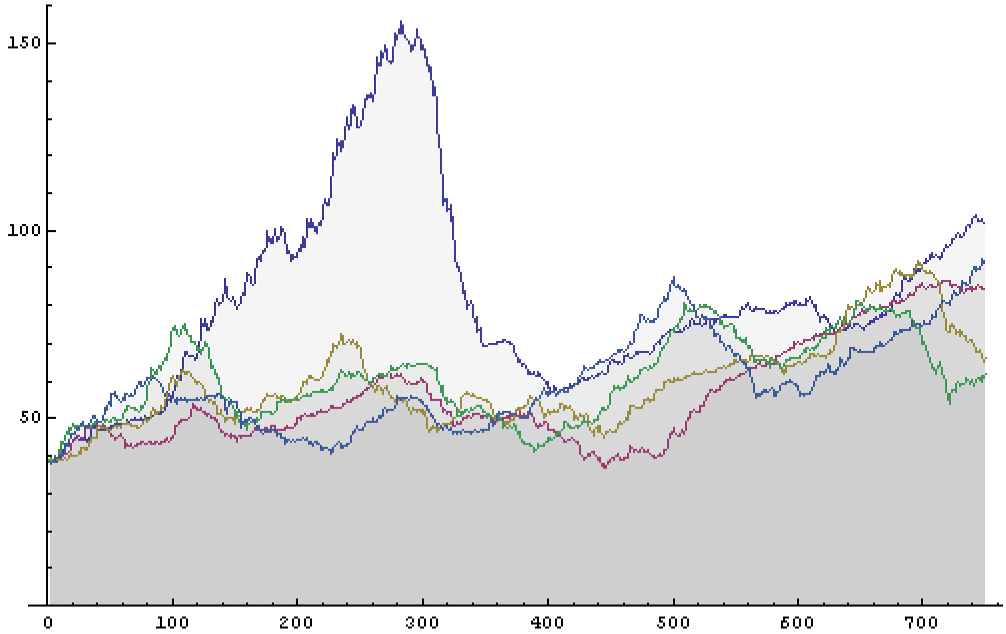


Figure 7.2.2 - 4 different price evolutions

Behaviour variety from different strategy sets

Reasonably stable around trend growth

ListLinePlot[Last@tworun, Filling -> Bottom, PlotRange -> {0, All},

 ImageSize -> 500, PlotLabel -> tworun[[1, All, 1 ;; 2]]]

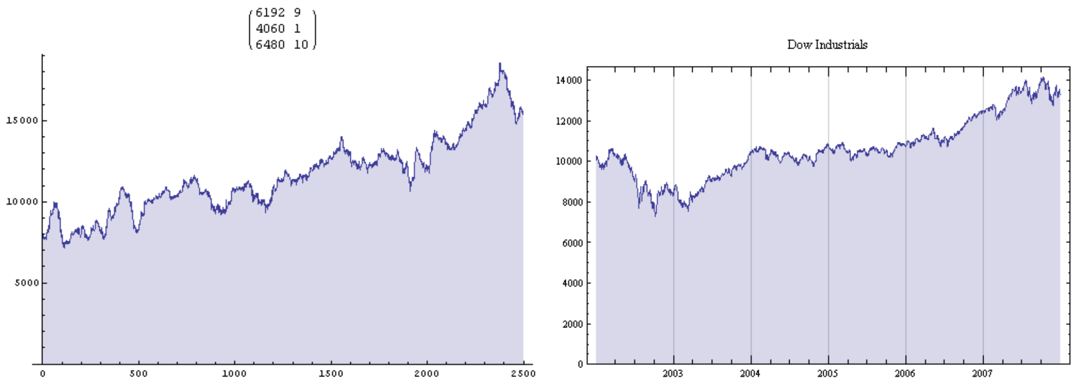


Figure 7.2.3 - NKS price evolution (Left) – Dow Jones (Right)

Choppy and directionless

ListLinePlot[Take[Last@threerun/250, -1250], Filling→Bottom, PlotRange→{0,All},

ImageSize→500, PlotLabel→ threerun[[1,All,1;;2]]]

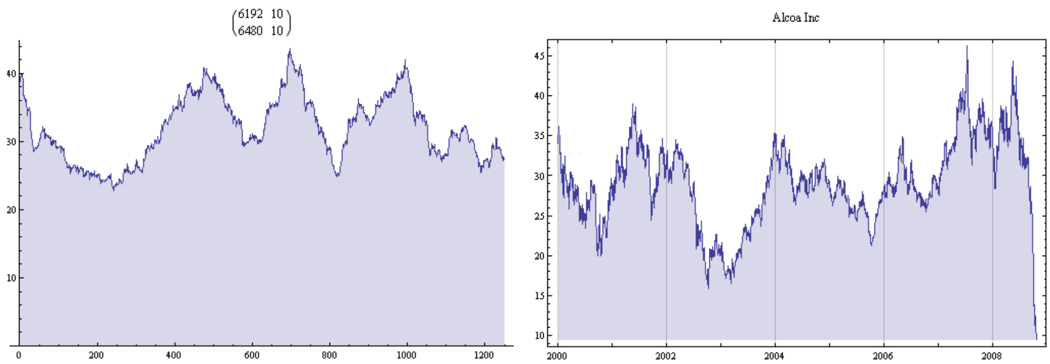


Figure 7.2.4 - NKS price evolution (Left) – Alcoa (Right)

Trend-following bubble

ListLinePlot[Take[Last@runfour/270, 1000], Filling -> Bottom, PlotRange -> {0, All},

ImageSize -> 500, PlotLabel -> runfour[[1, All, 1 ;; 2]]]

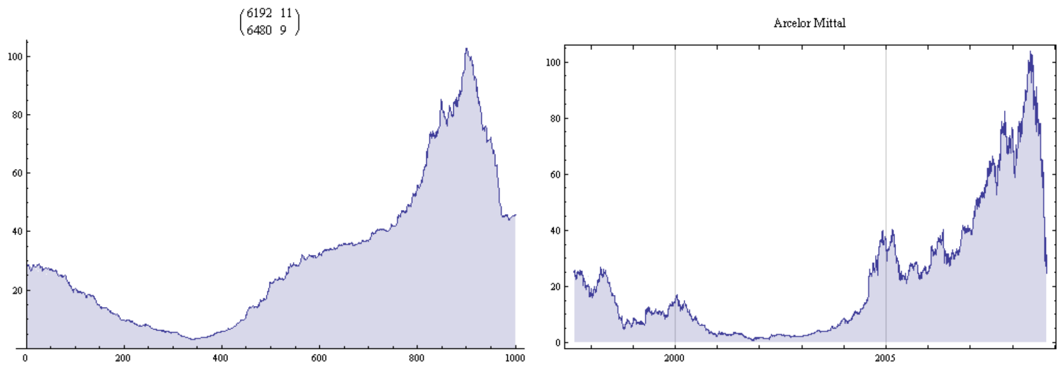


Figure 7.2.5 - NKS price evolution (Left) – Arcelor Mittal (Right)

7.3 Analysis and implications

The model can reproduce **stylized facts** of real markets that are absent from lognormal random walks. The internal mechanisms creating those stylized facts are at least plausible. These may be taken as evidence that non - uniform trading strategies and their different feedback effects on prices may be operative and important in real markets.

The wide variety of possible **behaviours** seen in the model suggest that real markets operate in only some subset of the space of possible strategy weights theoretically possible in this general a model. Some runs look like South Sea bubble series, many look like more common recent price series, some look very stable with just a little high frequency noise, etc. These regimes are sensitive to the overall levels of power of the different indicators. Some of this variety is undoubtedly present in real markets, and varies over time.

Trend following is a destabilizing indicator to trade on, and markets heavily influenced by trend following strategies naturally produce bubbles. Realistic bubbles are seen near the transition to trend - following dominance. When trend following outweighs all other strategies combined, one instead gets unrealistic sustained exponential behaviours, in either direction. Arguably even some of those are occasionally realistic, e.g. in hyperinflations.

Quite small weights of **level trading strategies** strongly stabilize markets, especially when enough news or random trading exists to occasionally outweigh trend following and break existing trends. At random breaks in trends, level traders steer a momentarily uncertain market toward rational, stable behaviour.

The main difficulty simply advocating level strategies to the exclusion of all other methods is that it is hard to rationally assess reasonable trend rates and levels. But this is much easier for large aggregates than for individual commodities or single companies. Long run aggregate stock prices are anchored to corporate profits which are anchored to GDP which is anchored to money supply etc. Thus it might be possible to meaningfully stabilize markets by rational level trading in broad market indices, on a modest scale.

All strategies other than trend following have some tendency to damp its effects. Direct anti-trend strategies simply reduce trend following weight one for one. Strategies merely insensitive to trend will occasionally weaken trend signals or counteract them, if large enough in overall weight for their random variations to match or exceed the scale of trend others are following. Level strategies have the strongest stabilizing effect.

Trend following strategies are not profitable in the long run in the bubble markets they create. This means eventually trend followers must lose their market-weight, eventually allowing other strategies to remain and to stabilize markets. But their scale may exceed anything one wants to rationally put up with.

In the short run trend following may appear profitable and indeed may exceed the profits of other strategies, especially before completing round trips to cash. For some early movers they may be profitable even over whole cycles. Along with their ease of use, these characteristics mean that trend following will remain endemic to markets. **Rational** investors therefore need means of detecting bubble periods, and market rationality cannot be relied on instead.

Anything that focuses incentives on longer time scales and absolute performance after full round trips to cash, away from relative performance on short scales and while still fully invested, will encourage profitable and stabilizing level strategies and reduce over-reliance on trend following strategies. In addition, predictability of average growth rates for level strategies (from e.g. macro and monetary stability) will tend to encourage more rational trading strategies.

It is our firm belief and markets provide many examples that markets operate in **regimes**, regimes "caused" by behaviour of market participants following their strategies. Our model is something that could bring light to these changes of regimes. It is clear that some market states provide "equilibrium" markets maybe a Orstein-Uhlenbeck equilibrium, other markets follow different behaviours consisting in bubbles and bursts, meaning very strong secular trends that change abruptly in what we call non-stable markets, markets that can reach a disrupted state.

8. Conclusions and further work

We think our NKS Artificial Market Model offers an adequate balance of describing how real agents behave and interact as well as the market paths and stylized facts are agree perfectly with real markets. The model is using deterministic and stochastic components that are able to mimic market complicated behaviour.

We have made not only the simulation but also the analytical effort using a simple equation model that has for some cases well known analytical solutions: Geometric Brownian Motion, Orstein-Uhlenbeck models and mixtures of both. So model output and insights might be well reconciled with derivatives pricing and stochastic calculus techniques.

A tremendous effort has been made by econophysicists and applied mathematicians to find a model in which is possible to use the apparatus of stochastic calculus, dynamical systems, time series analysis and scaling to understand better the nature of financial markets.

The extension of the work may bring us to:

- **Estimation of parameters** in real world markets. Feasible for some of the parameters we have introduced in the model as fundamental levels (analysts views), trend following levels,etc... Clearly given the high number of degrees of freedom this task should be done very carefully avoiding **overparametrization**. Reducing then the degrees of freedom and finding observables is a task needed.

- The equation I labelled **Simple Market Equation** is very rich in its behaviour and much more work could be done in terms of studying the dynamics of it. Because it links the agents' behaviour and dynamic and stochastic processes theory has a great interest for the researchers.

- **Predictability**. This Artificial Market is predictable in some regimes and some state of the parameters. It is intuitive to figure out when with our model. We define predictability as finding techniques to obtain better risk-return ratios than the market. Non-Linear techniques provide a rich variety of tools to crack the markets in terms of the risk-return. Detecting the regimes mentioned before is a key aspect of predictability as different things should be done in different regimes.

- Developing **bubble and burst indicators** is a form of prediction, given the current state of the world much more work on that aspect will be devoted not only for investment purposes but also for macroeconomic policies and regulations. As a hint , our opinion is that markets that are far from an "equilibrium" risk-return ratios are firm candidates to be in a "bubble" state in one sense or another meaning "excess bubble" or "defect bubble".

This work has been written and developed during the exceptional, rare, activated, fat-tail, black-swan events of October 2008. It is clear after all the events of 2008. In the light of these extraordinary events it is clear that much more work is needed to understand market dynamics as is very difficult for the financial system to handle this extreme events caused by several things and that can be understood by using artificial markets that are able to mimic real market stylized facts and behaviour.

In the figure we can see the S&P 500 from 1st January 1990 to 24th October 2008.

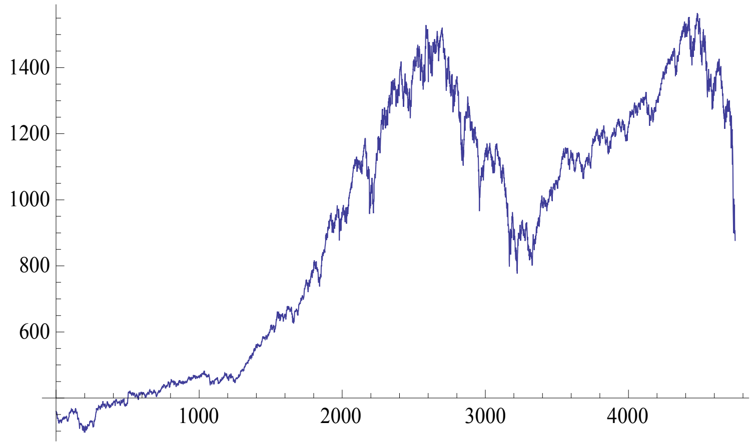


Figure 8.1 S&P500 from 1st jan 1990 to 24th oct 2008

I think at that point in time the most relevant effects have been studied and a complete set of tests is available on request.

As the model is very rich in terms of variety of traders that could be studied, different parameters that could be used, different stochastic models for news and trading orders please do not hesitate to contact me for further information and experiments. I will be more than glad to collaborate with other researchers.

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